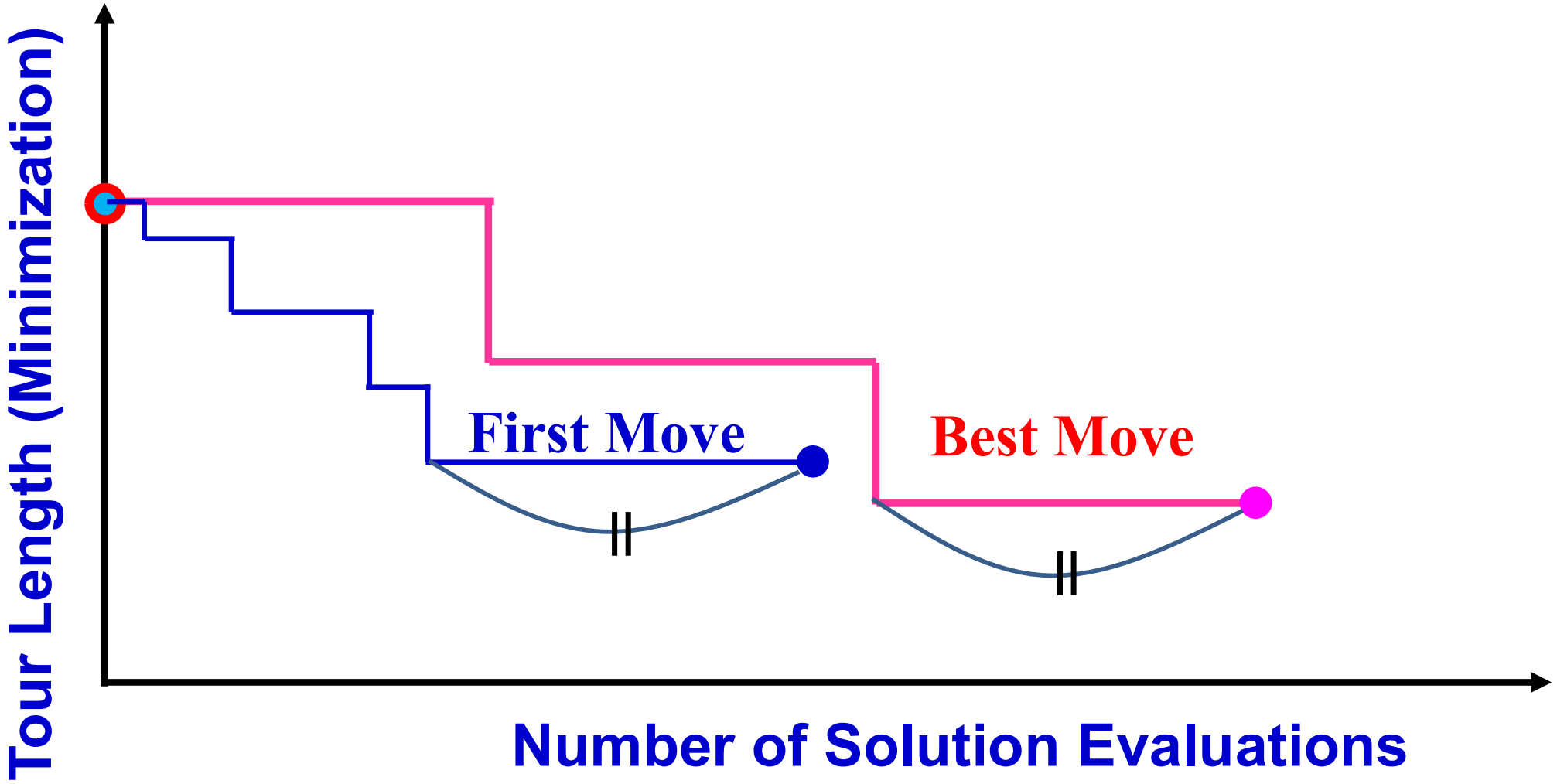
# Optimization Methods

1. Introduction.
2. Greedy algorithms for combinatorial optimization.
3. LS and neighborhood structures for combinatorial optimization.
4. **Variable neighborhood search, neighborhood descent, SA, TS.**
5. Branch and bound algorithms, and subset selection algorithms.
6. Linear programming problem formulations and applications.
7. Linear programming algorithms.
8. Integer linear programming algorithms.
9. Unconstrained nonlinear optimization and gradient descent.
10. Newton's methods and Levenberg-Marquardt modification.
11. Quasi-Newton methods and conjugate direction methods.
12. Nonlinear optimization with equality constraints.
13. Nonlinear optimization with inequality constraints.
14. Problem formulation and concepts in multi-objective optimization.
15. Search for single final solution in multi-objective optimization.
16: Search for multiple solutions in multi-objective optimization.

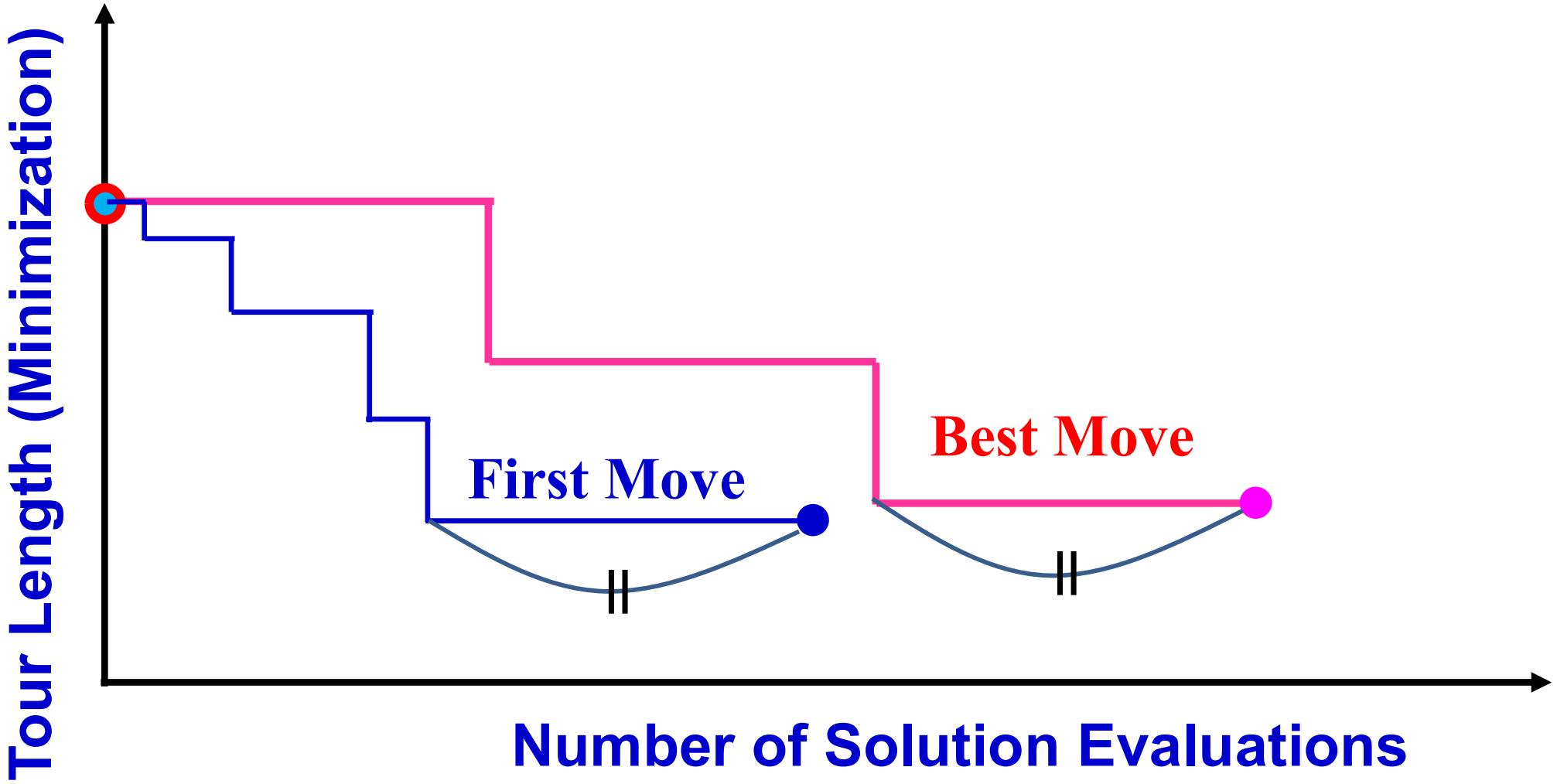
# Best Move and First Move: Which is better ?




(1) In general, the first move needs less computation load.

(2) With respect to the final solution quality, we cannot say which is better between these two strategies.

# Best Move and First Move: Which is better ?

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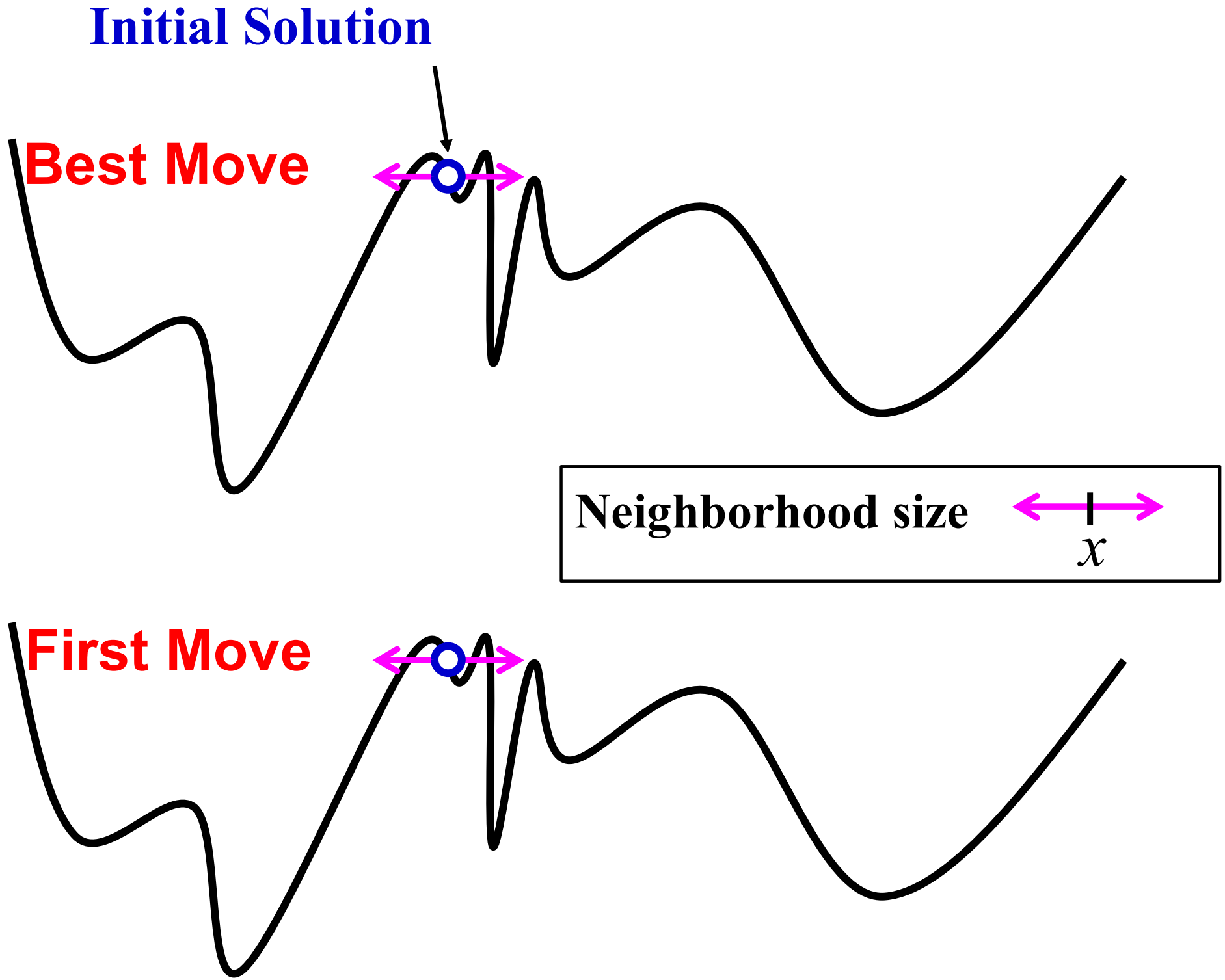
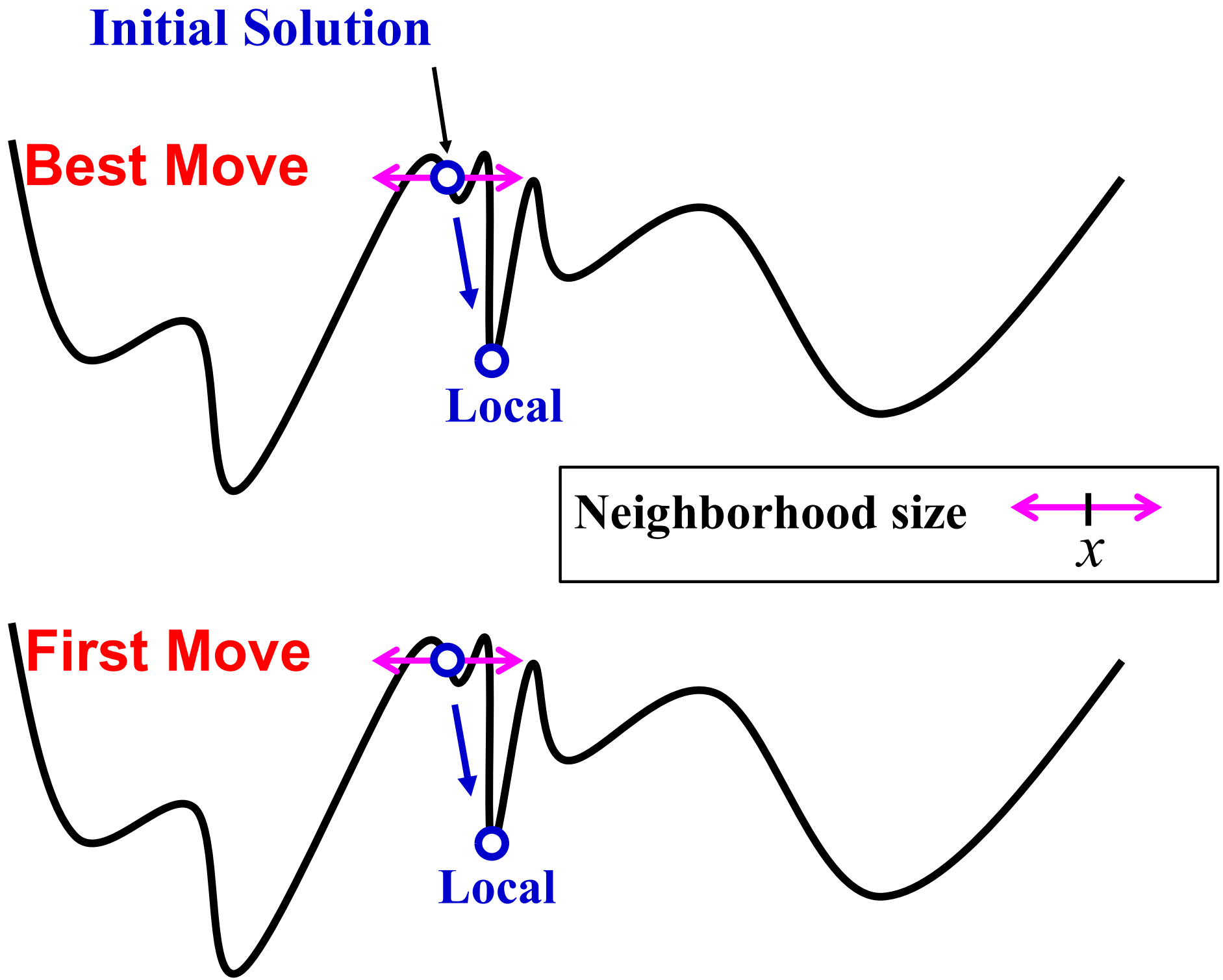
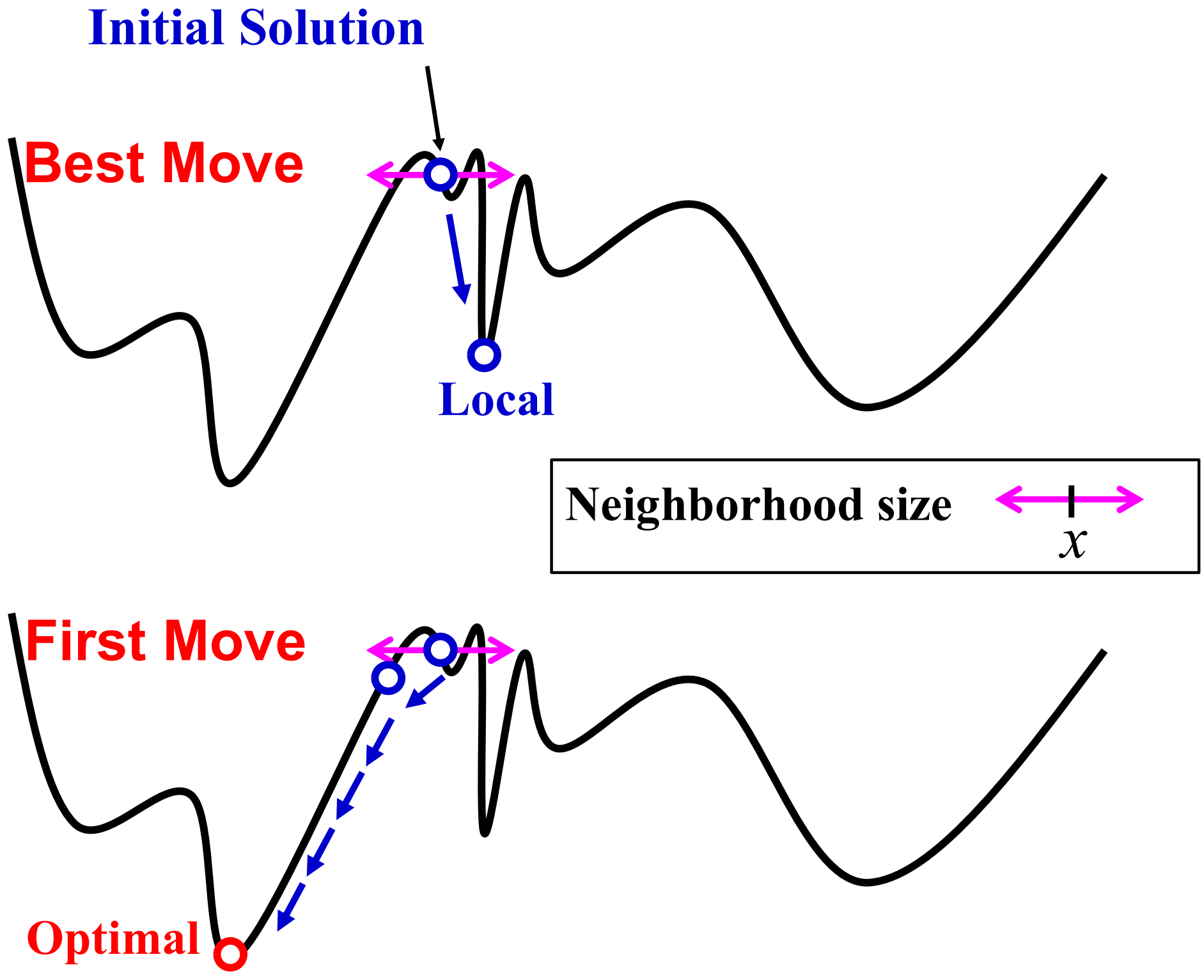

Initial Solution
Best Move
Neighborhood size
x
First Move


Initial Solution
Best Move
Local
Neighborhood size
x
First Move
Local


Initial Solution
Best Move
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Neighborhood size
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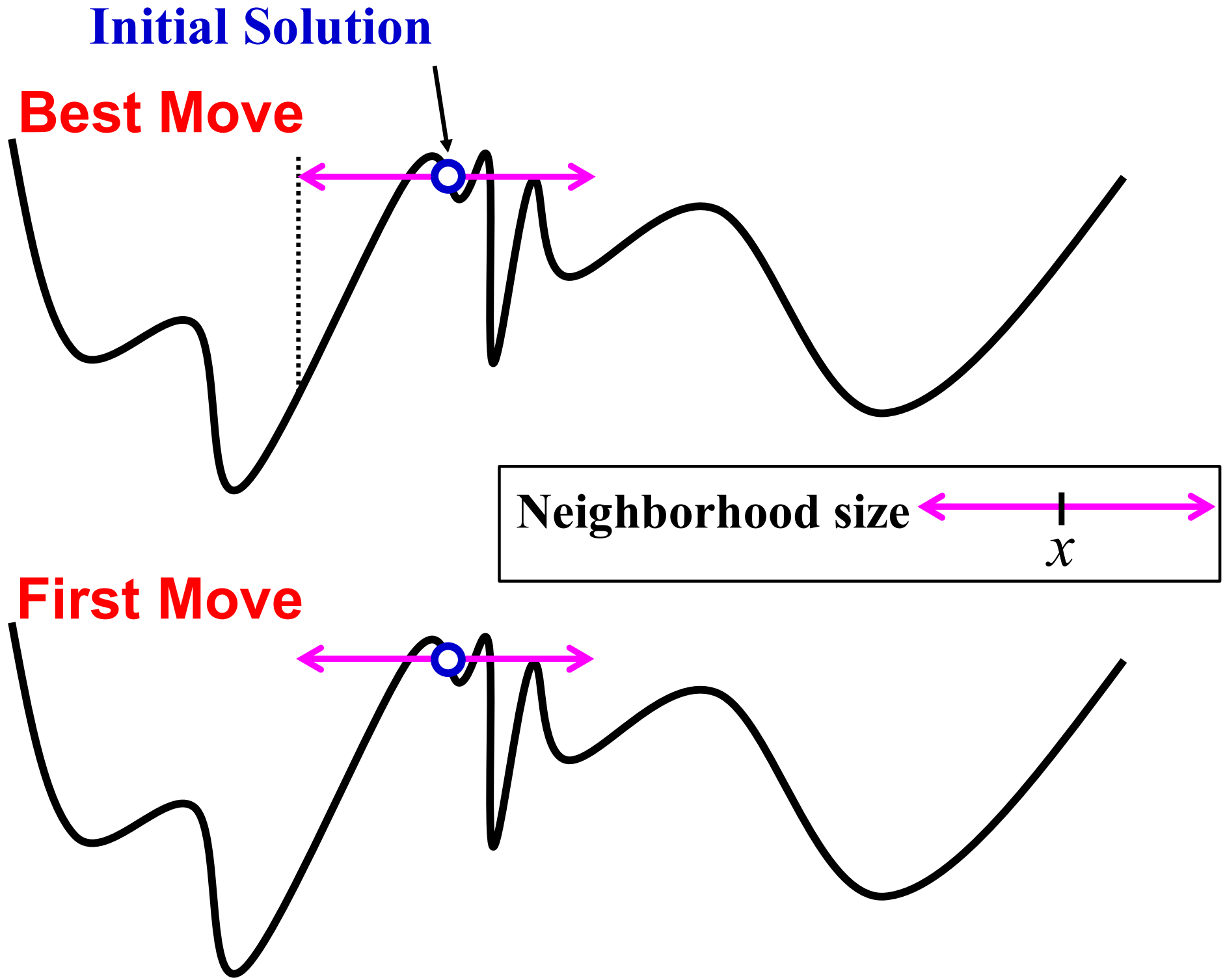
Optimal


Initial Solution
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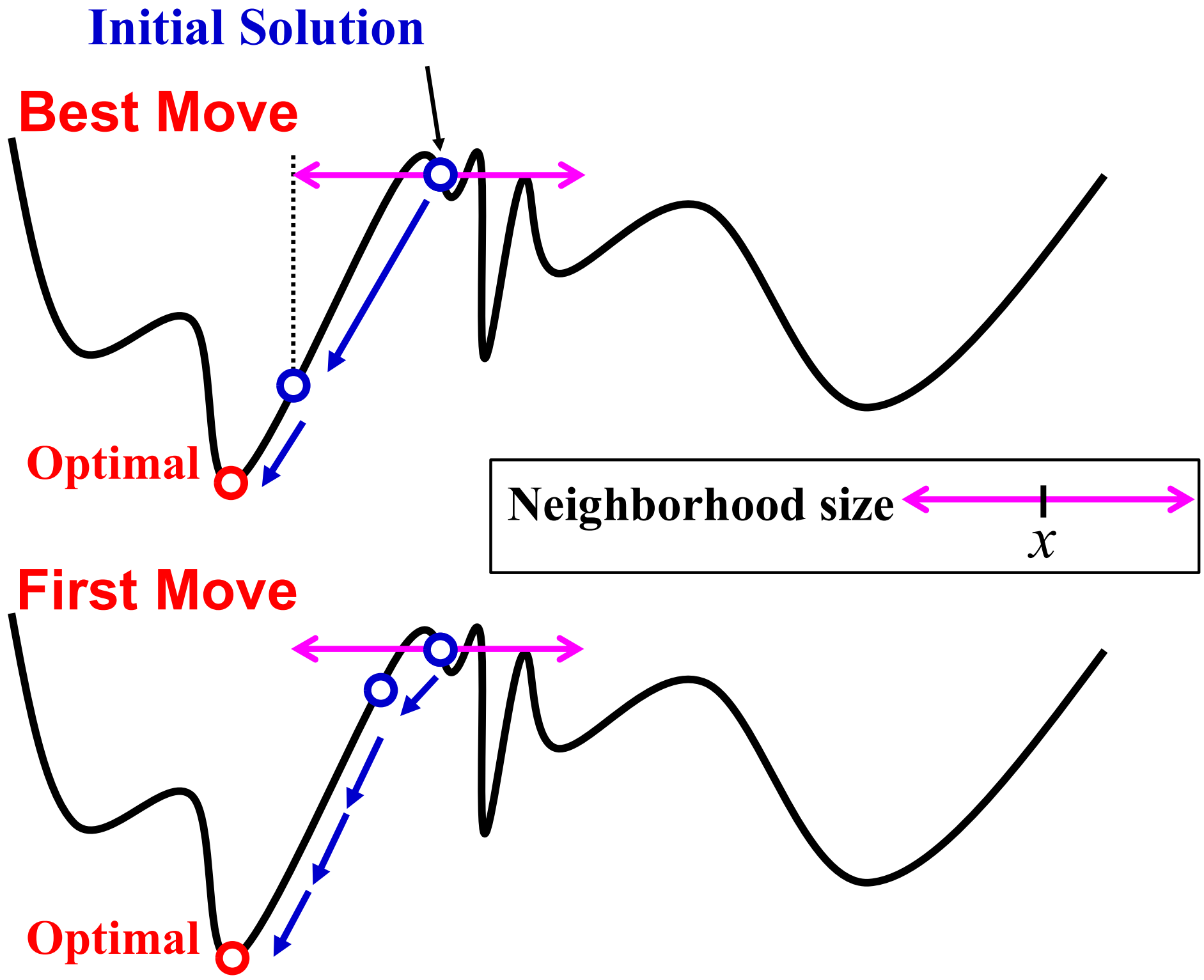
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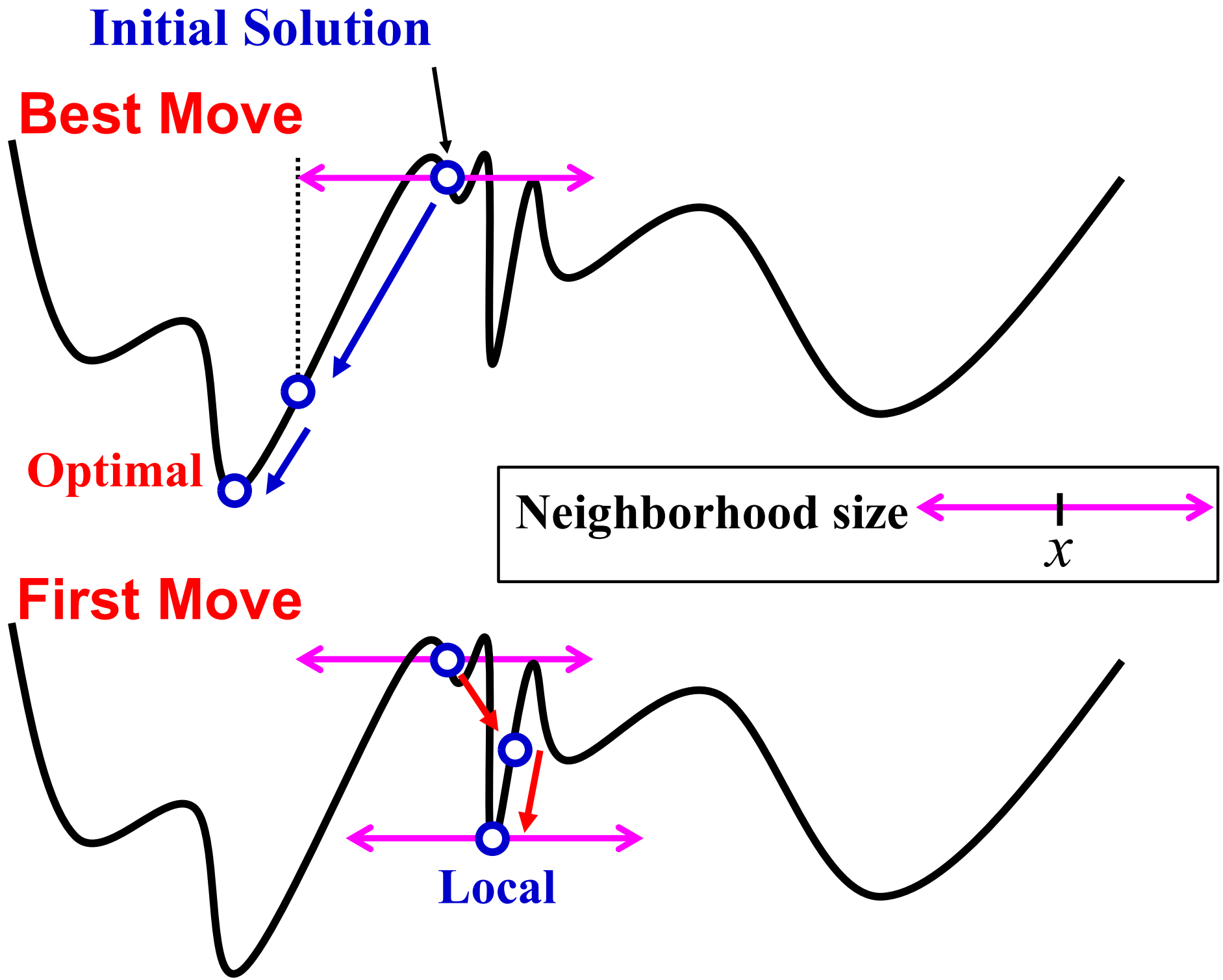
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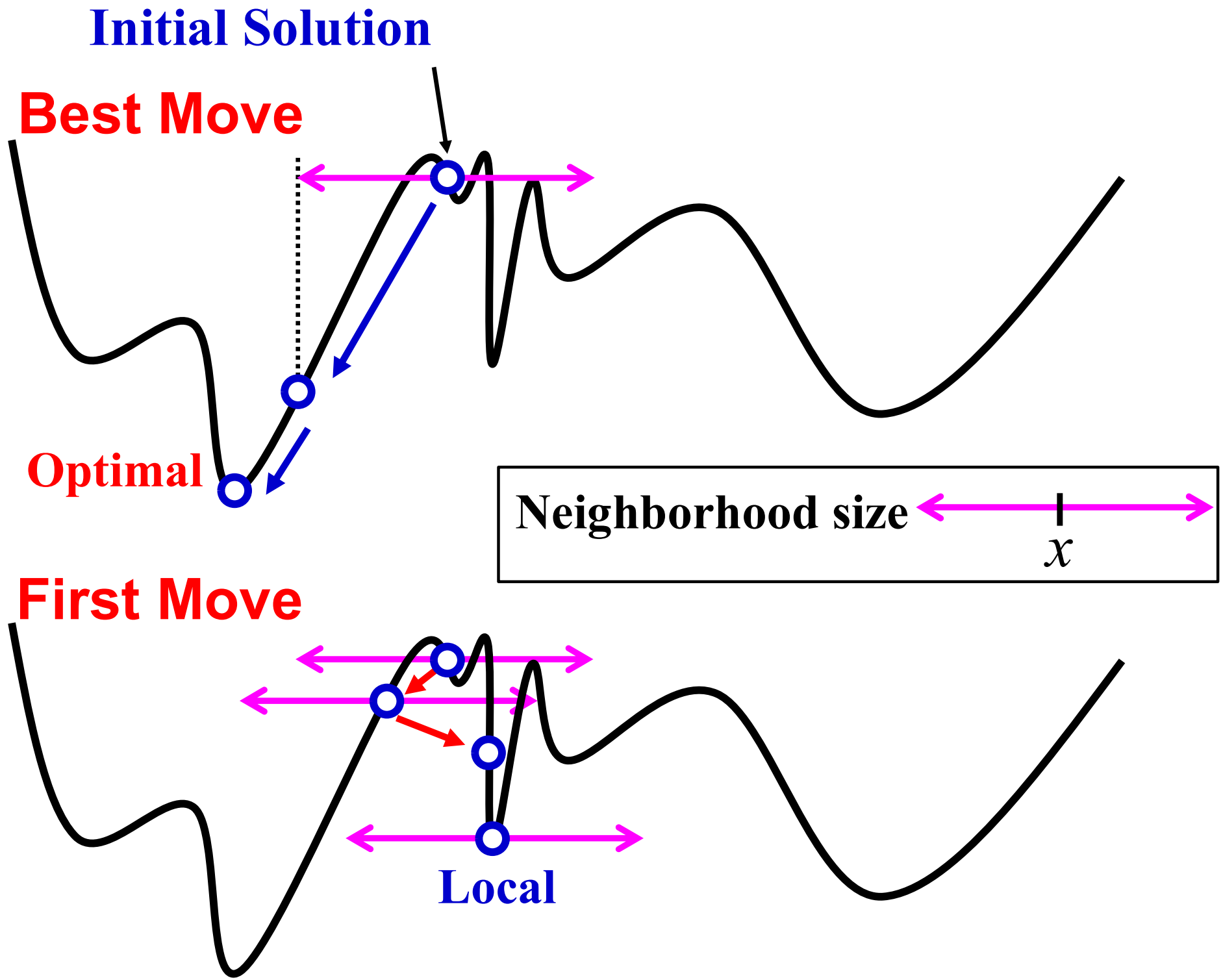
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Best Move
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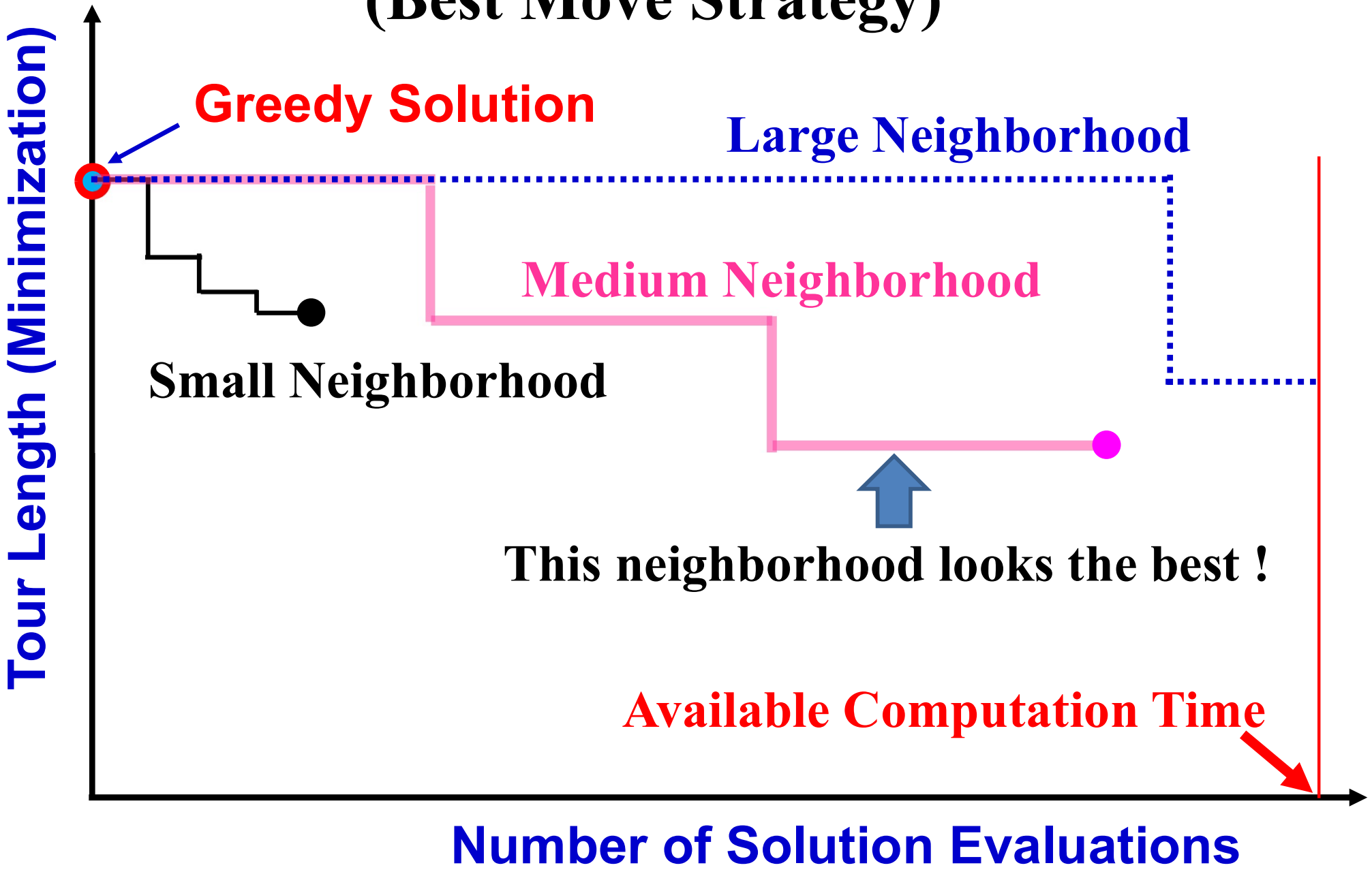
# Behavior of Local Search
## (Best Move Strategy)



# Behavior of Local Search
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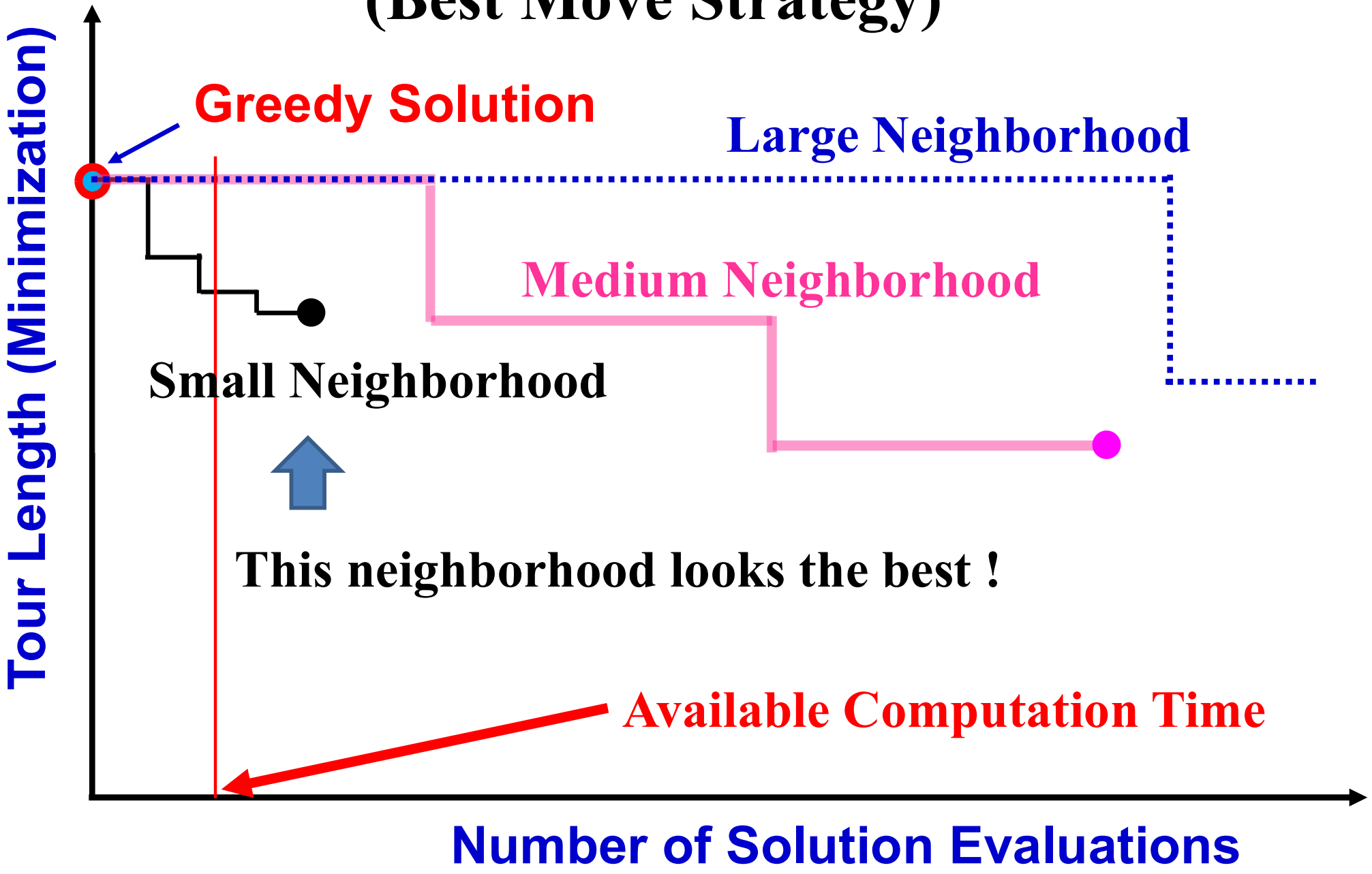

# Behavior of Local Search
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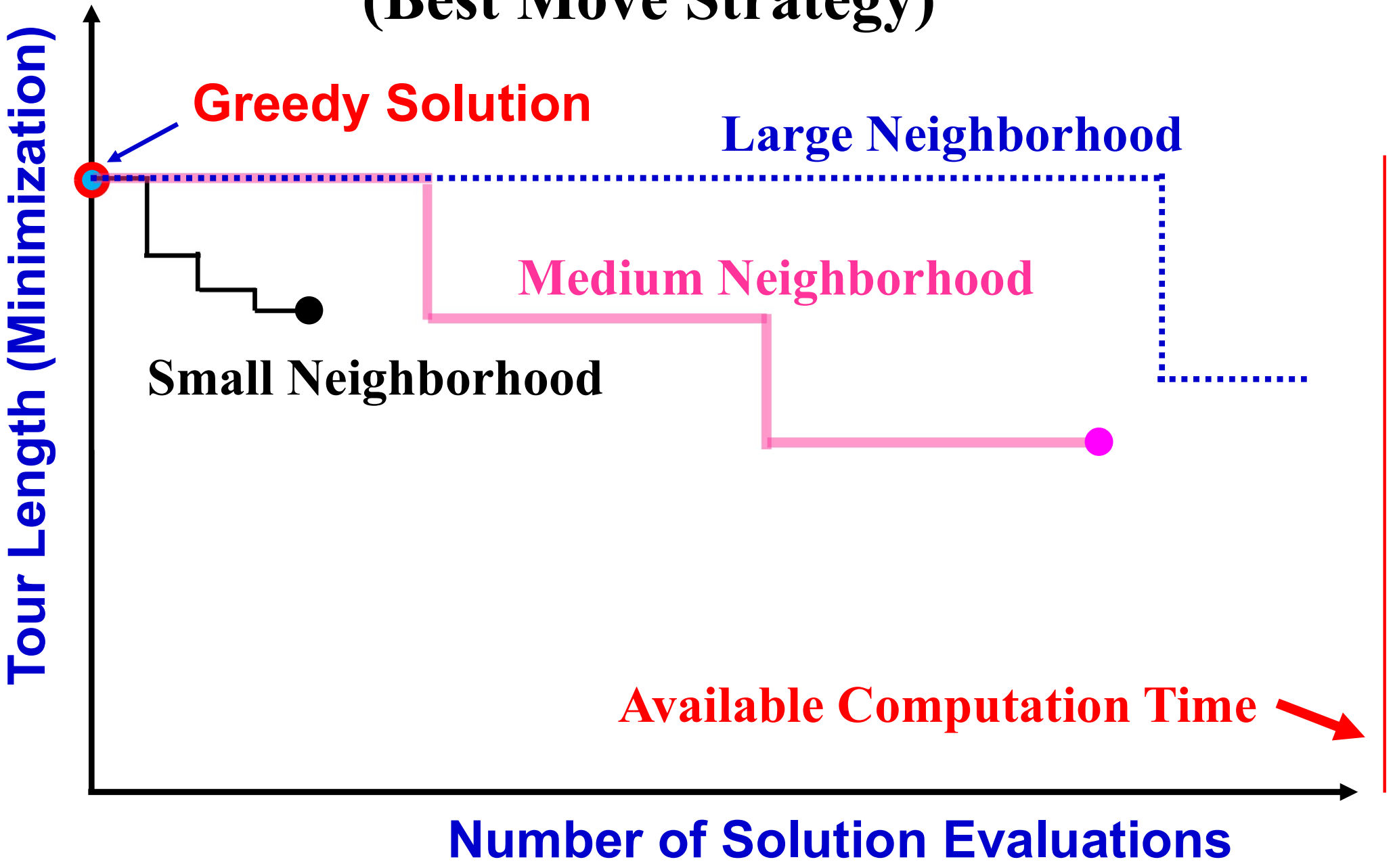


# Local Search

**Implementation Issues**

(1) Specification of an initial solution

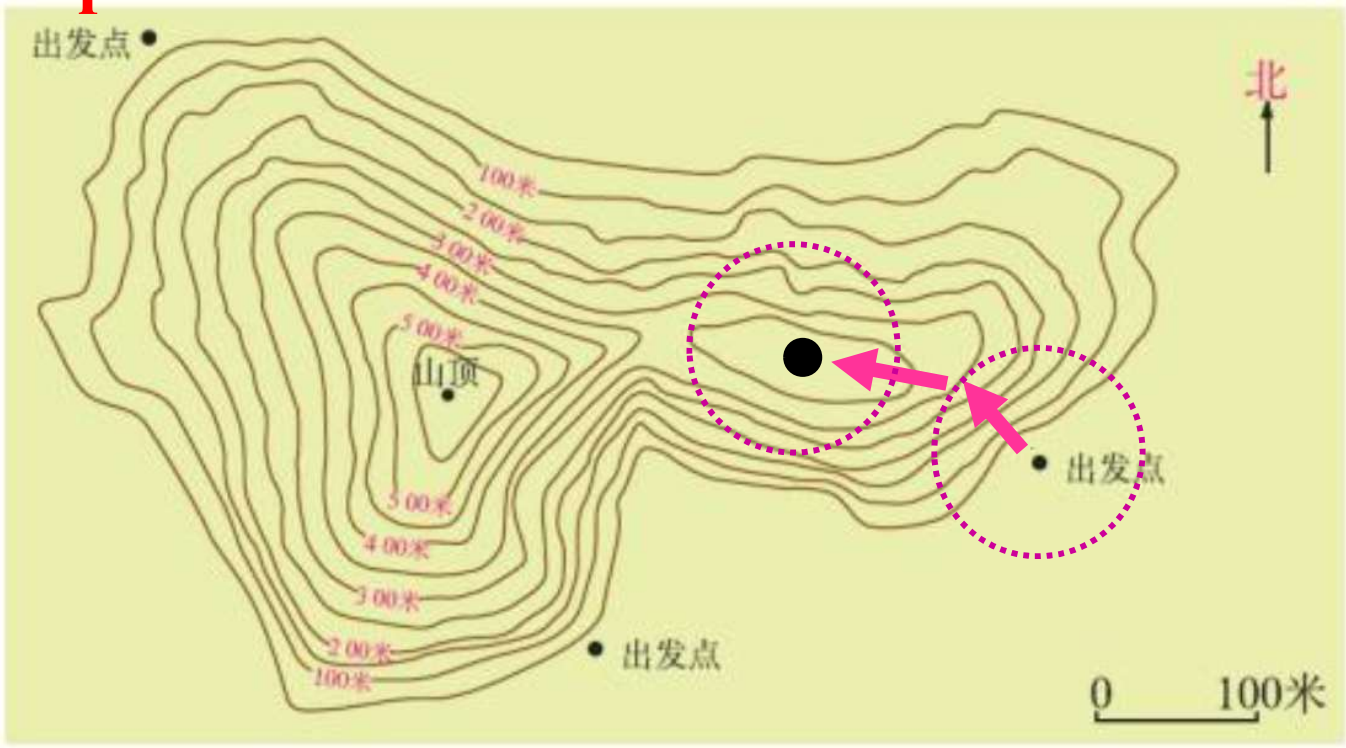
(2) Specification of a neighborhood structure

(3) Choice between the first move and the best move

⇩

**Local Solution**

**Question: How can we improve local search ?**



# Local Search

**Implementation Issues**

(1) Specification of an initial solution

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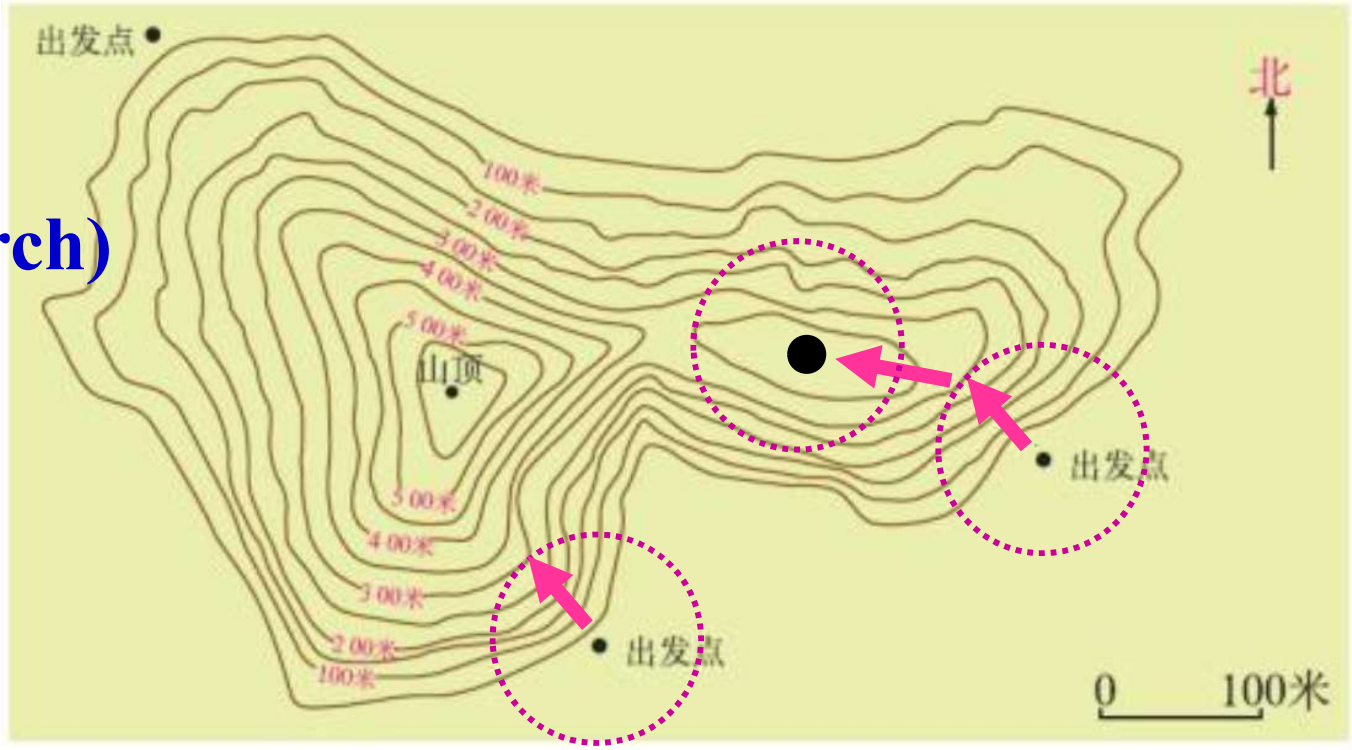
**Local Solution**

## Next Step

**- Restart**

**(Iterated Local Search)**



# Iterated Local Search

## Iterate LS from Different Initial Solutions

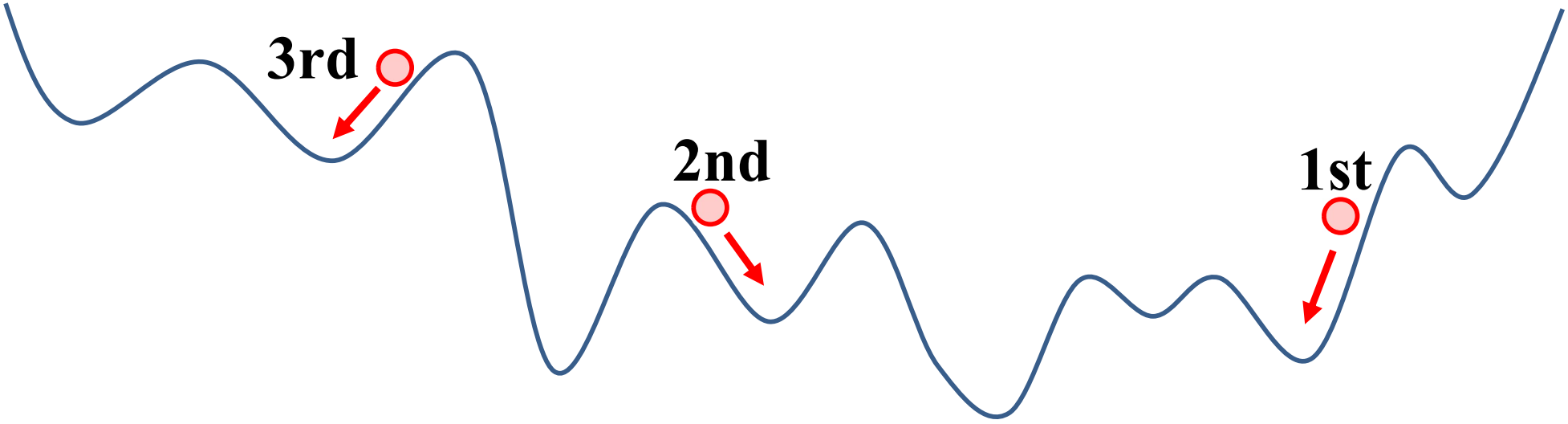
Iterate the following steps:

(1) Generate an initial solution $x$.

(2) Iterate the following steps:

(i) The first move or the best move (if a neighbor $y$ of the current solution $x$ is better than $x$, replace $x$ with $y$).

(ii) If no better solution exists in the neighborhood of $x$, terminate the current execution of LS (i.e., restart with a new initial solution $x$ in (1)).



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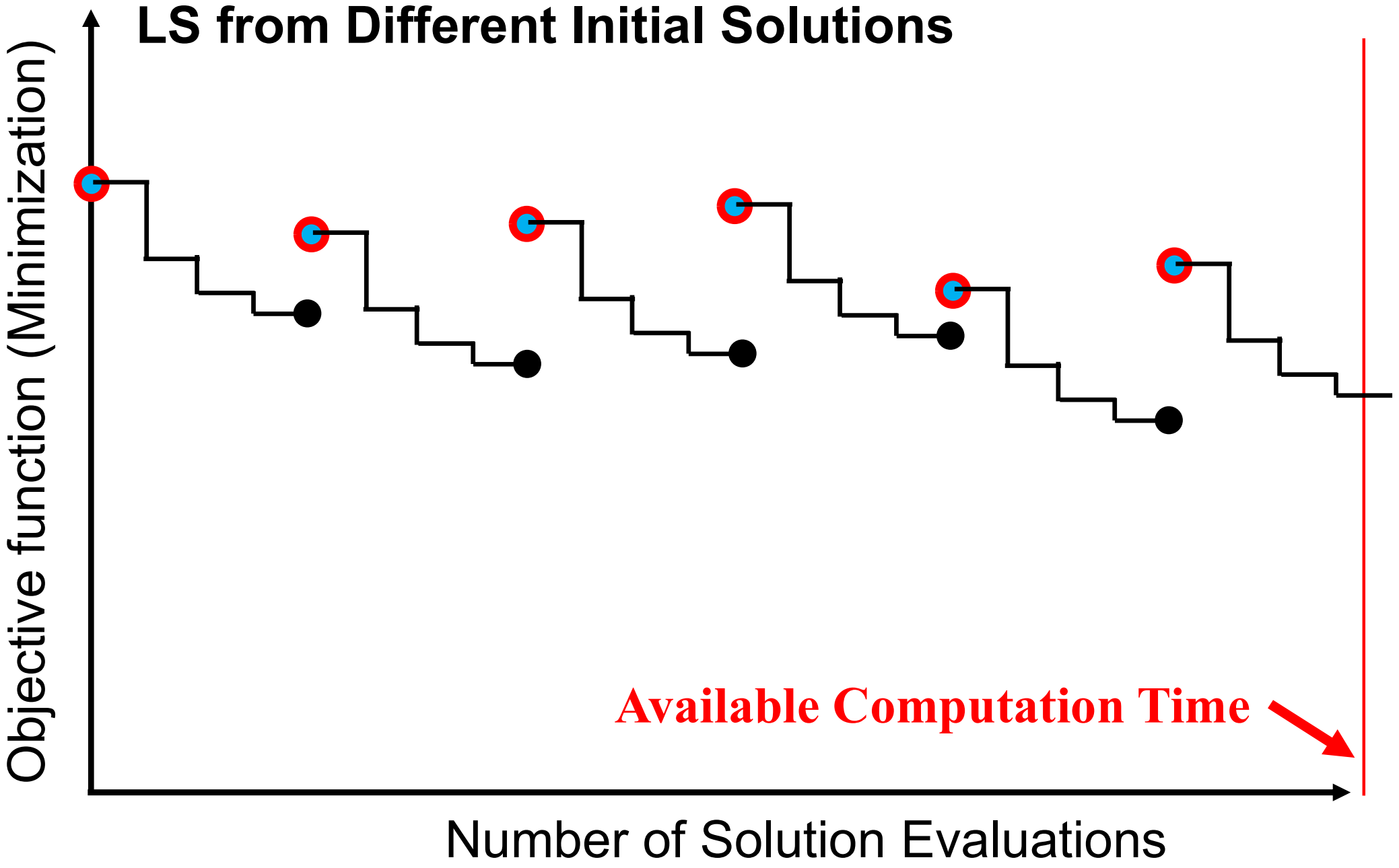

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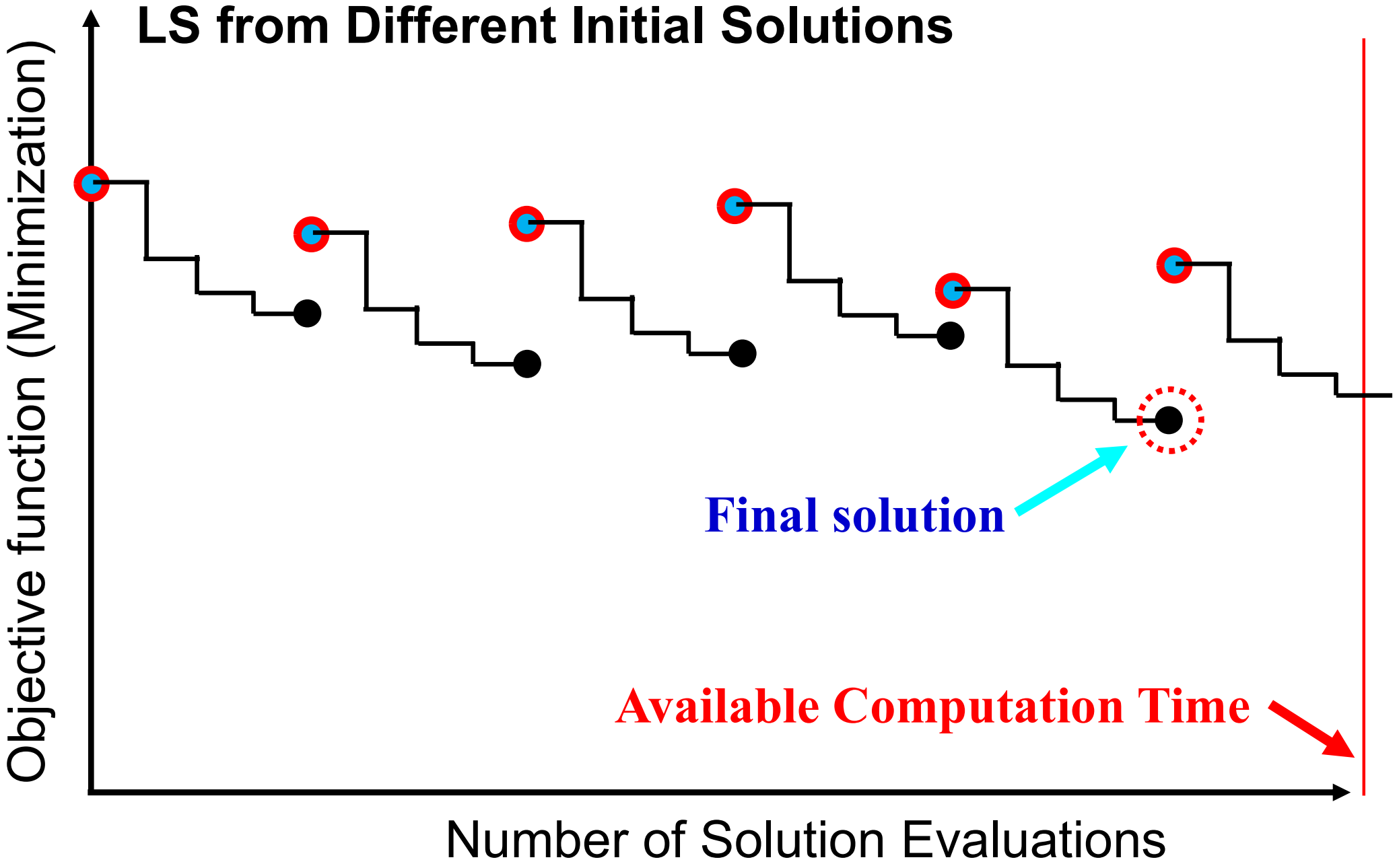

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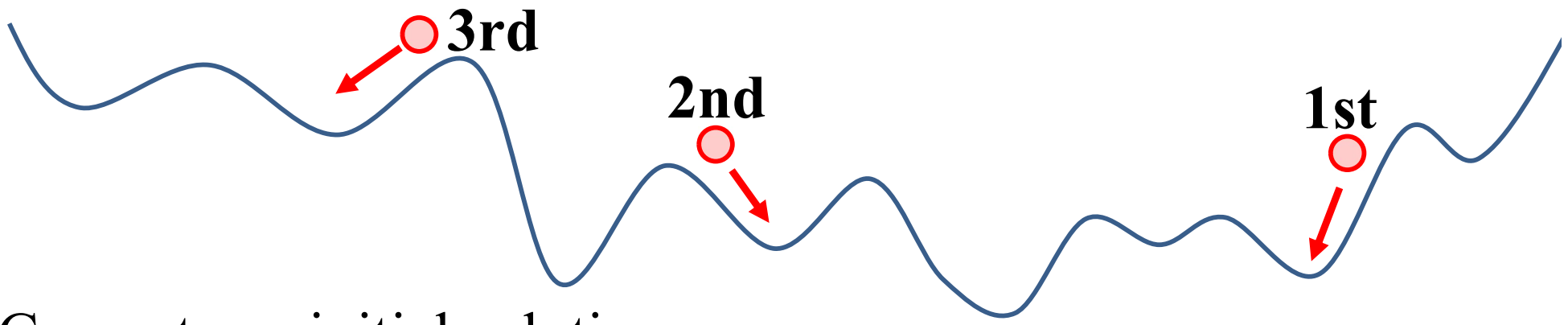
**Discussions:**

How to specify an initial solution for each LS run in (1)?

- In the first LS run: **Greedy solution.**
- In the other LS runs: **??????????** .

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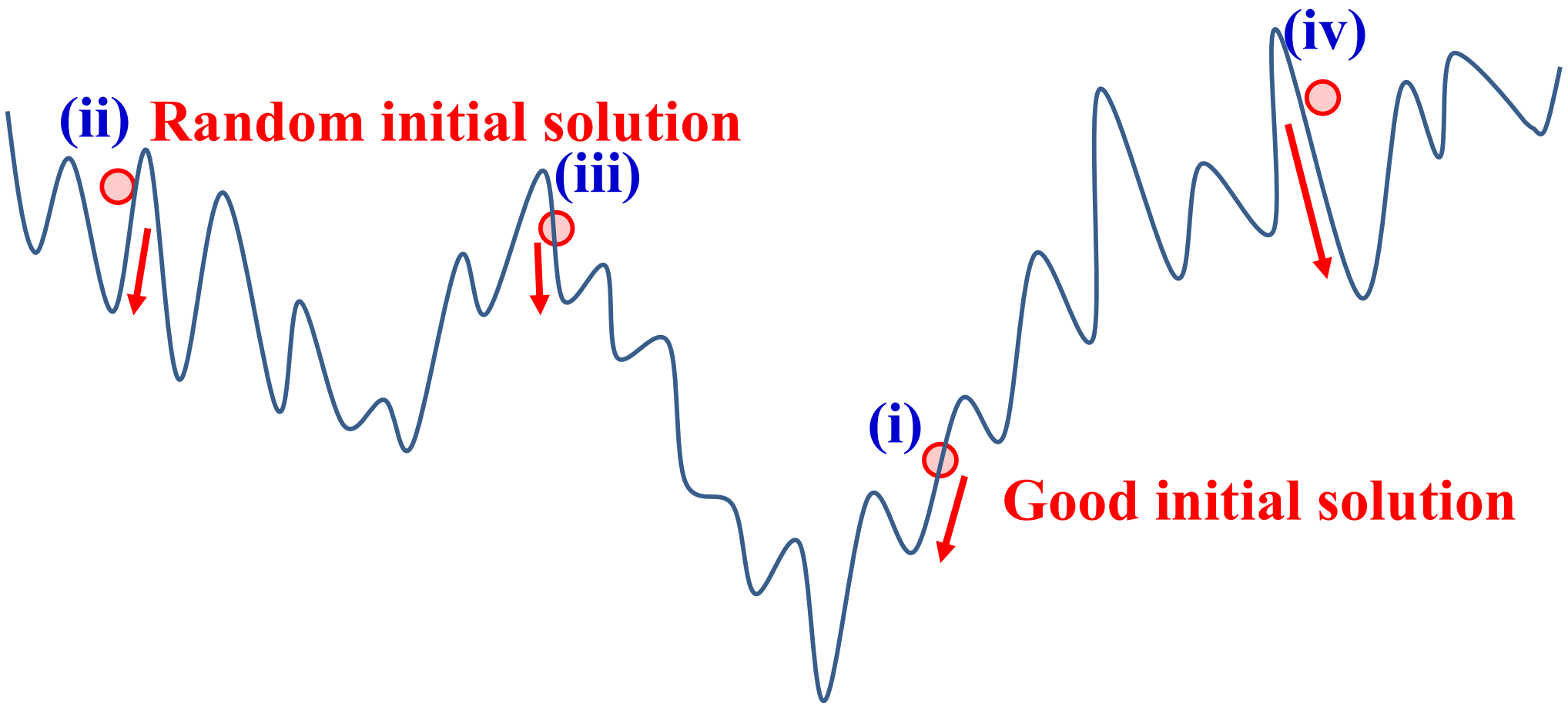
# Main Issue

## Choice of Initial Solutions:

- First initial solution: **Greedy solution**.
- Other initial solutions: **Random solutions**.



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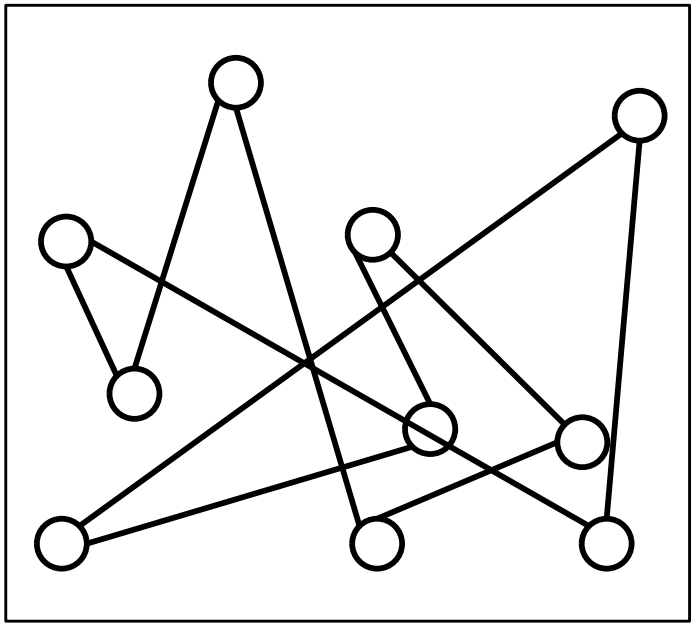

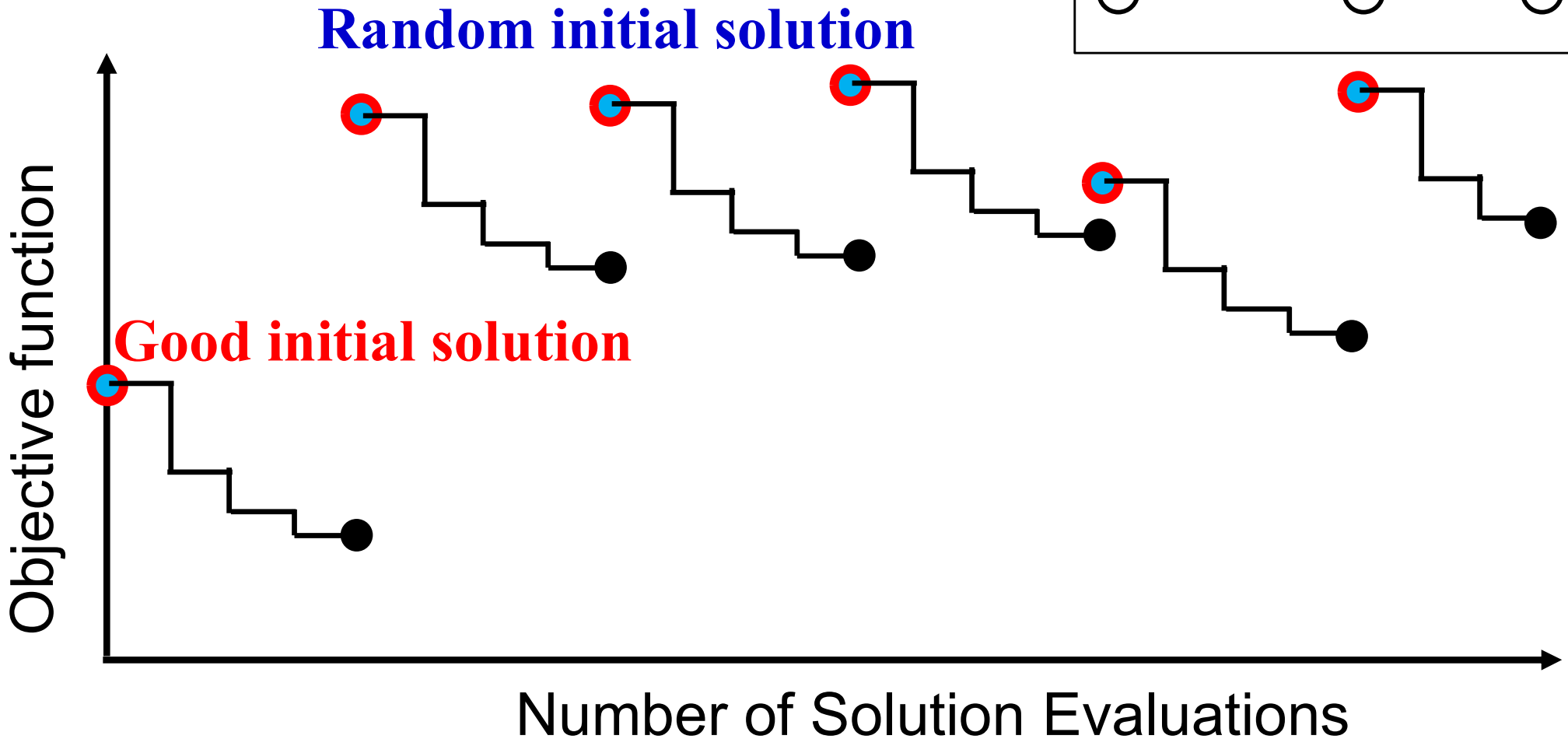

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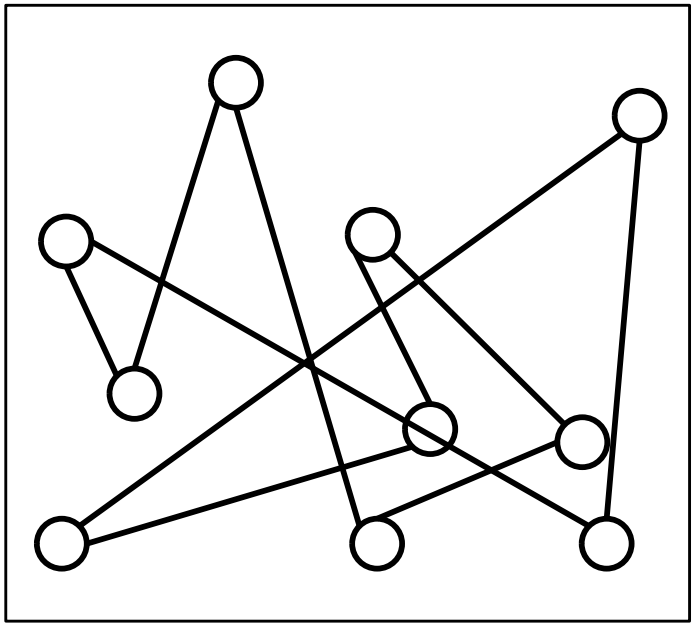

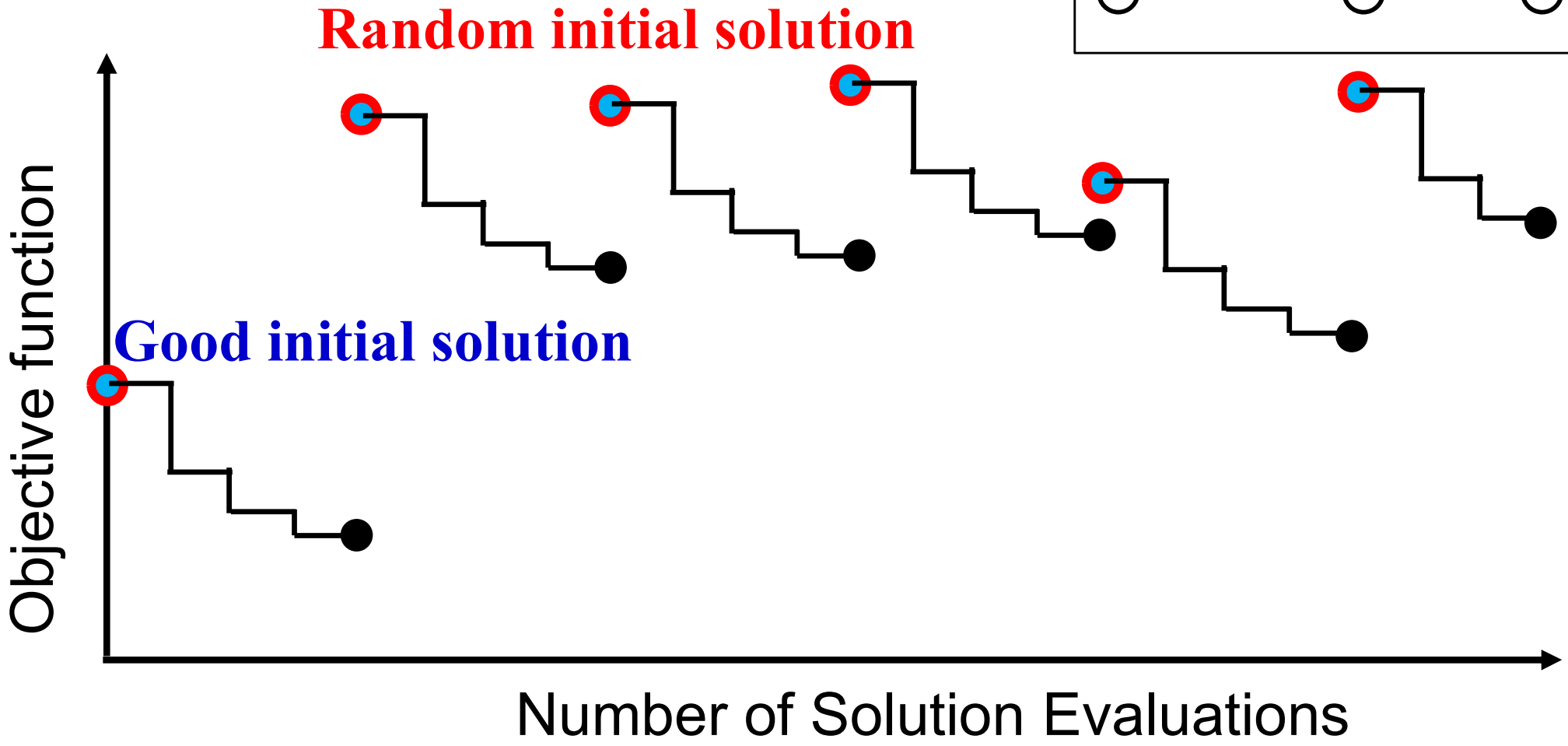

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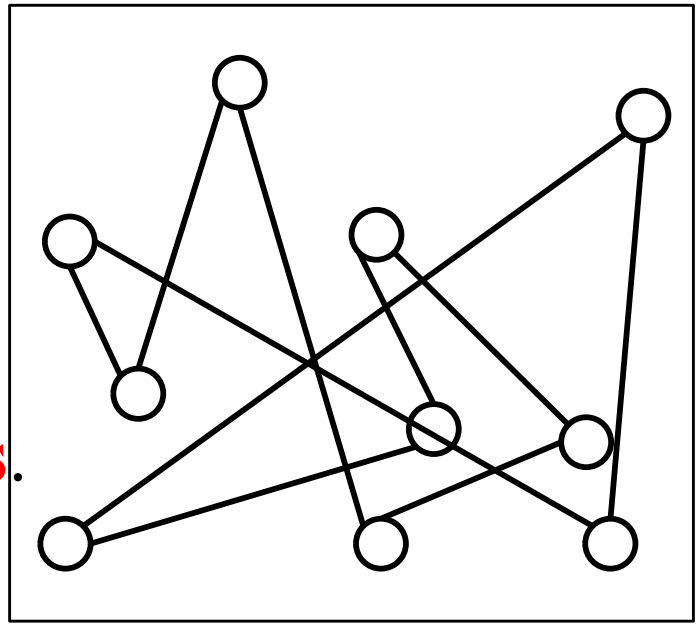

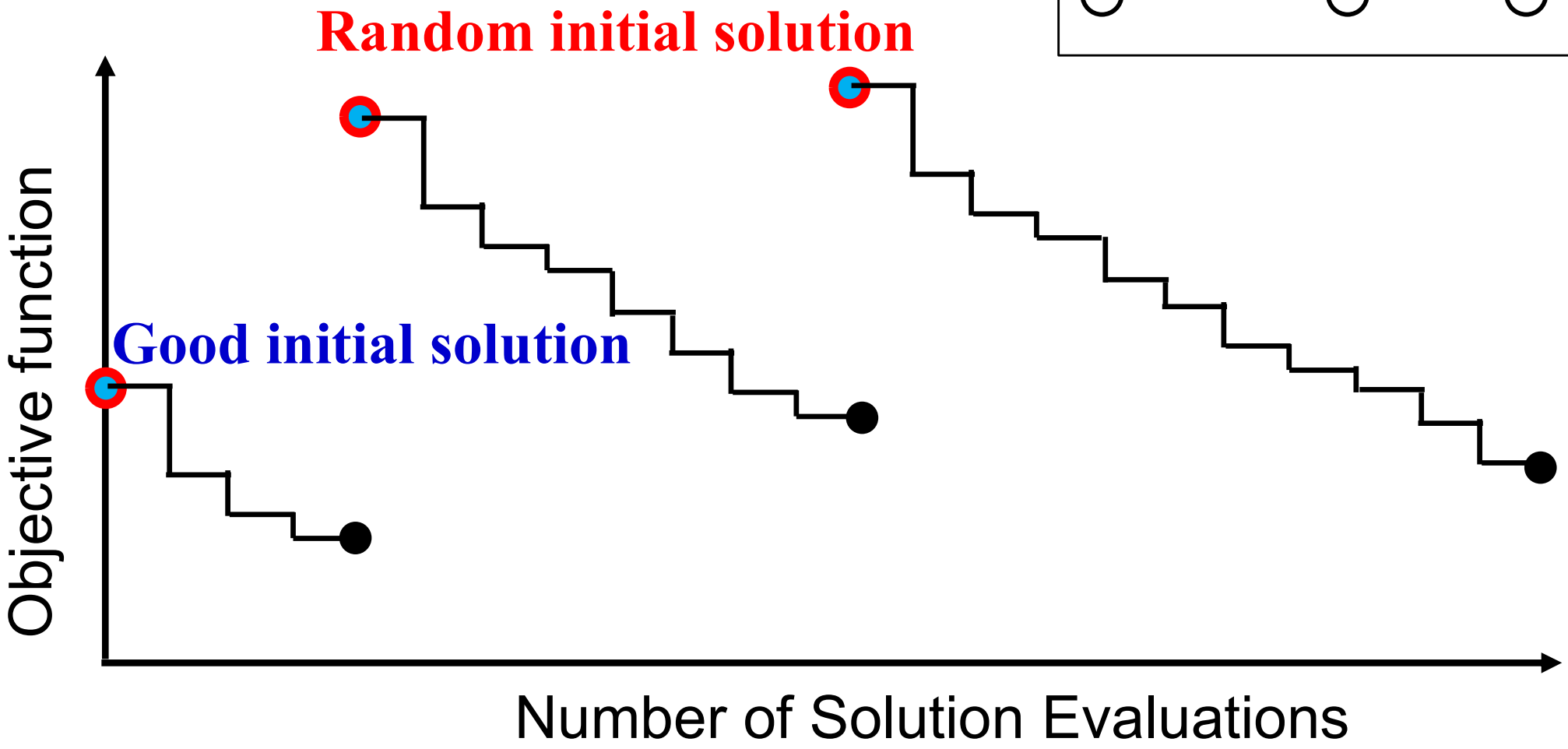

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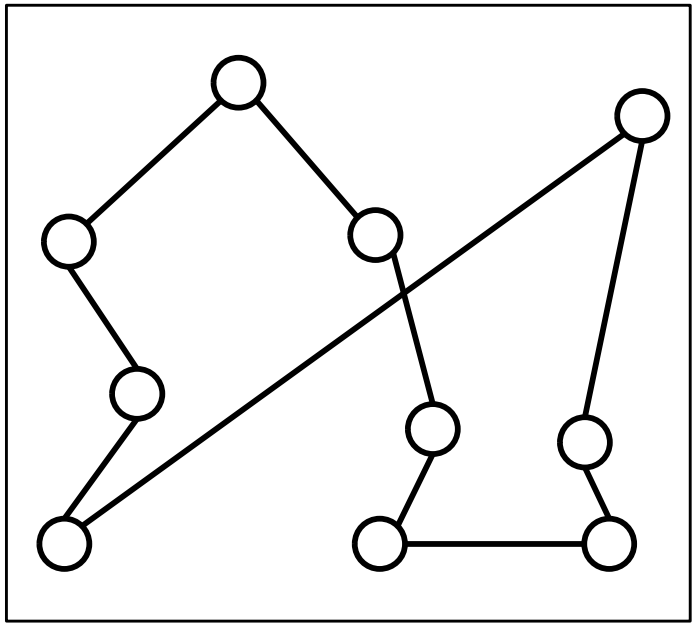
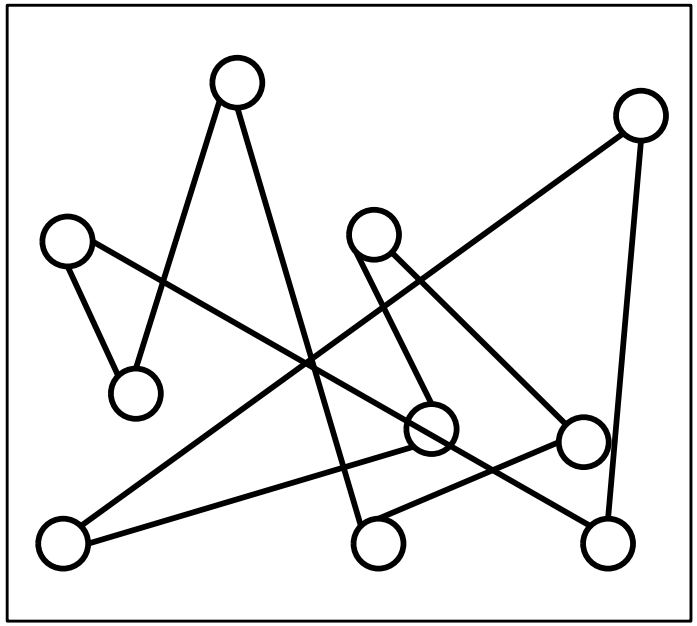


Good greedy solution



Poor random solution

# Main Issue

## Choice of Initial Solutions:

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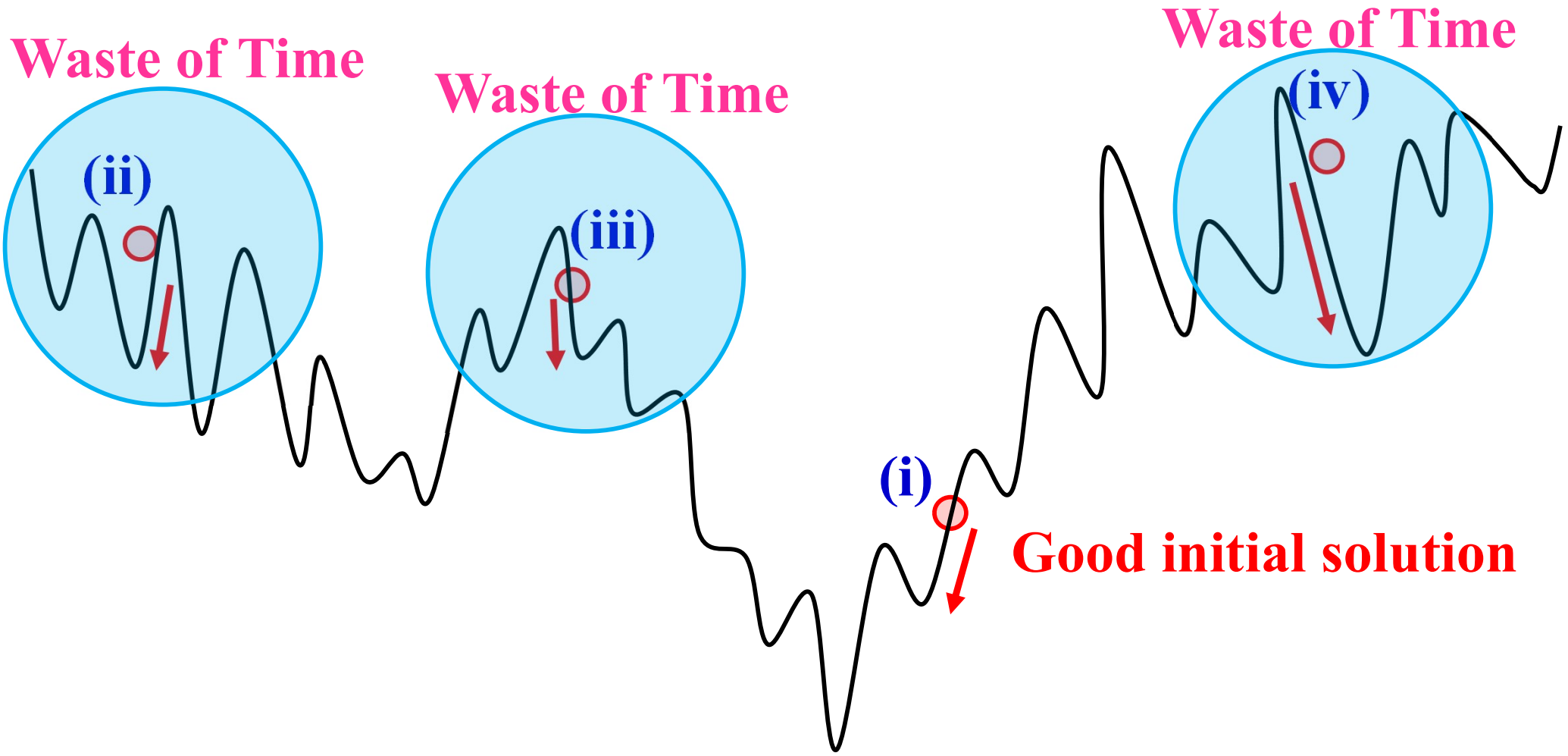

# Main Issue

Any other good ideas ?

## Choice of Initial Solutions:

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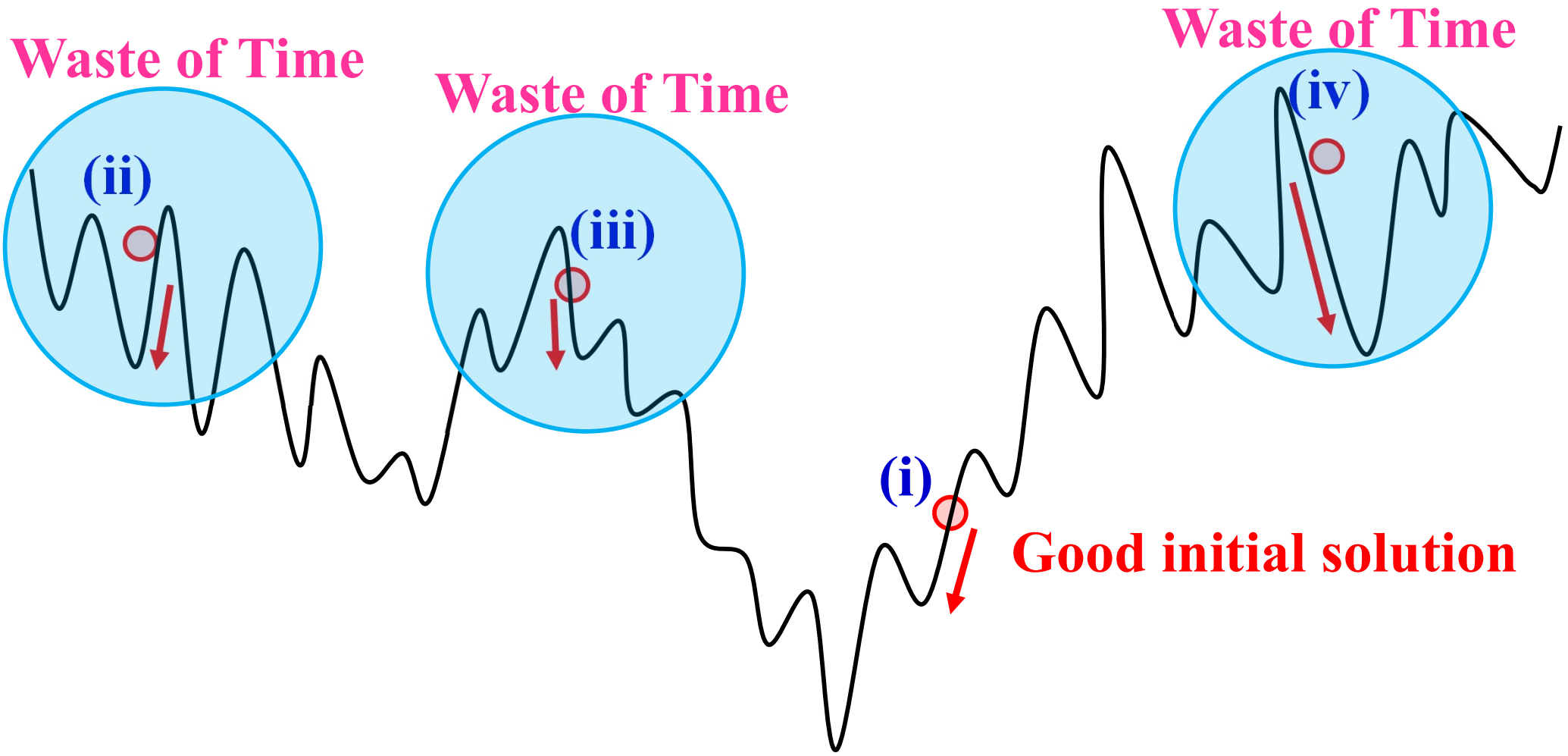
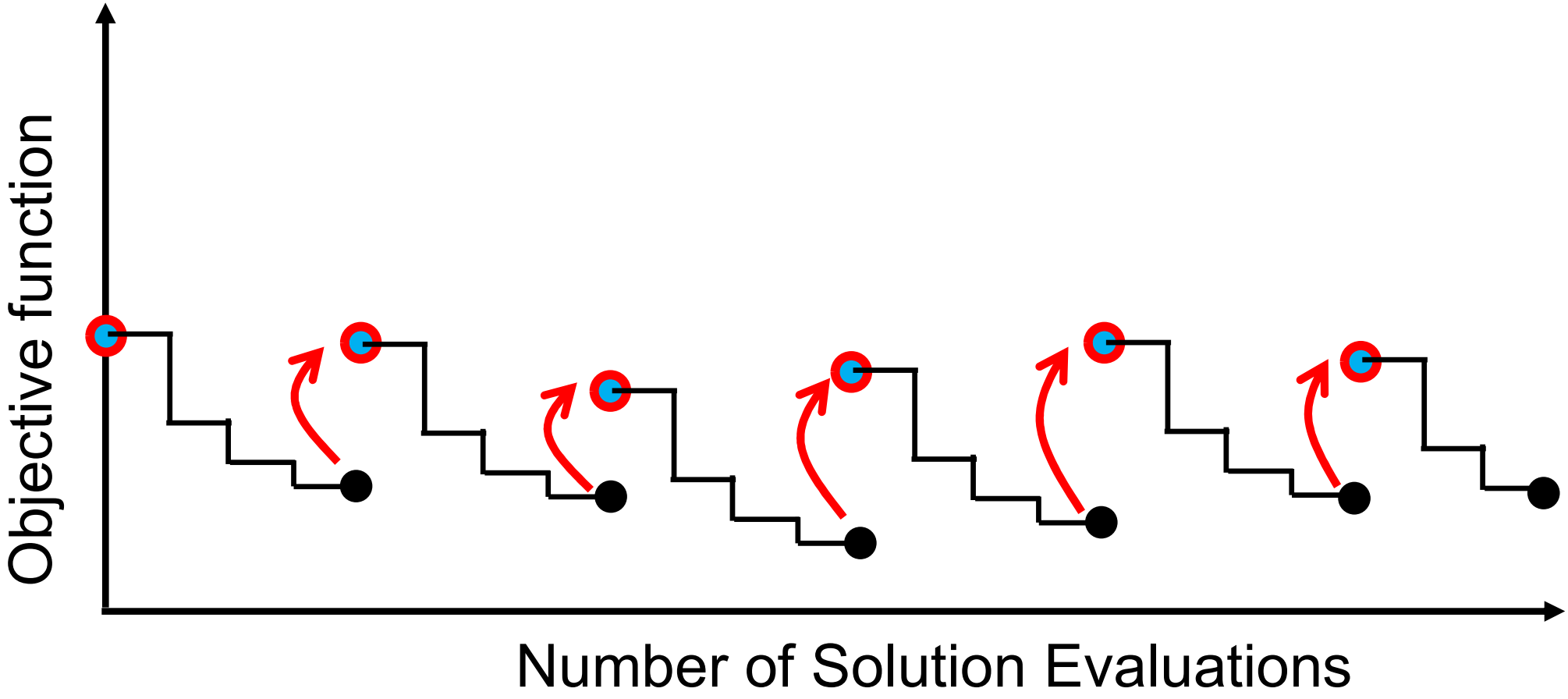


# One Idea

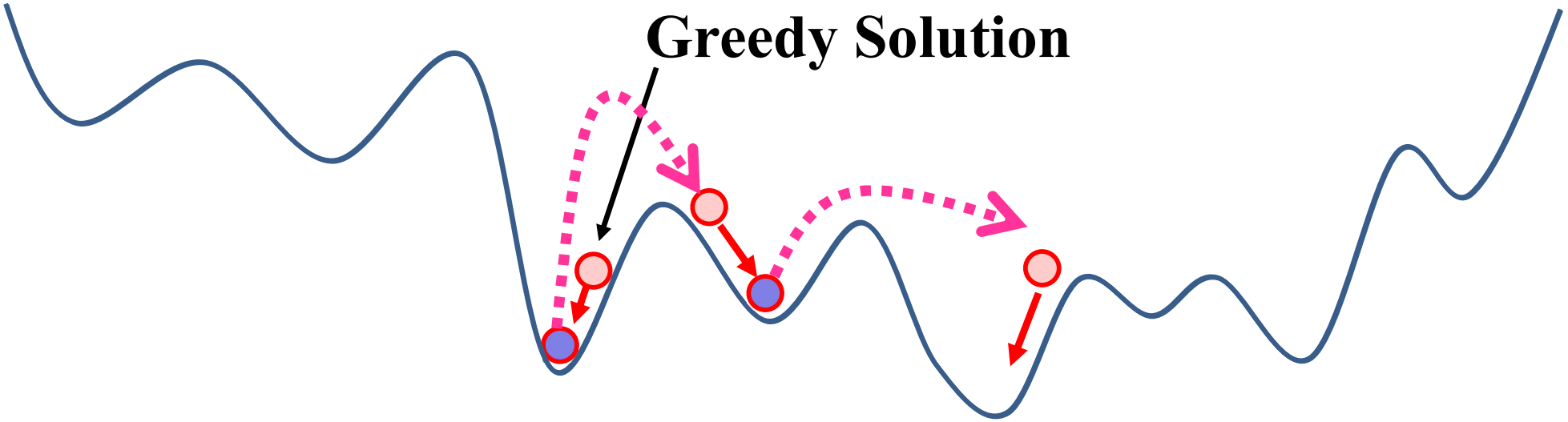
## Use of the Current Solution

- First initial solution: Greedy solution.
- Other initial solutions: Generate from the final solution of the previous LS run

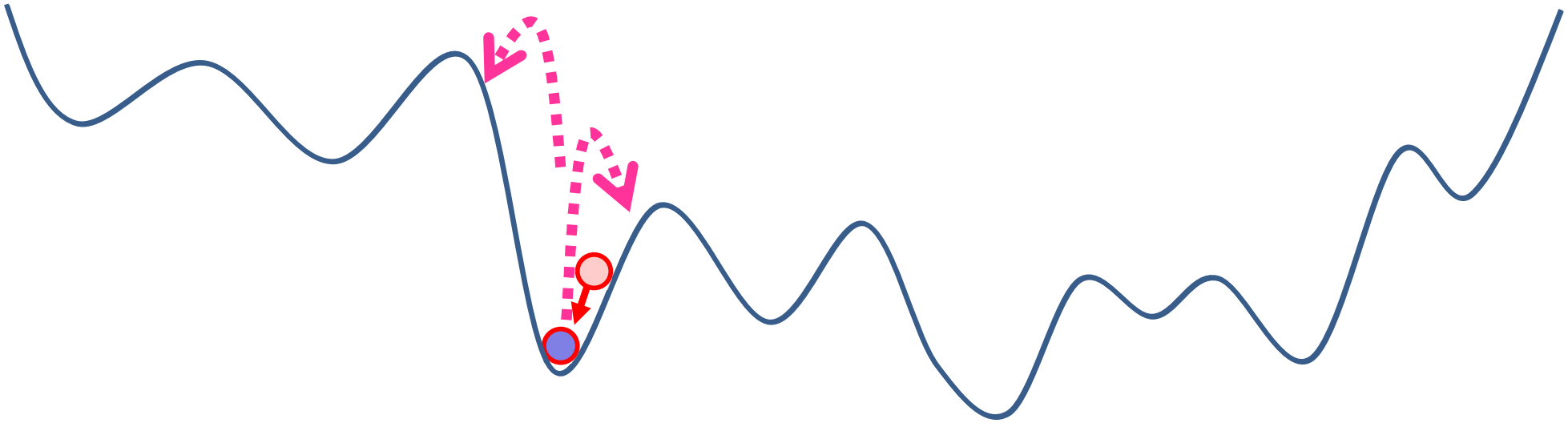


# How to specify initial solutions

## A new solution form the final solution

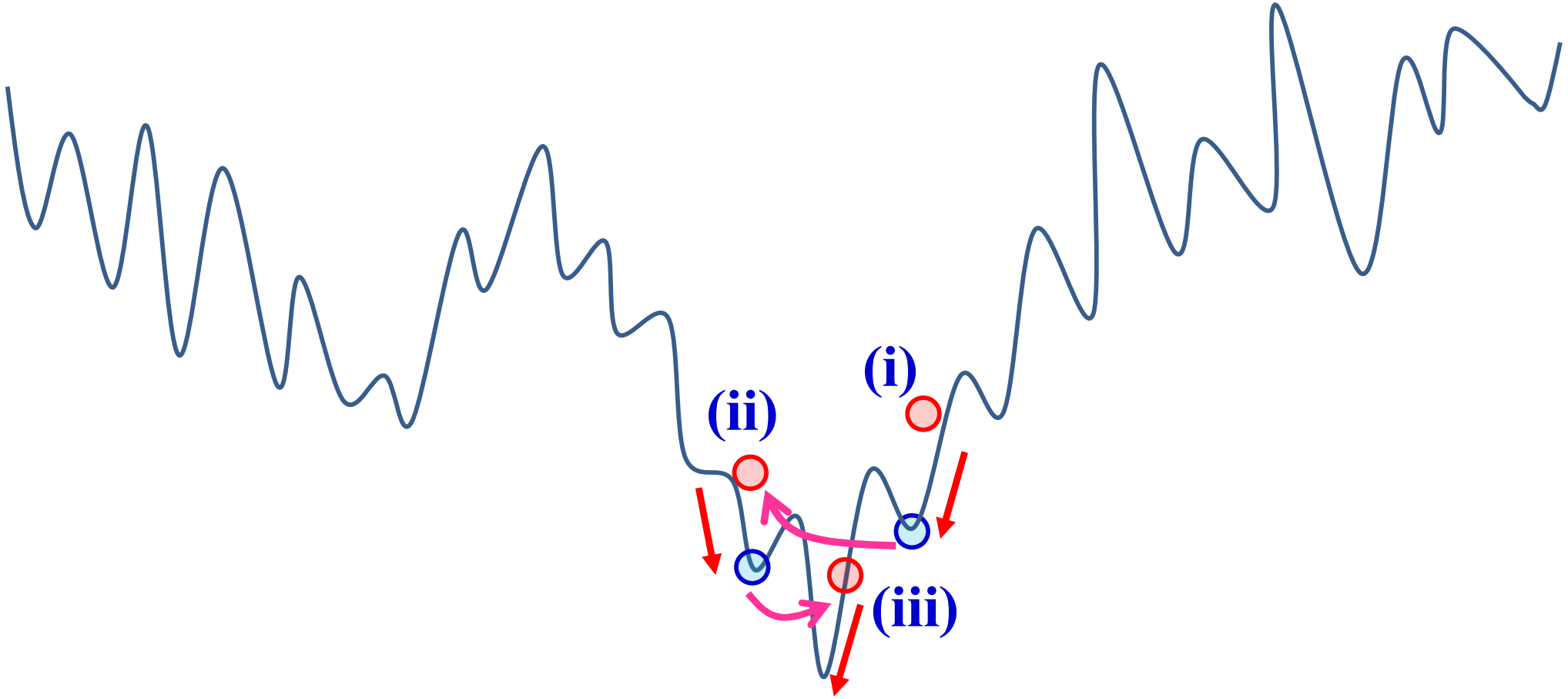



**The following undesirable situation can happen.**



# Use of the final Solution

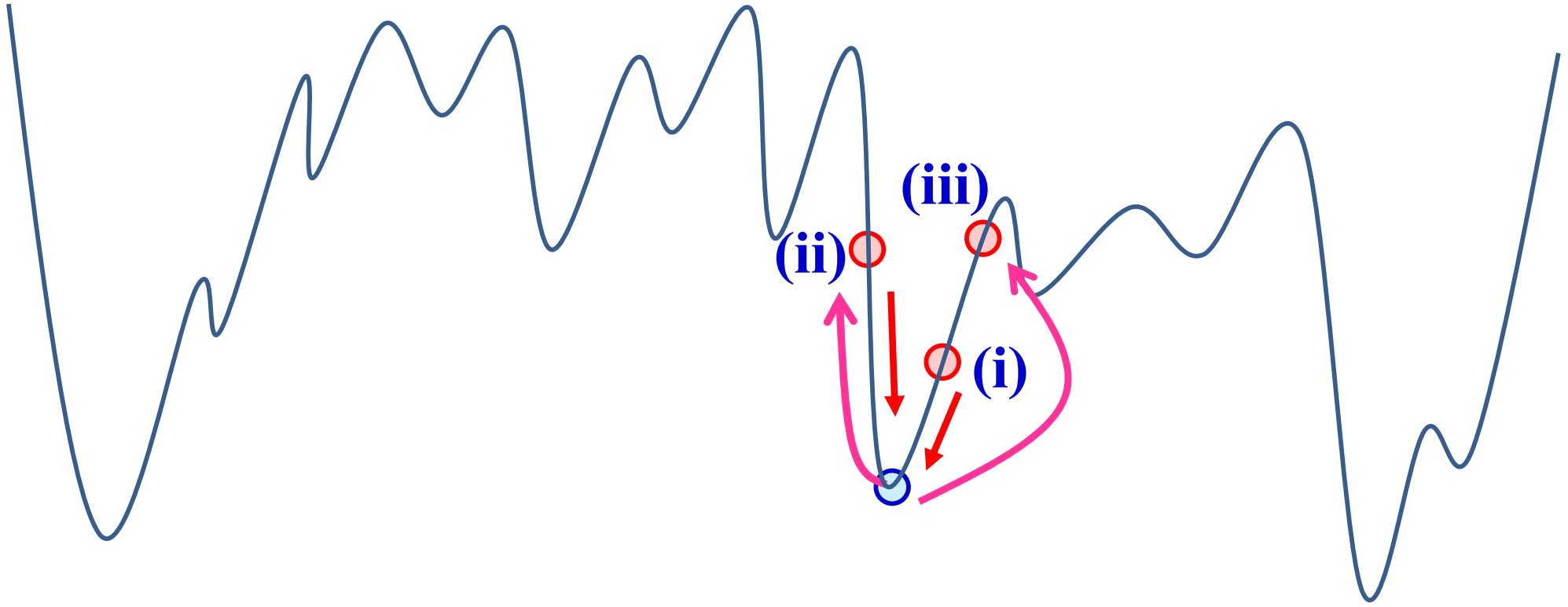
- First initial solution: Greedy solution.
- Other initial solutions: Generate from the final solution




The optimal solution can be efficiently searched by local search..

# Use of the final Solution

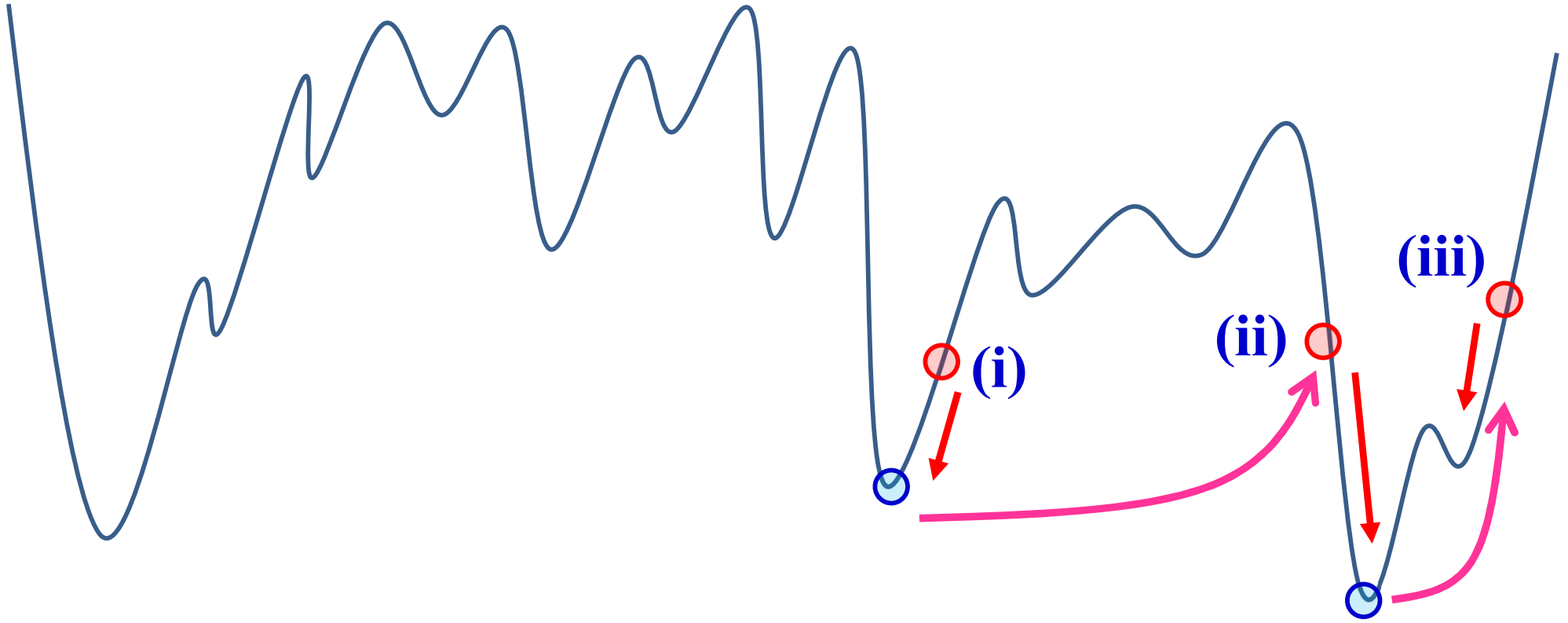
- First initial solution: Greedy solution.
- Other initial solutions: Generate from the final solution




The same solution can be obtained from different initial solutions.

# Good Initial Solutions

- Close to a good local solution
- Search for a different local solution



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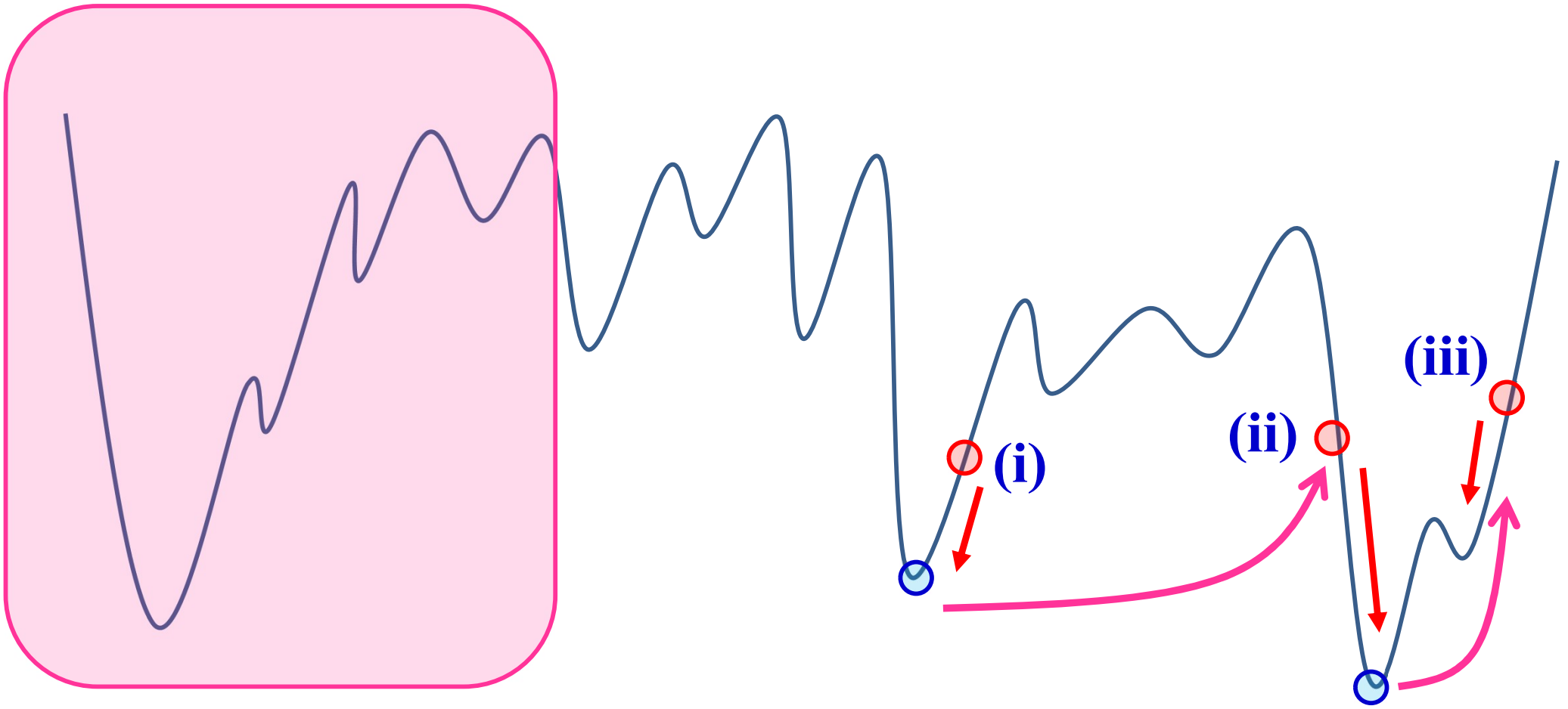
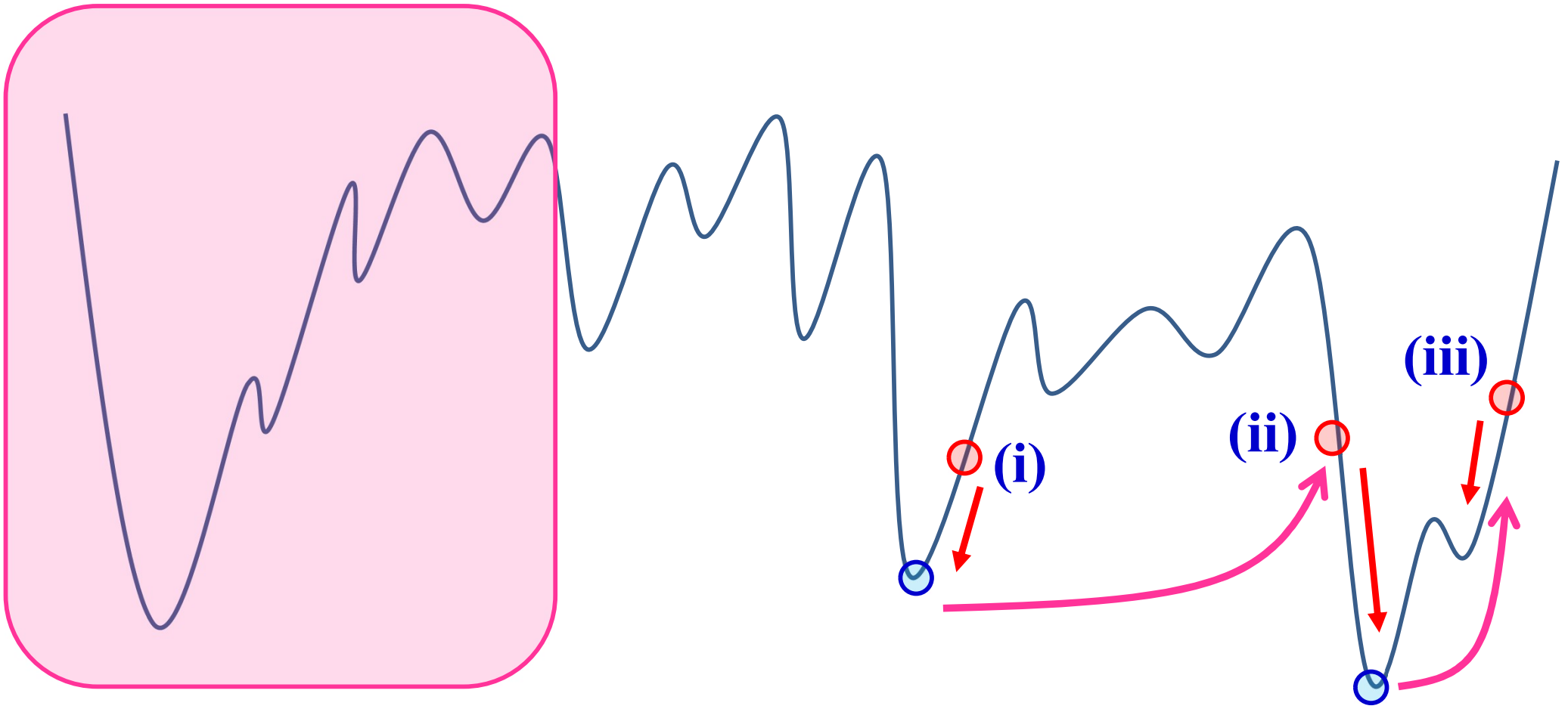



**The search of this region is difficult.**

# Initial Solutions

Random: Wide search region & Inefficient search

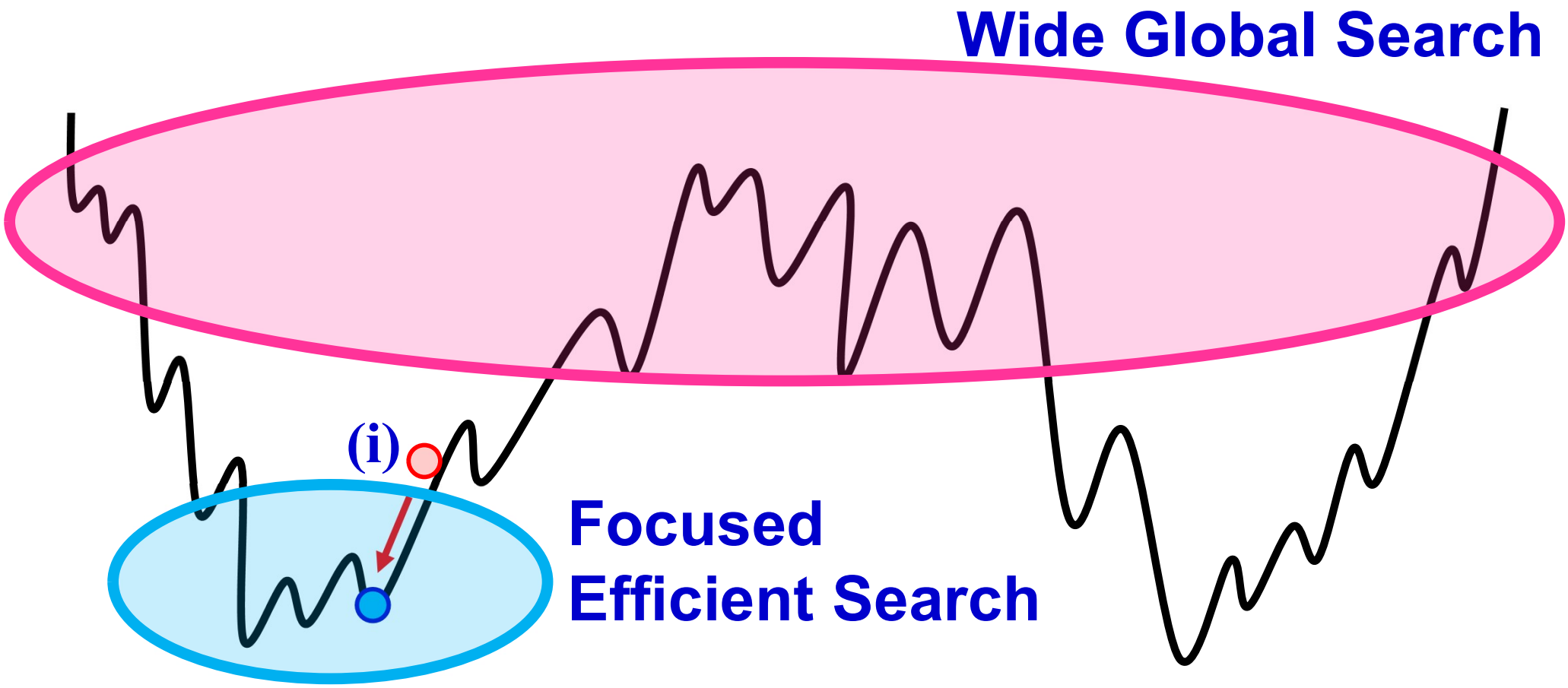
Focused: Narrow search region & Efficient search




**The search of this region is difficult.**

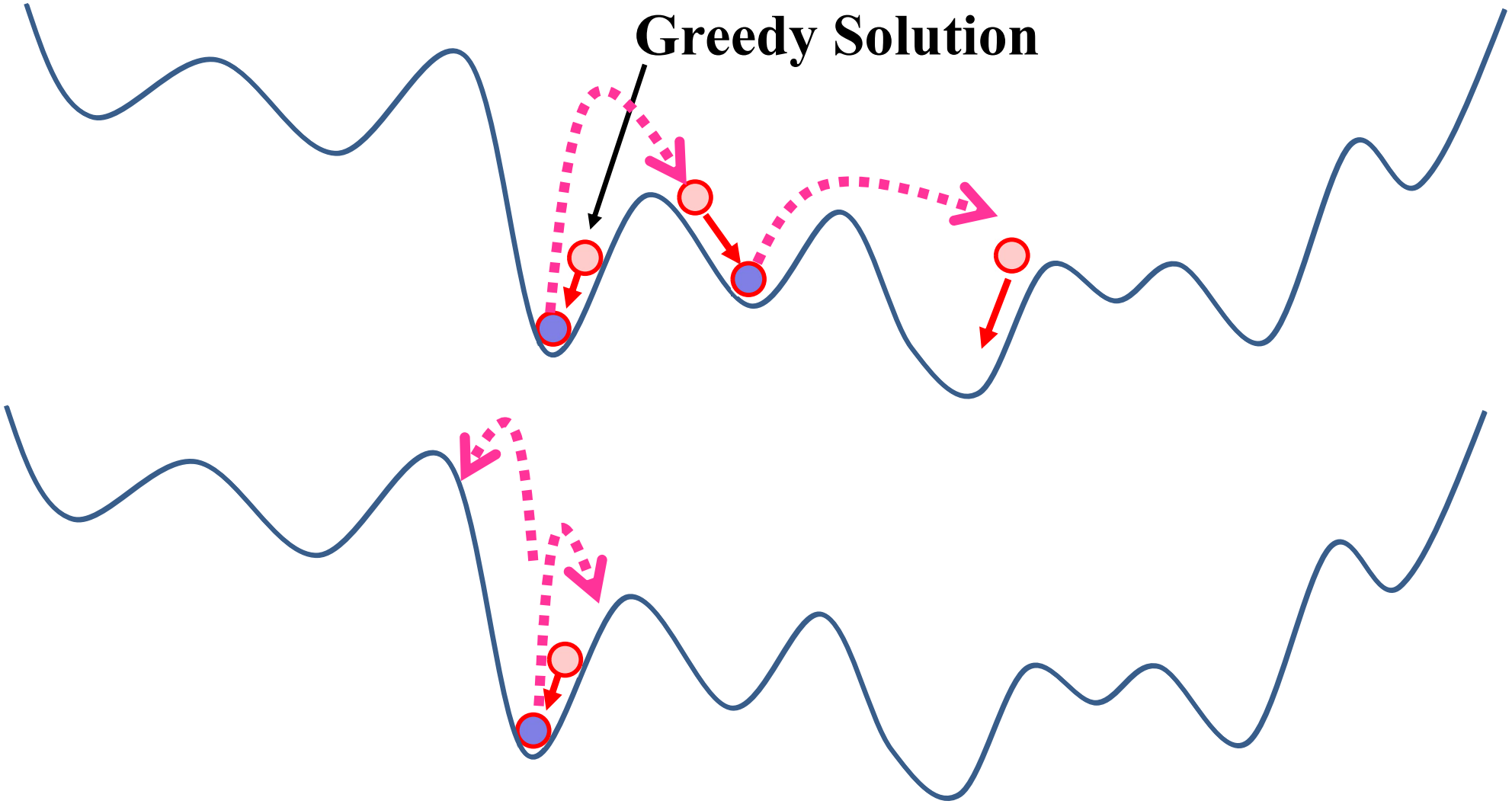
# Optimization Algorithm Design:

**Find a good balance between the wide global search and the focused efficient search** (the good balance depends on the problem size and the available computation time)



## General Guideline:

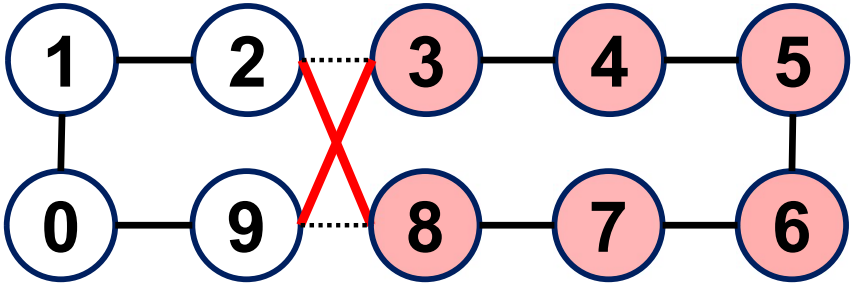
**Generate good initial solutions which are not likely to go back to the same final solution.**

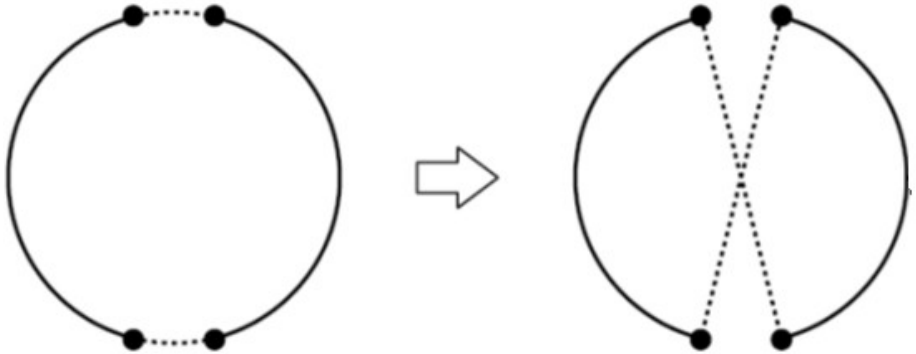


# Lab Session Task 1

## Local search

- Neighborhood structure: Inversion (two-edge change)



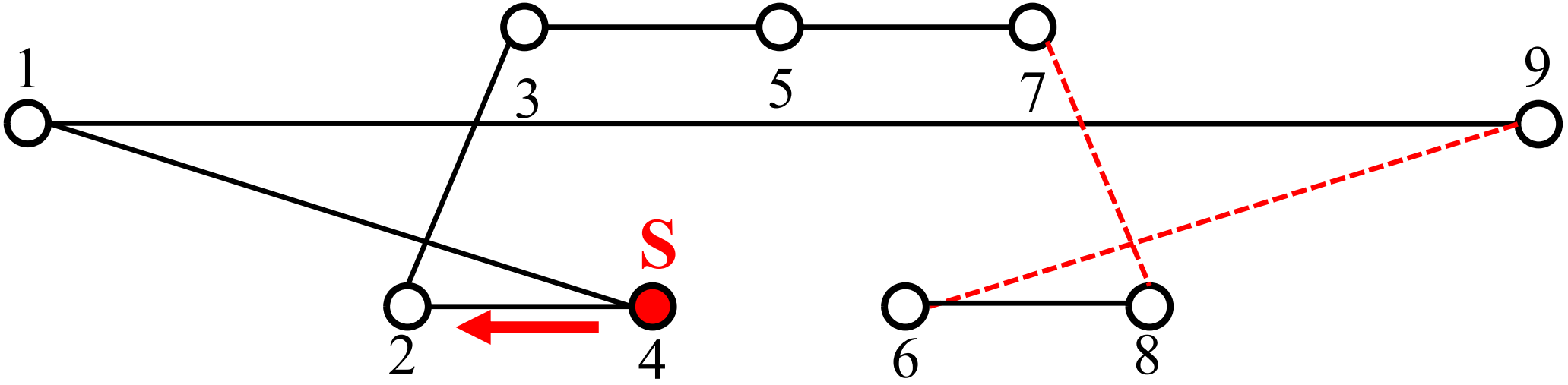
## Creation of a new initial solution:

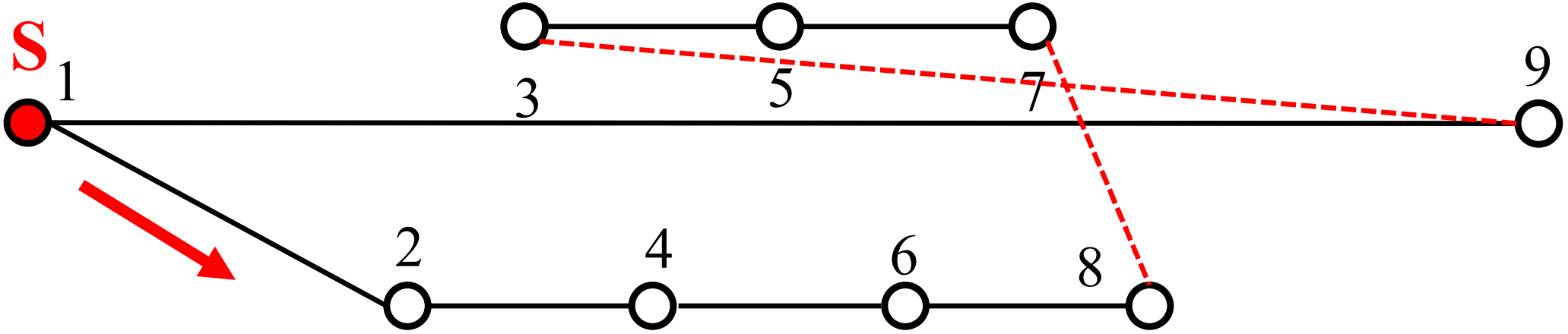
**- 1st initial solution: Greedy solution**

### Task 1

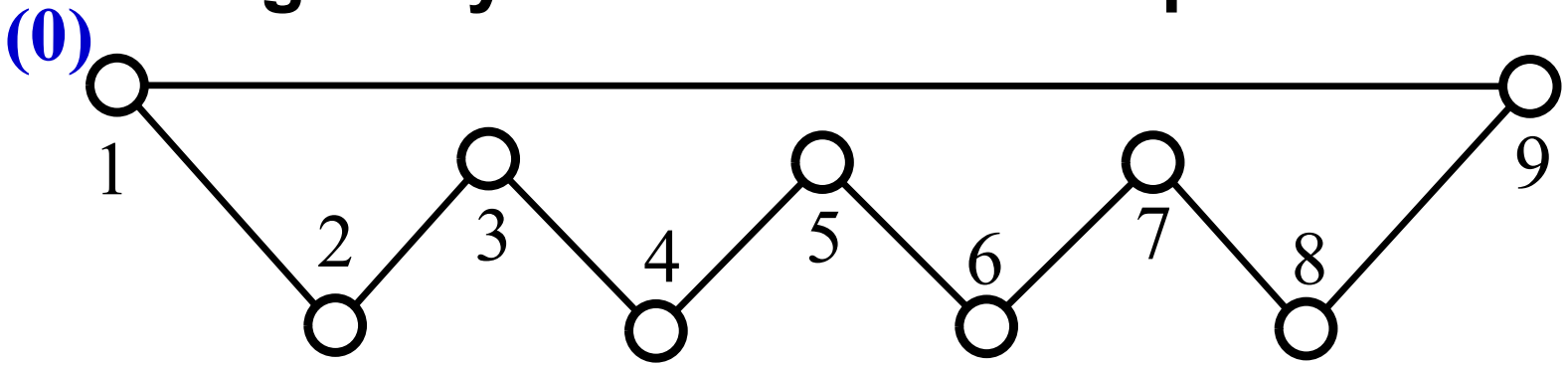
Create a TSP problem where a **non-optimal** greedy solution is obtained from an initial city and the obtained greedy solution cannot be improved by inversion-based local search. It is enough to show that one greedy solution cannot be improved. You do not have to show that any greedy solution from an arbitrary initial city cannot be improved.

**TSP Problem: In many cases, greedy solutions can be improved by inversion-based local search.**

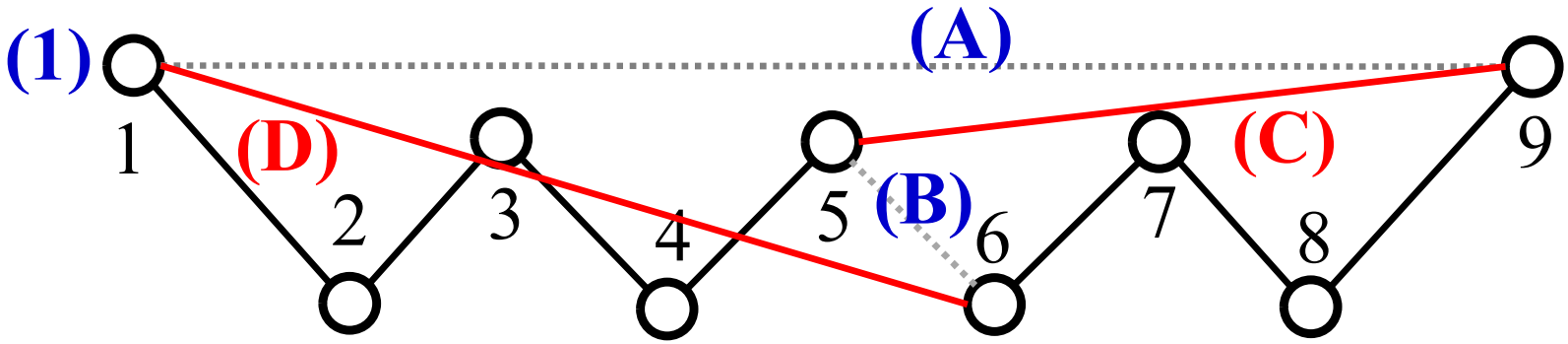


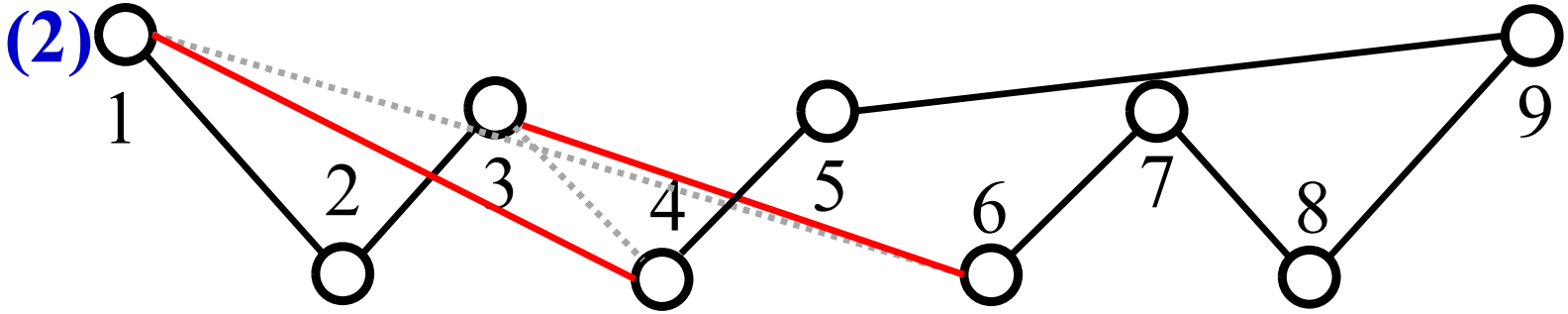
**This greedy tour looks local optimal**

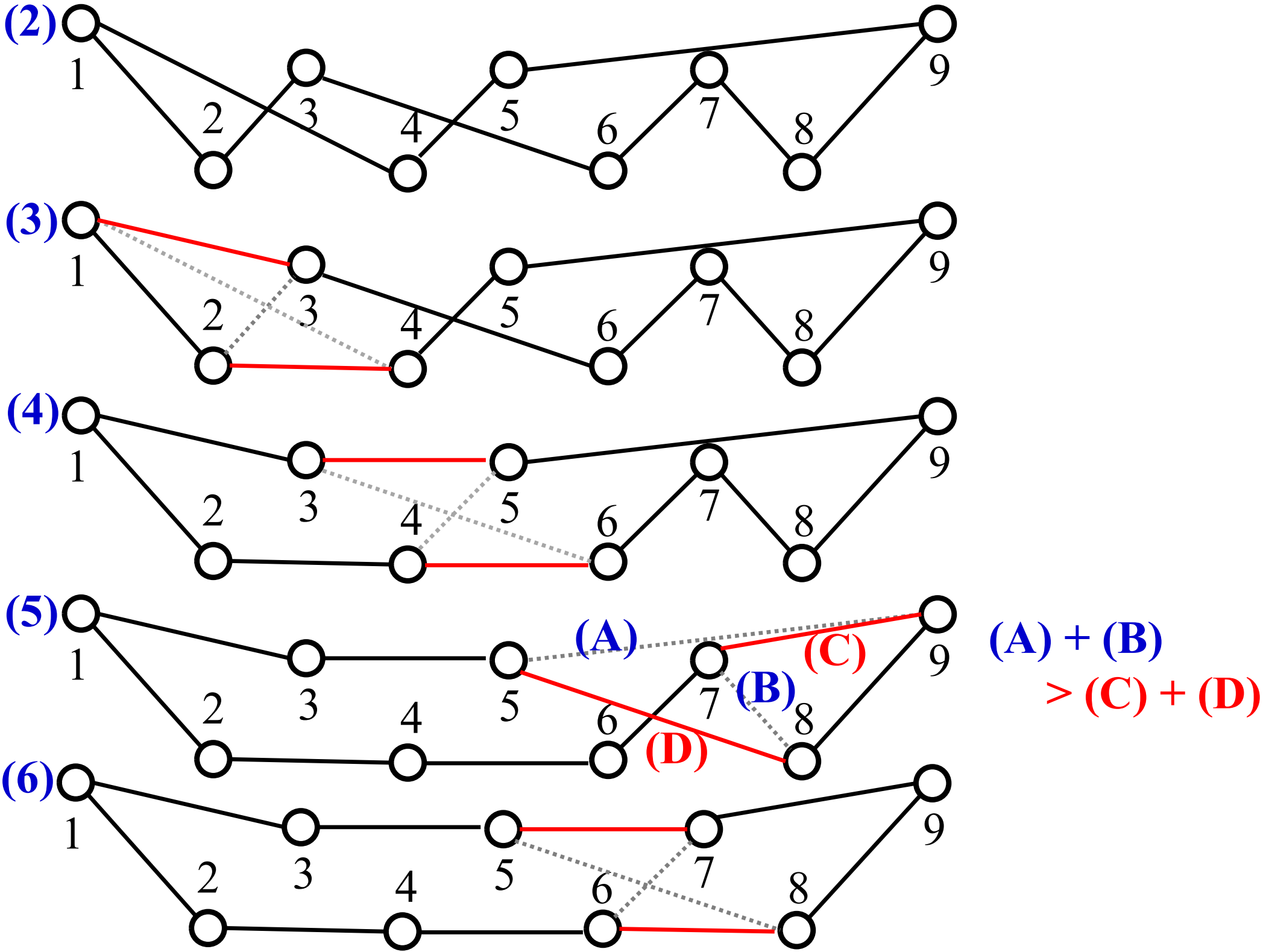



**However, this can be improved by inversion-based LS.**




**(A) + (B) > (C) + (D)**




(2)
1
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(3)
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9
(4)
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9
(A)
(B)
(C)
(D)
(6)
1
2
3
4
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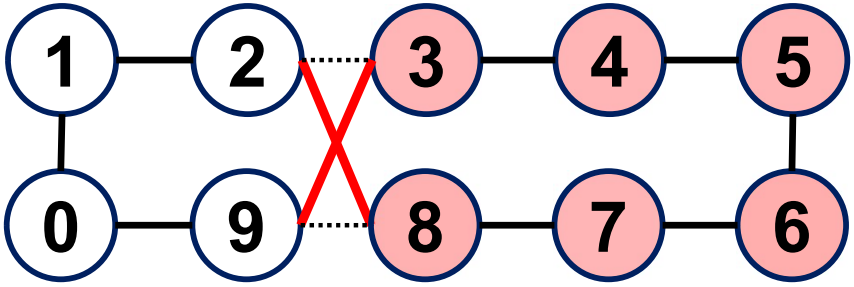
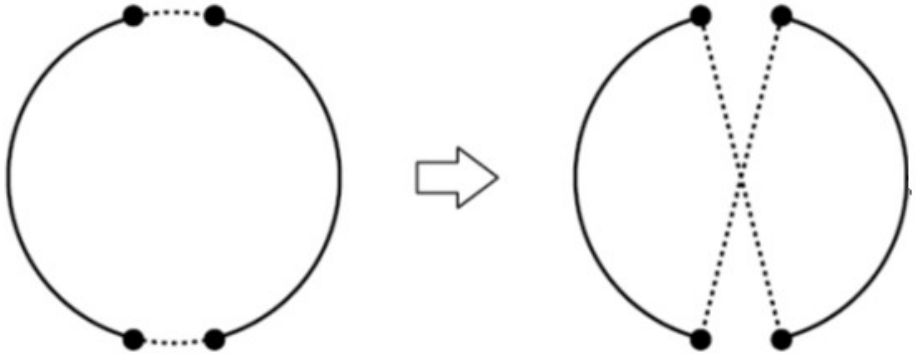
7
8
9
(A) + (B)
> (C) + (D)

# Lab Session Task 2

## Local search

- Neighborhood structure: Inversion (two-edge change)

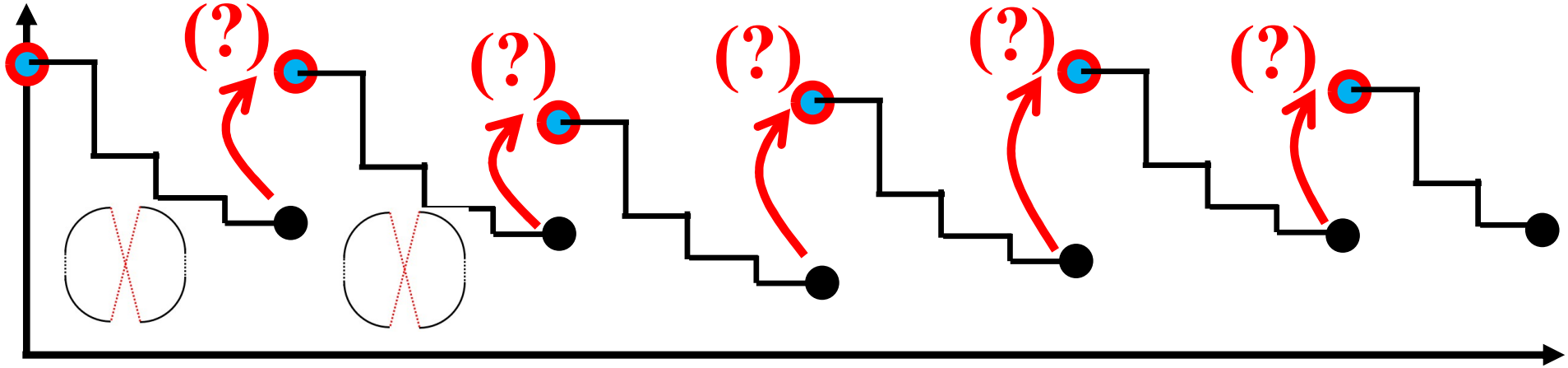


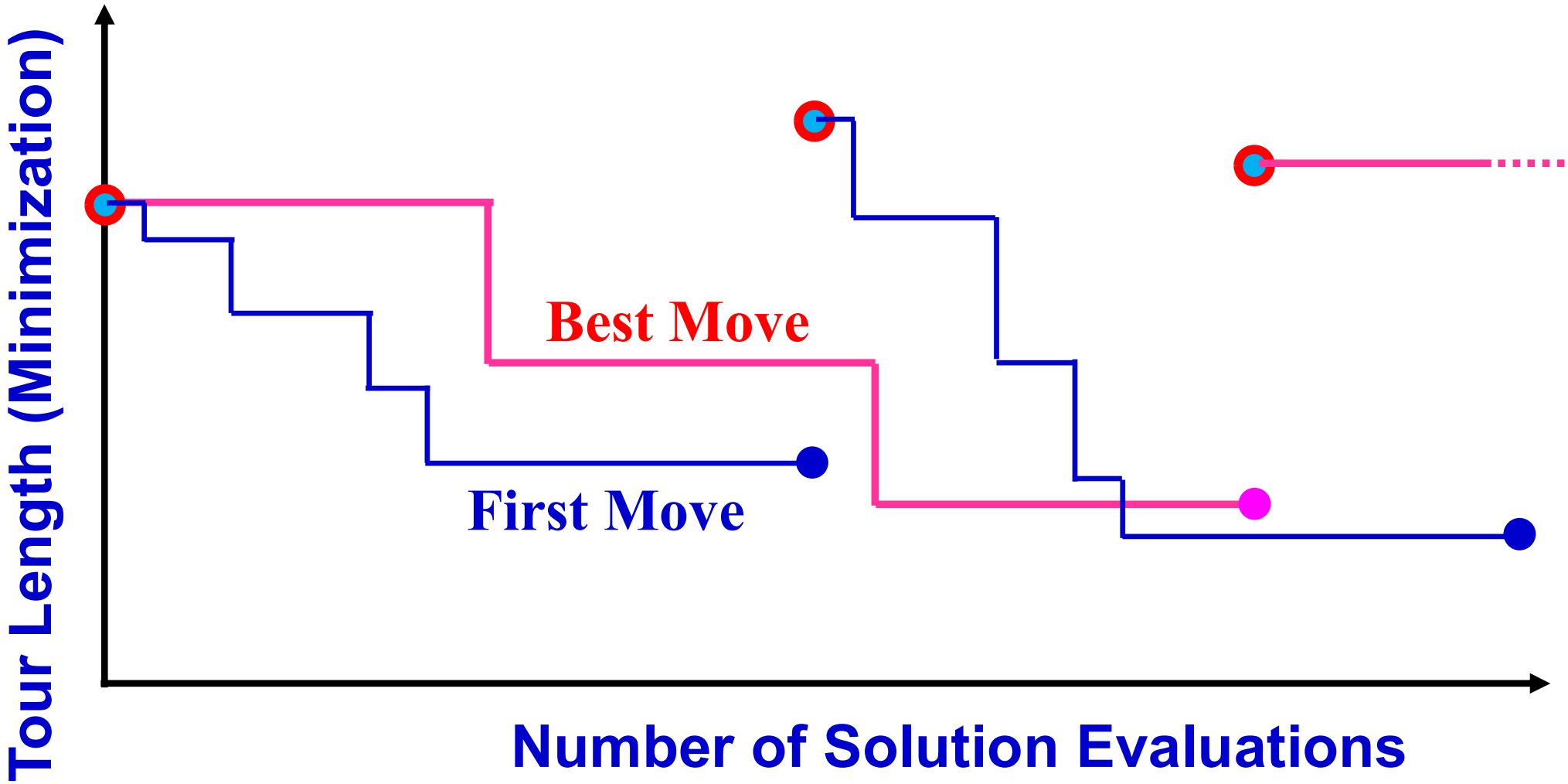
## Creation of a new initial solution:

**- 1st initial solution: Greedy solution**

**- Other initial solutions: Explain your own idea about how to create an initial solution for each local search run (Task 2).**

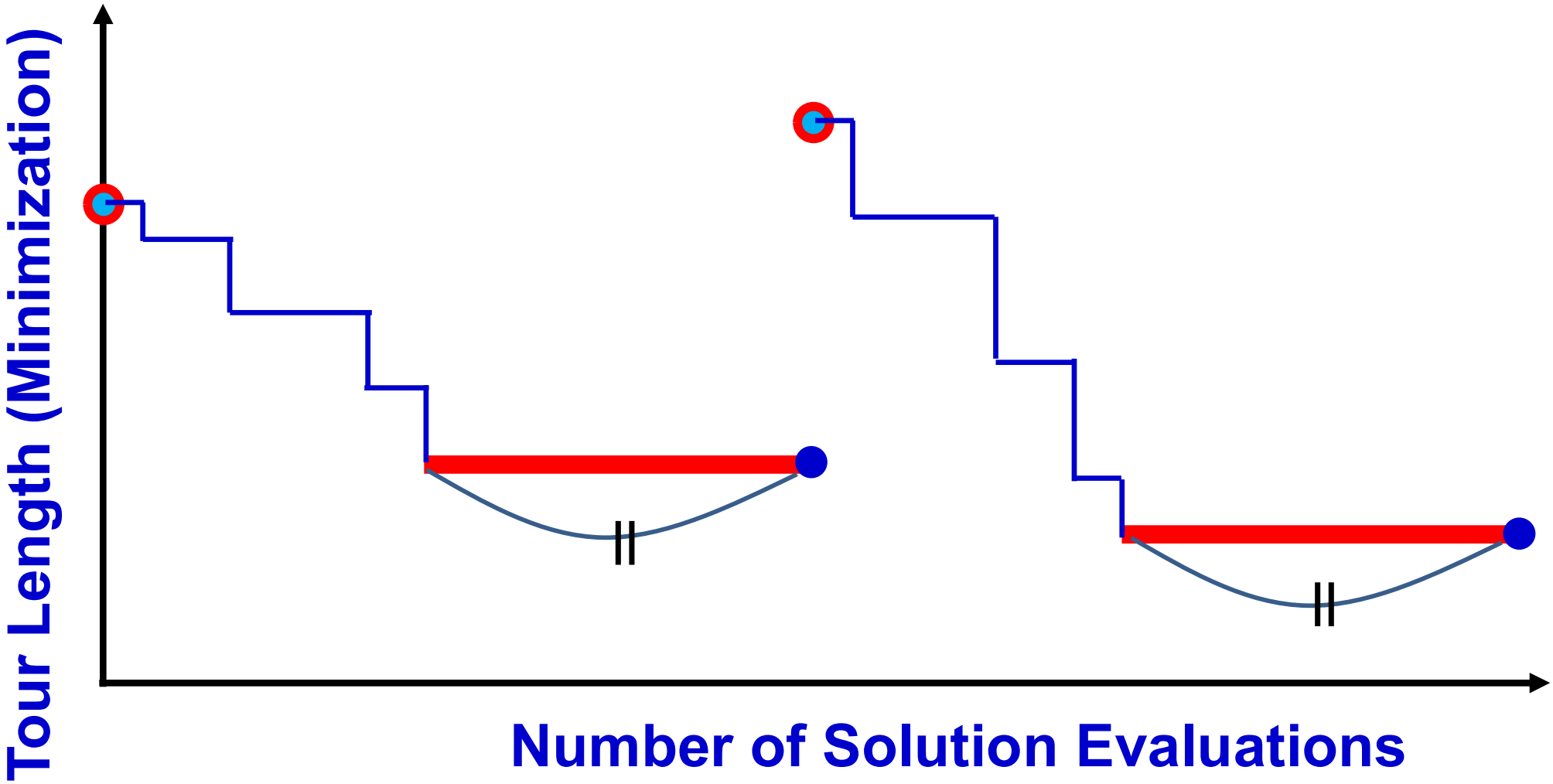


# Best Move or First Move ?



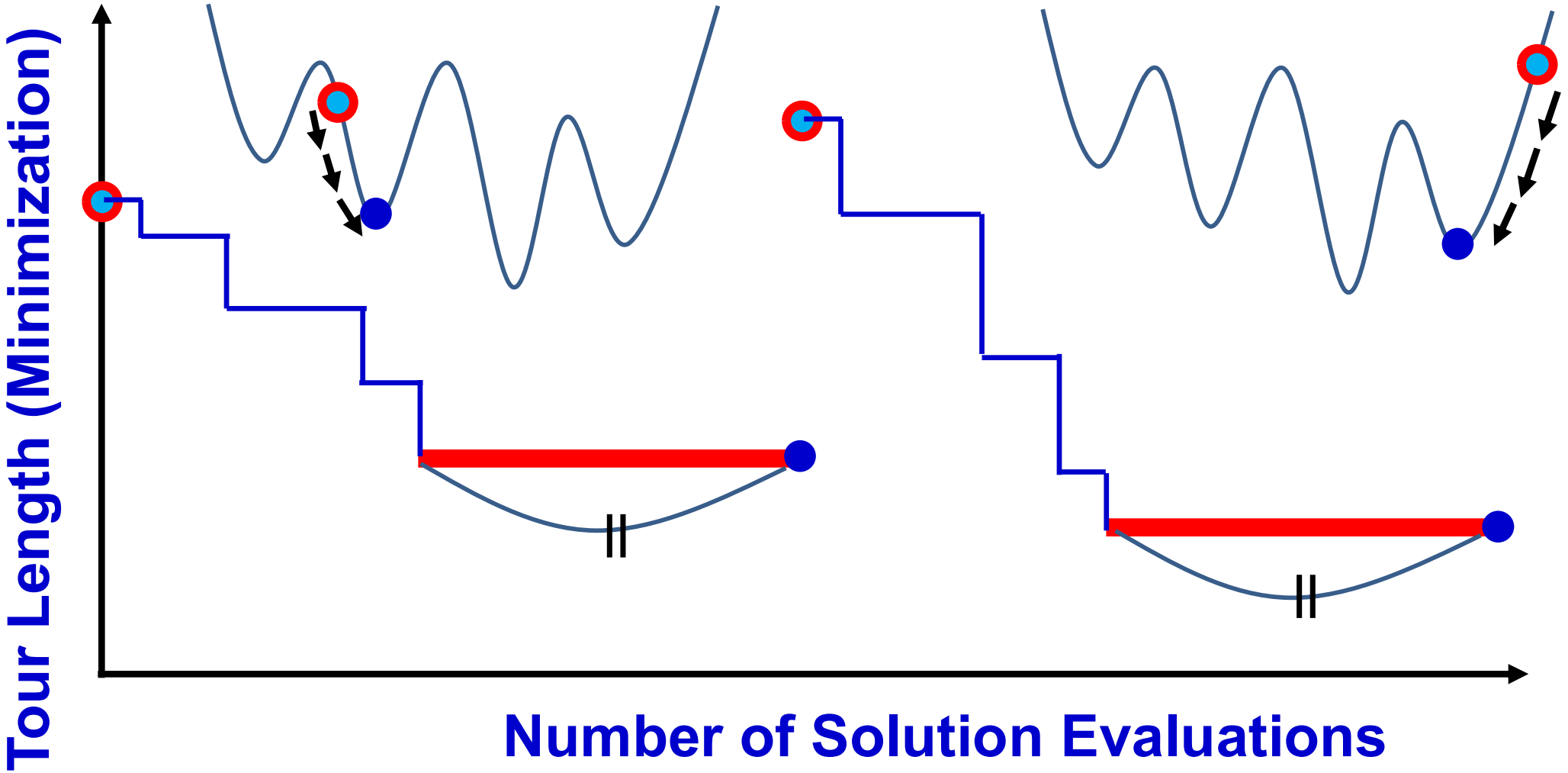

It may be a good idea to use the first move for examining more initial solutions (i.e., for finding more local solutions)

# Early Termination




Even in the first move local search, all neighbors are examined before the termination of local search. This usually spends a long computation time with no performance improvement, which looks waste of time.

# Early Termination

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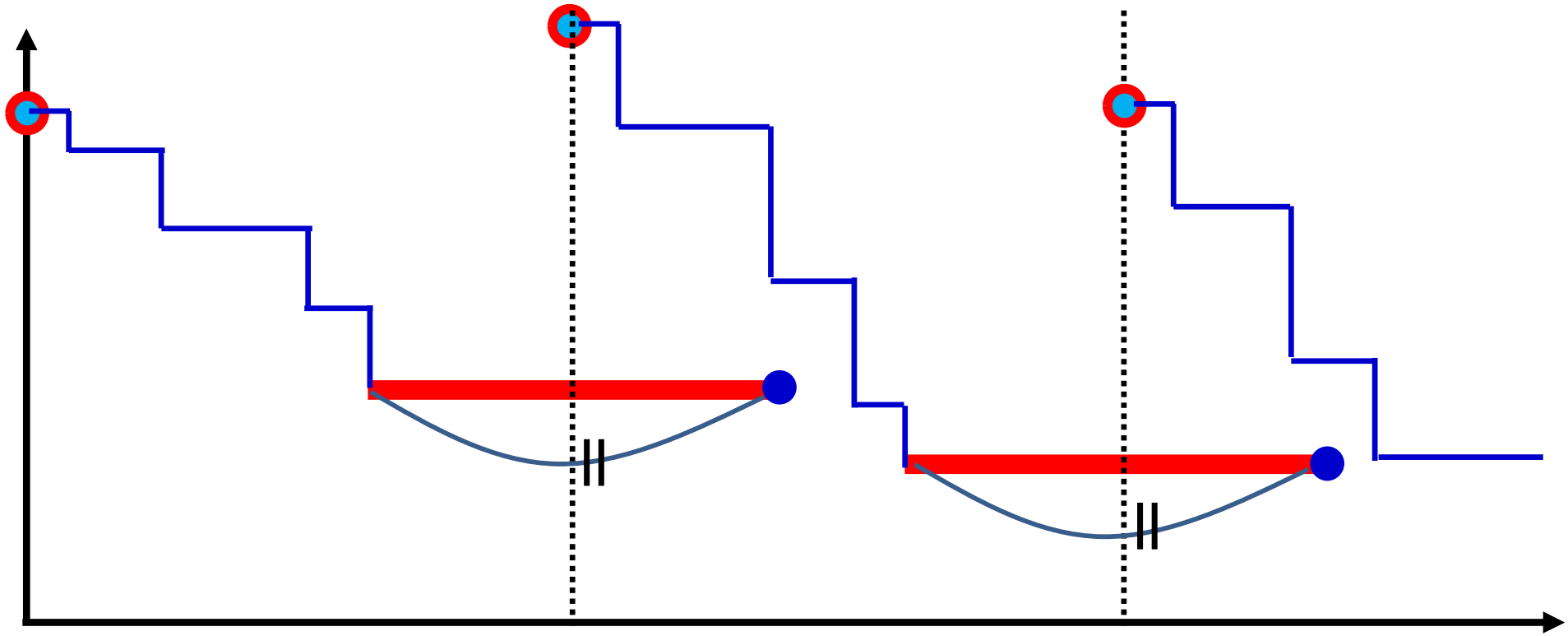
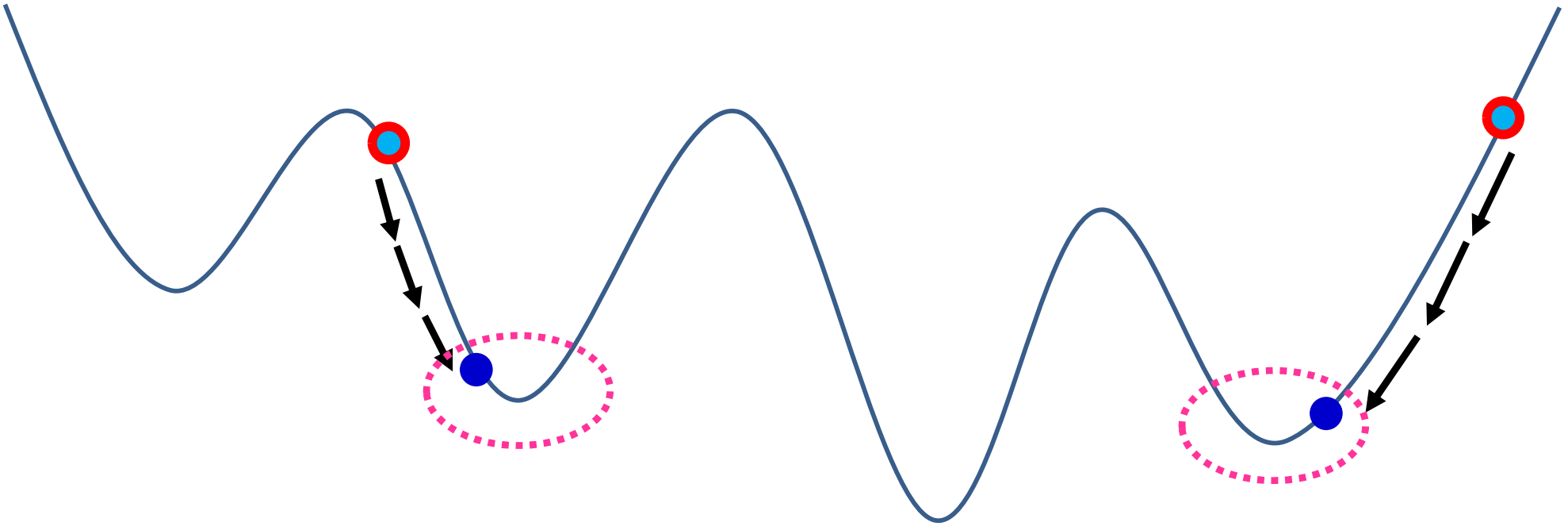
# Early Termination

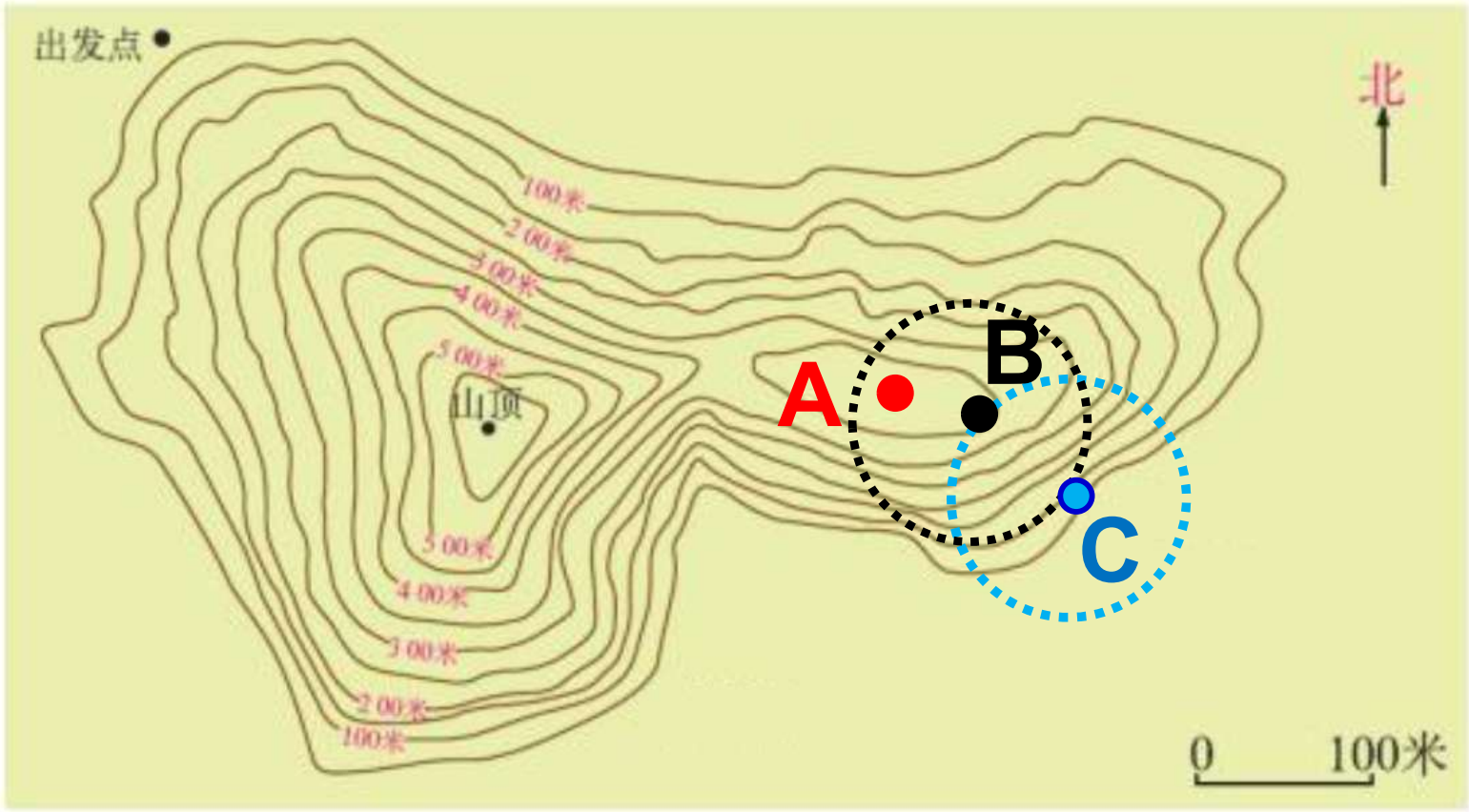


It may be a good idea for efficient search to terminate the local search before examining all neighbors (e.g., restart from a new initial solution after examining 100 neighbors or 20% of neighbors).

# Early Termination



One negative effect is that the local search will terminate before finding a local solution. The question is which is a better search strategy between (i) to carefully search for a local solution around the current solution, and (ii) to quickly search for better solutions from many initial solutions. Usually, the improvement by local search around a local solution is not large. Thus, (ii) is more efficient in many cases.



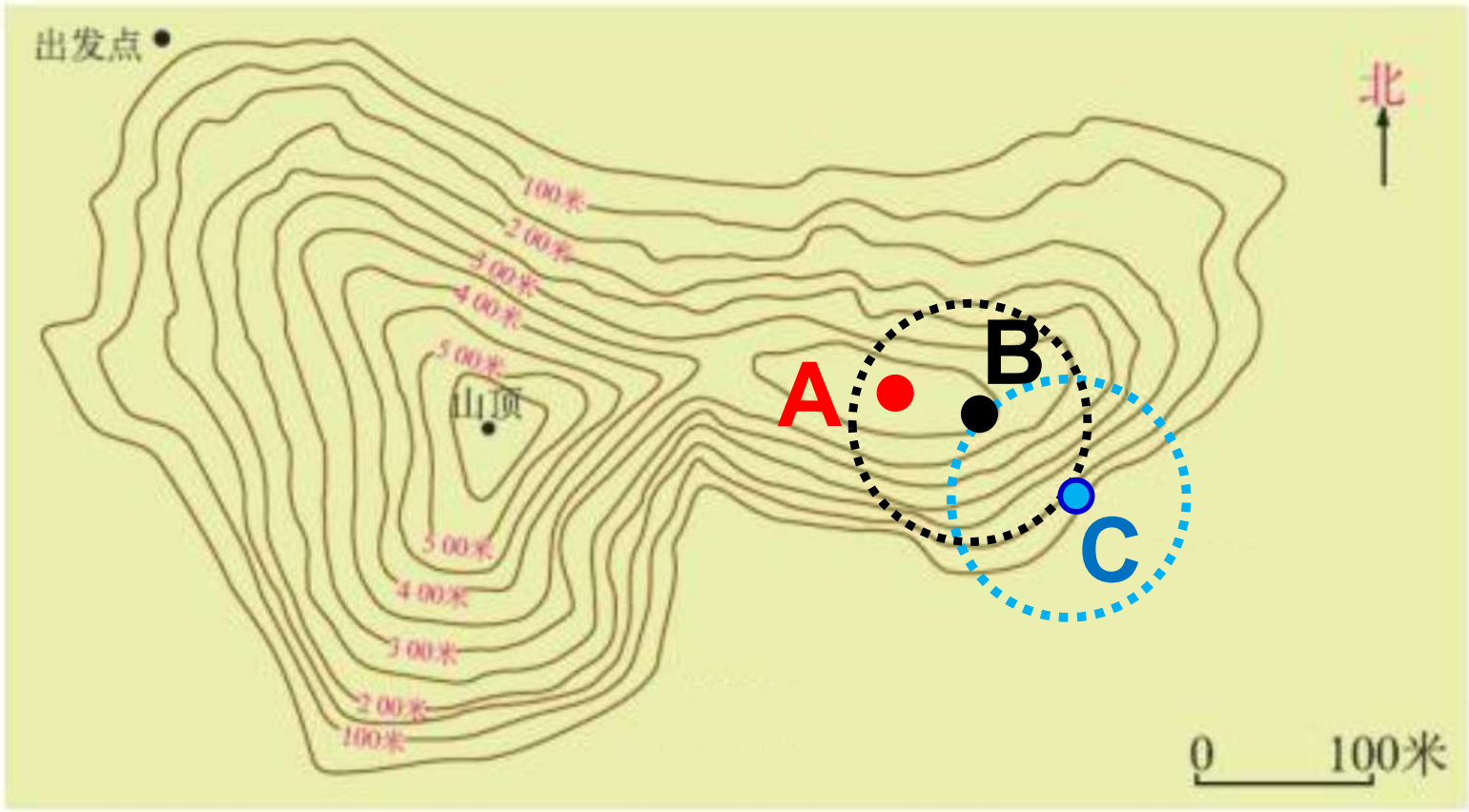

**Around the local solution A: Current Solution B**

(i) It is not easy to find a better solution

==> Early termination before finding the local solution.

(ii) The improvement by local search is small.

==> Early termination is not a big problem.




**Around the local solution A: Current Solution B**

(i) It is not easy to find a better solution

==> Early termination before finding the local solution.

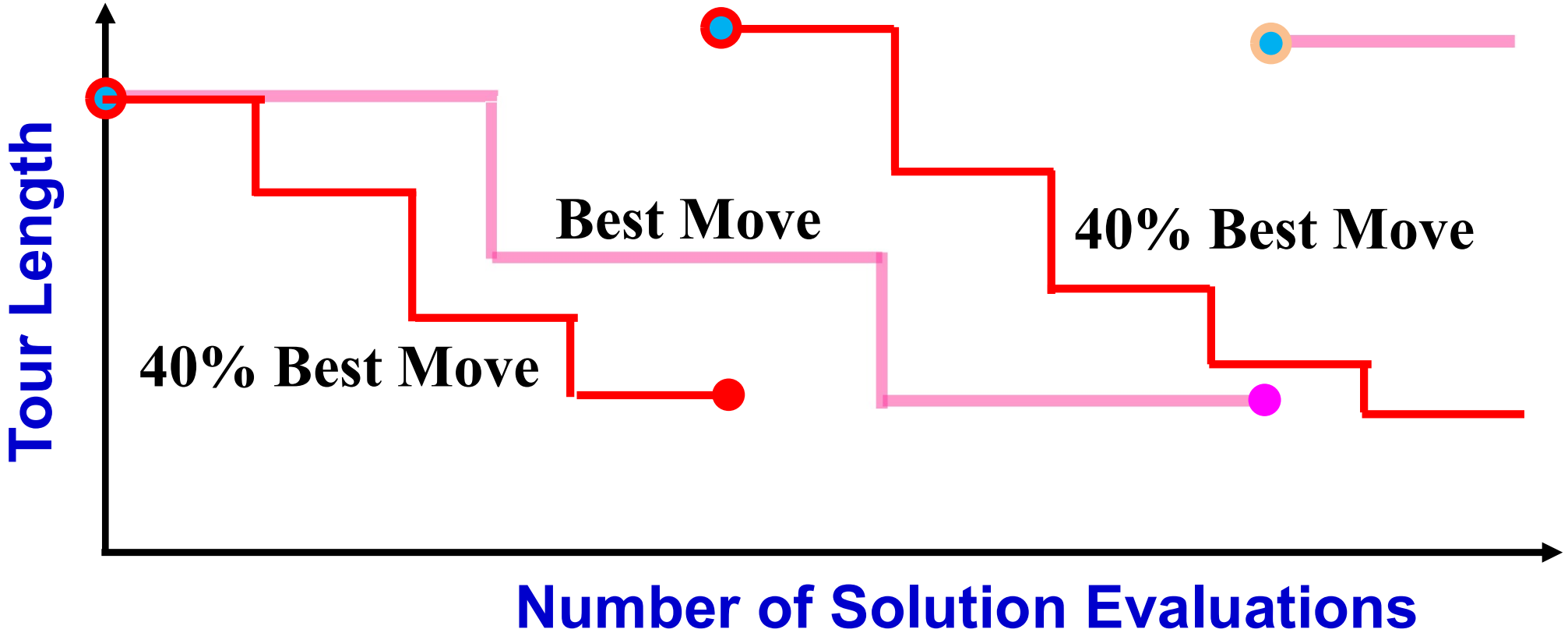
(ii) The improvement by local search is small.

==> Early termination is not a big problem.

**At solution C, it is easy to find a better solution.**

# Best Move with Early Termination

(i) Examine $\alpha$ % of neighbors.
(ii) Choose the best neighbor among them.
(iii) If the best neighbor is better than the current solution, move to the best neighbor. Otherwise, restart the local search from a new initial solution.



# Local Search

**Implementation Issues**

(1) Specification of an initial solution

(2) Specification of a neighborhood structure

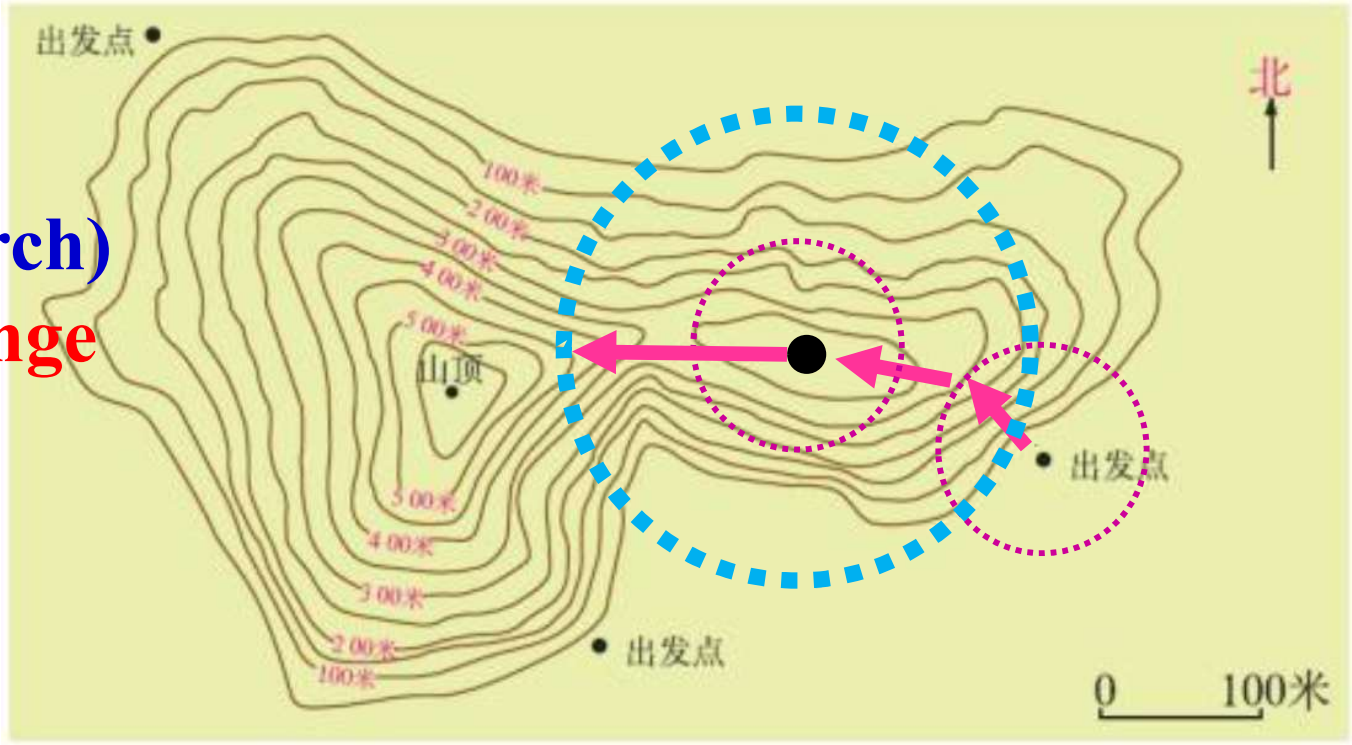
(3) Choice between the first move and the best move

⇩

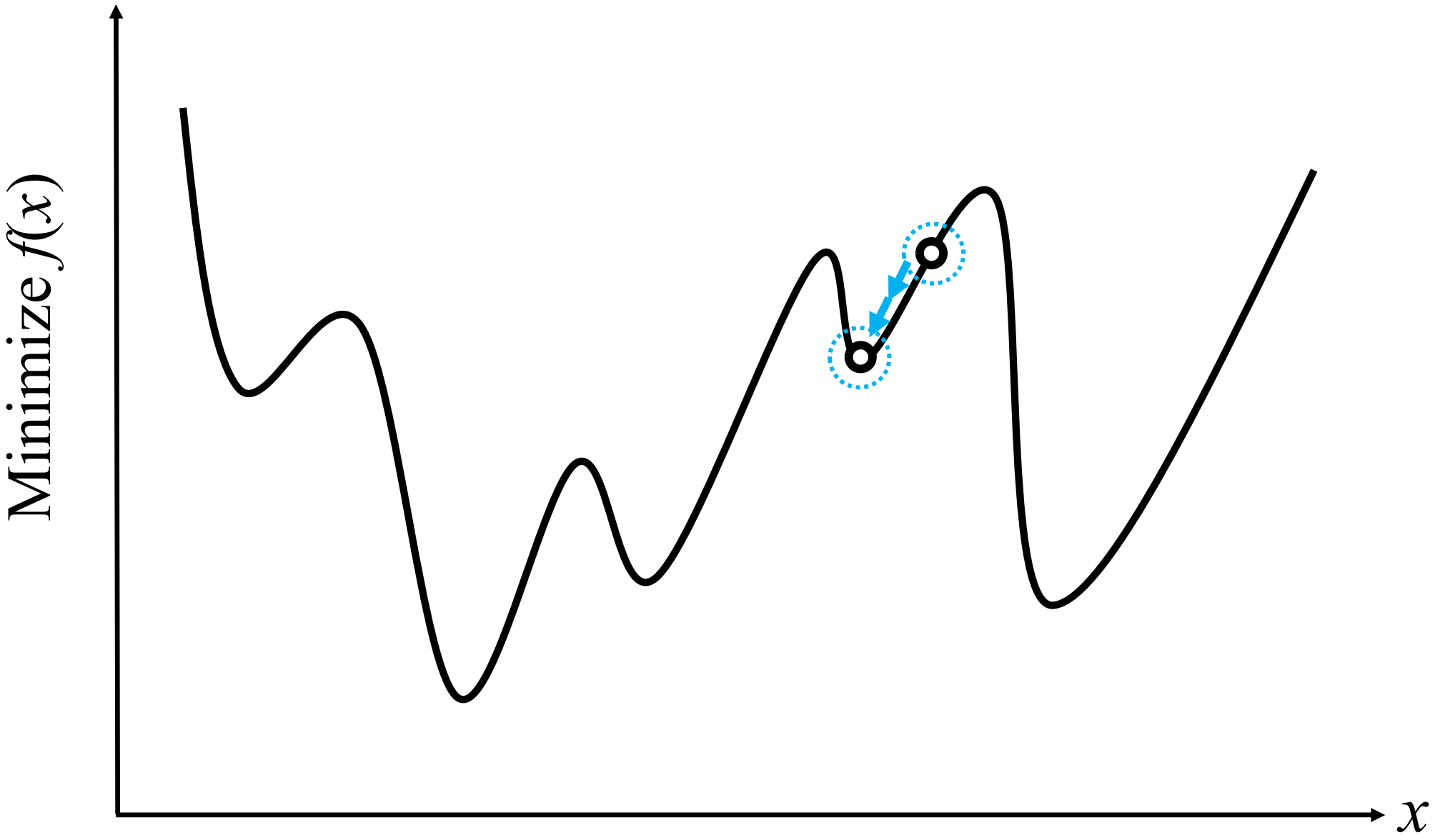
**Local Solution**

## Next Step

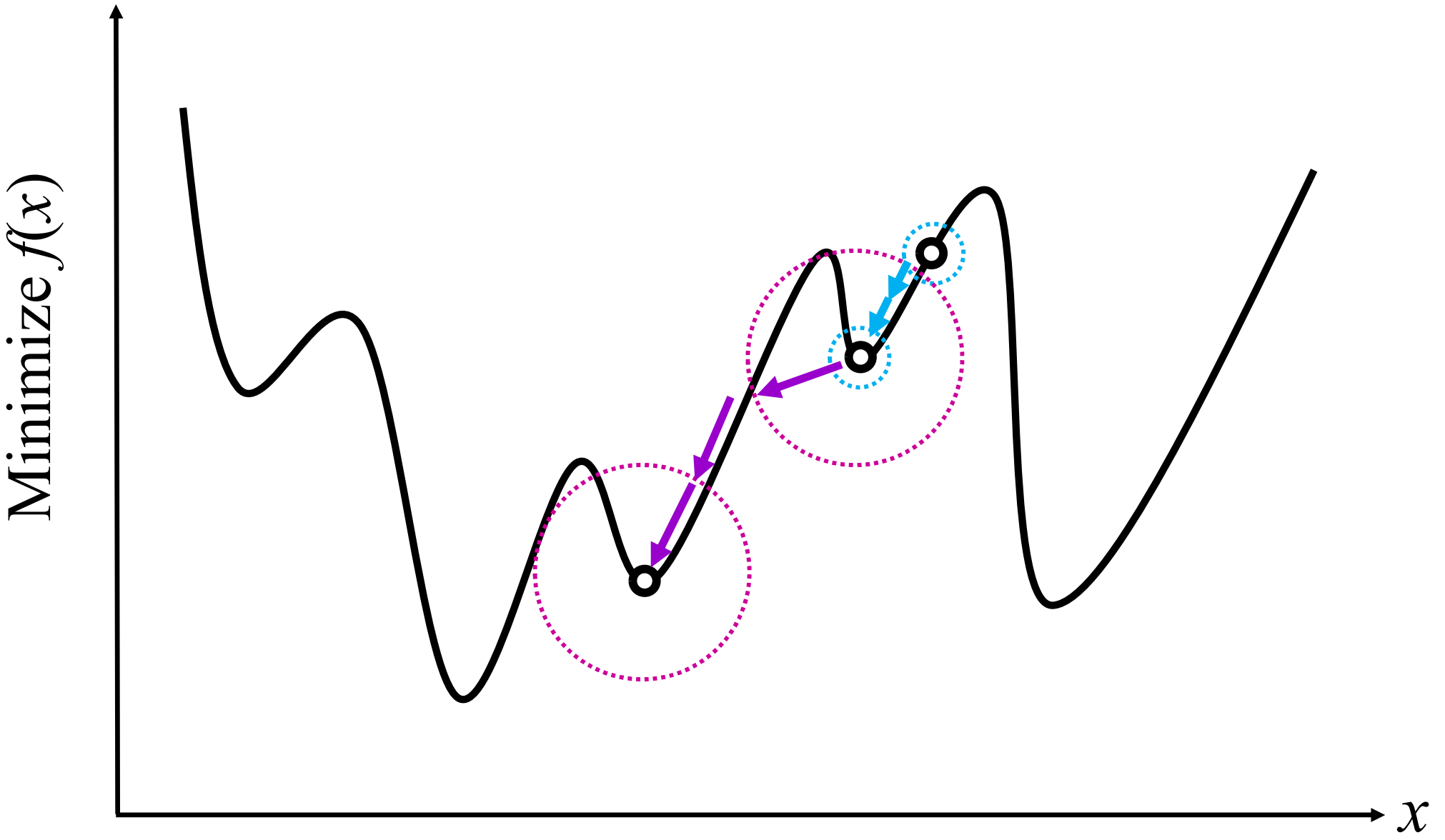
- **Restart**
  **(Iterated Local Search)**
- **Neighborhood Change**



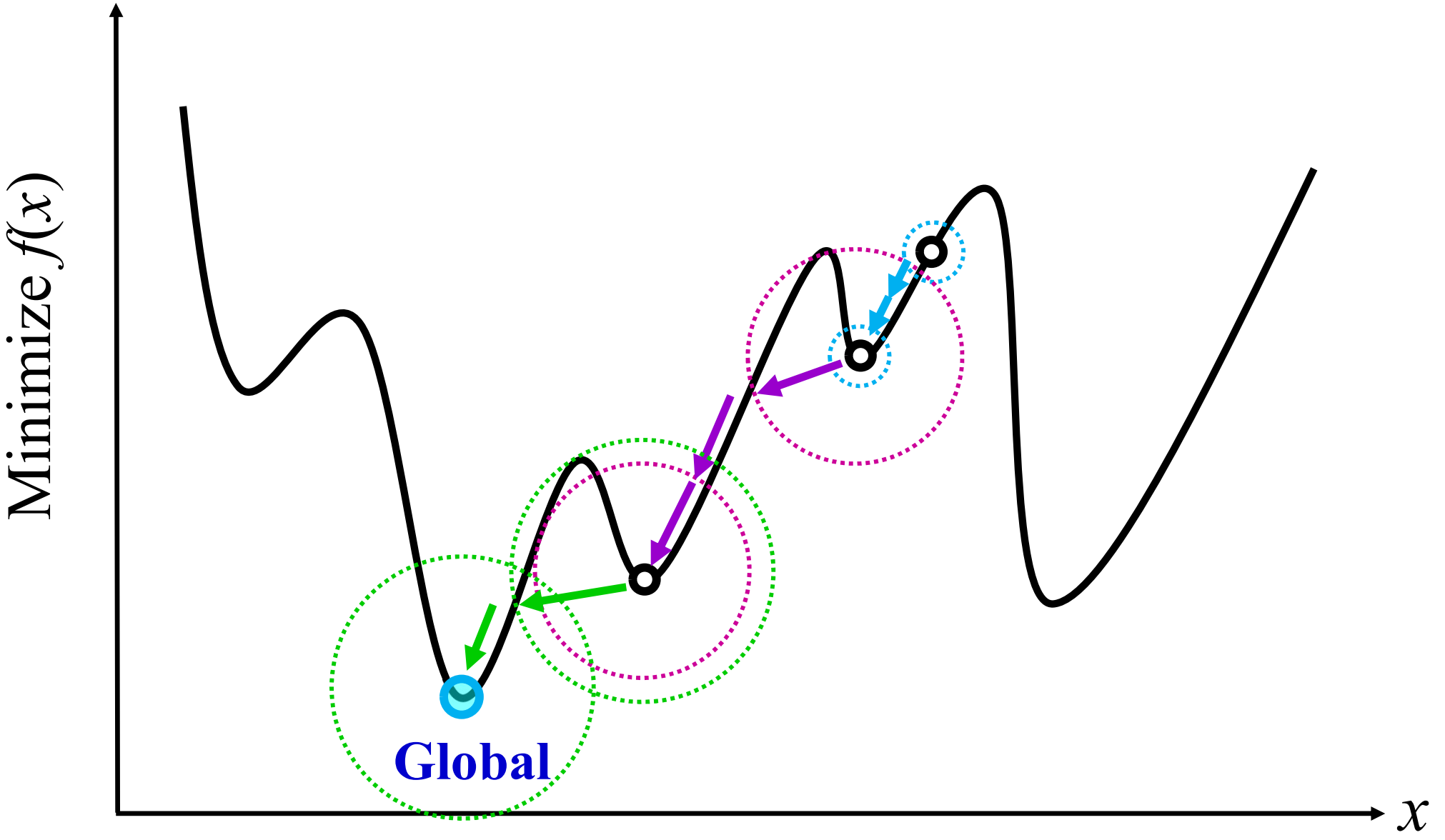
# Local Search with Neighborhood Change

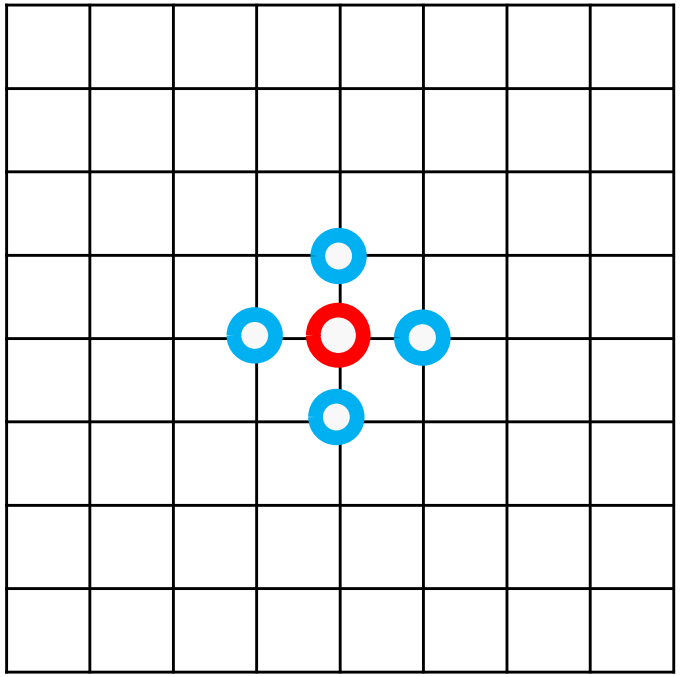


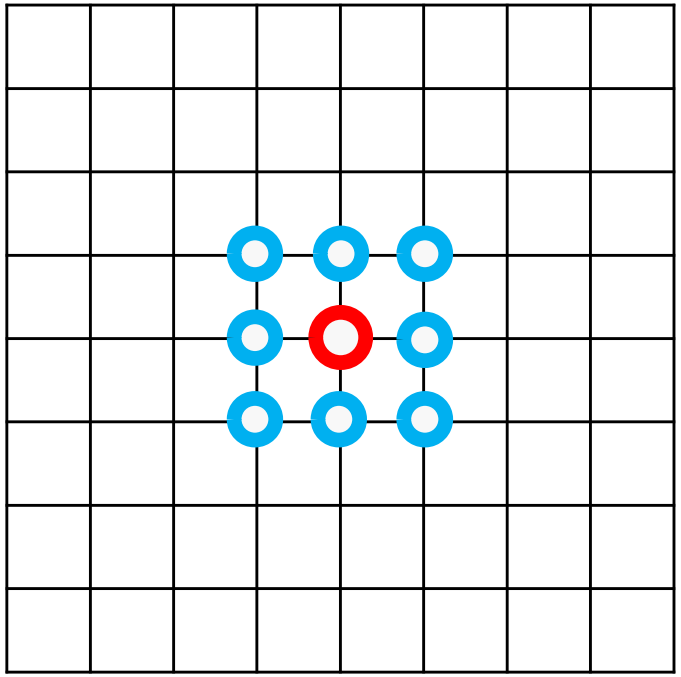
# Local Search with Neighborhood Change



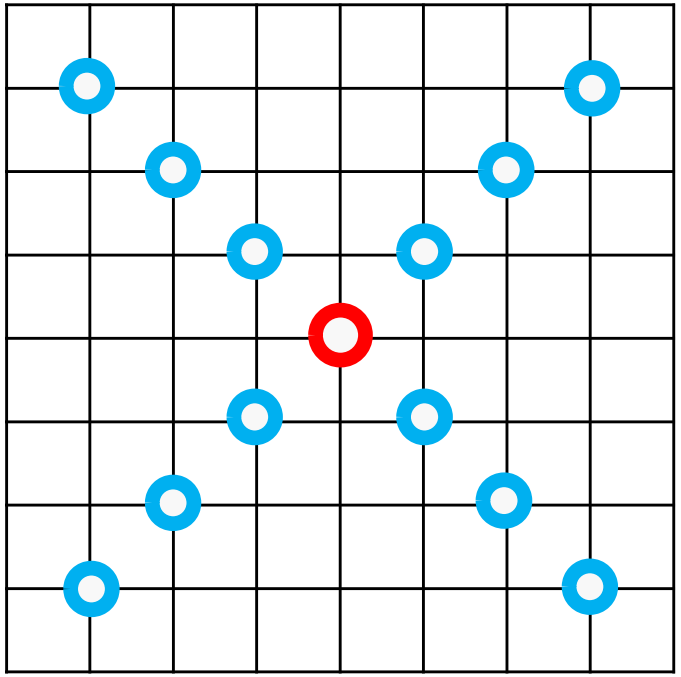
# Local Search with Neighborhood Change

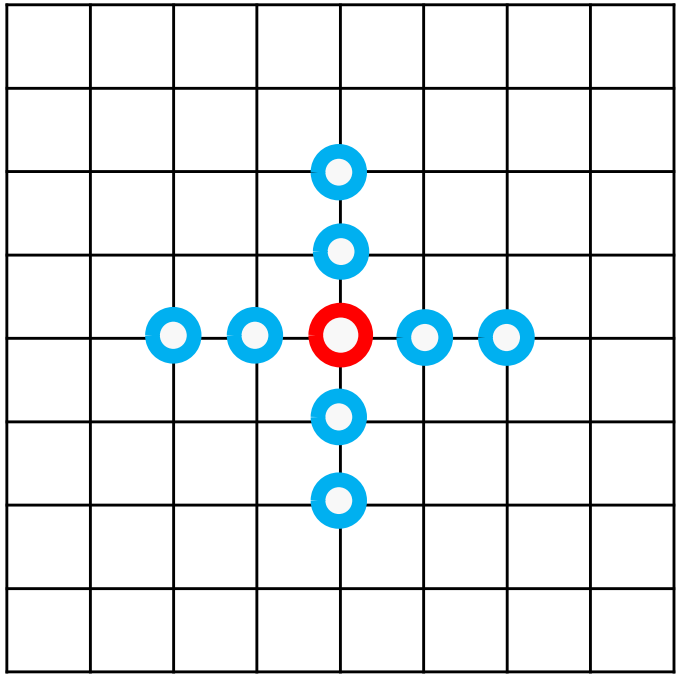






# Local Search with Neighborhood Change





## Adjacent two-city change

Size: $n$

| 0 | 1 | 2 | 3 | 4 | 5 | 7 | 6 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|

## Arbitrary two-city change

Size: $n(n-1)/2$

| 0 | 1 | 2 | 3 | 4 | 5 | 9 | 7 | 8 | 6 |
|---|---|---|---|---|---|---|---|---|---|

## Insertion (Shift)

Size: $n(n-3)$

| 0 | 1 | 2 | 3 | 4 | 5 | 7 | 8 | 6 | 9 |
|---|---|---|---|---|---|---|---|---|---|

## Inversion (Two-edge change)

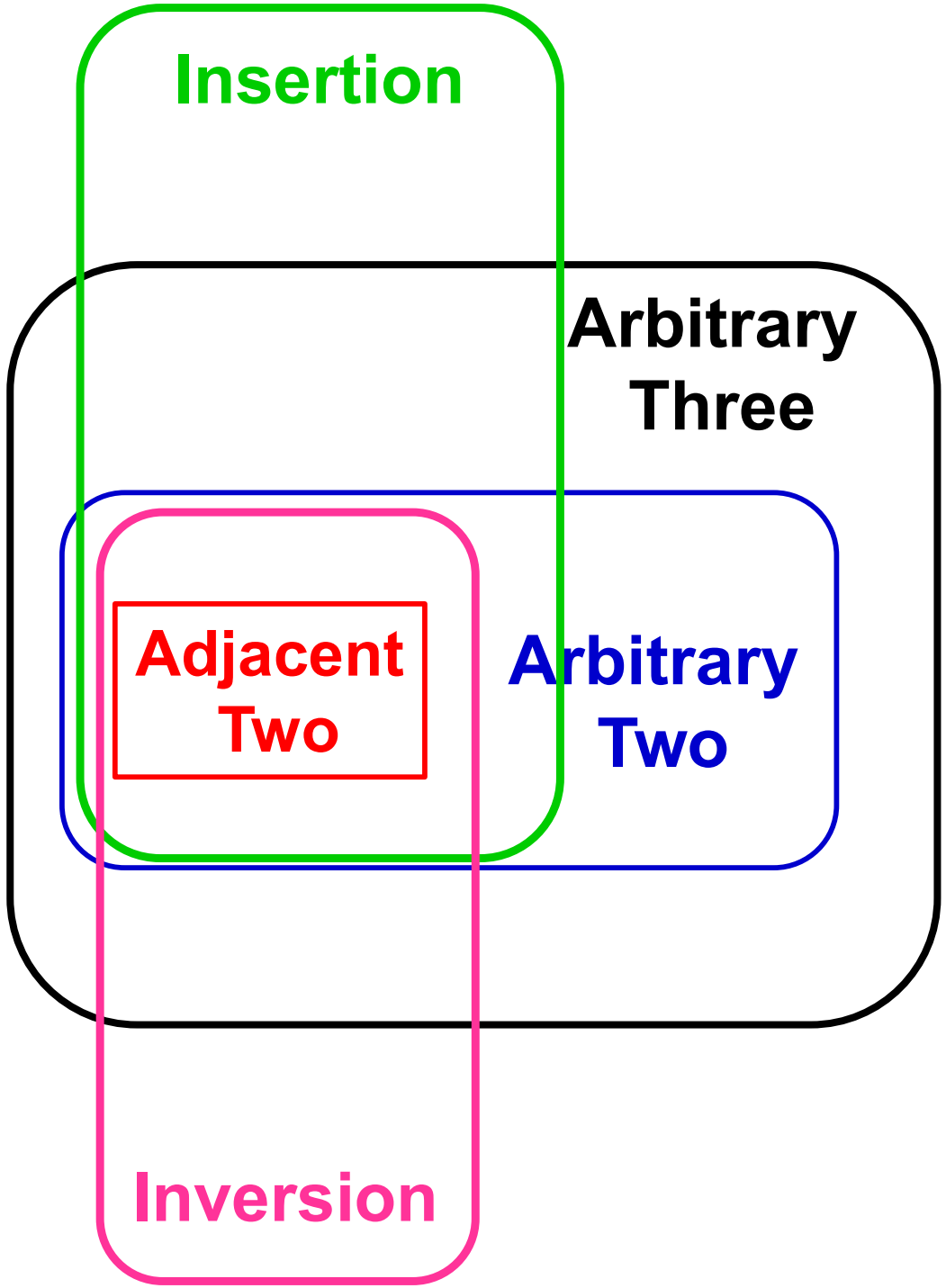
Size: $n(n-3)/2$

| 0 | 1 | 2 | 8 | 7 | 6 | 5 | 4 | 3 | 9 |
|---|---|---|---|---|---|---|---|---|---|

## Arbitrary three-city change

Size: $n(n-1)(2n-1)/6$

| 0 | 1 | 2 | 6 | 4 | 5 | 9 | 7 | 8 | 3 |
|---|---|---|---|---|---|---|---|---|---|



# Variable Neighborhood Search

Google Scholar

## Pierre Hansen

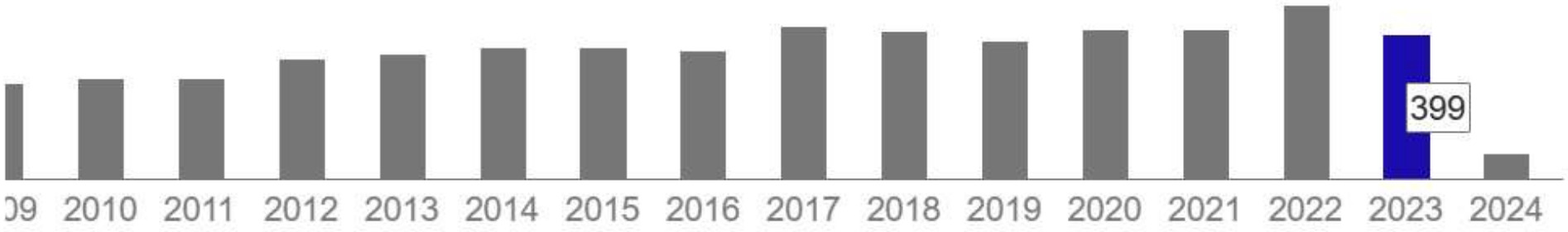
FOLLOW

GERAD, HEC Canada
Verified email at hec.ca - Homepage

Graph Theory VNS

| TITLE | CITED BY | YEAR |
|---|---|---|
| Variable neighborhood search<br>N Mladenović, P Hansen<br>Computers & operations research 24 (11), 1097-1100 | 6547 * | 1997 |

Cited by 6547



# Variable Neighborhood Search Algorithm

European Journal of Operational Research 130 (2001) 449–467

Invited Review

## Variable neighborhood search: Principles and applications

Pierre Hansen *, Nenad Mladenović

# Variable Neighborhood Search Algorithm

*P. Hansen, N. Mladenović / European Journal of Operational Research 130 (2001) 449–467*

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Fig. 1. Steps of the basic VNS.

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P. Hansen, N. Mladenović / European Journal of Operational Research 130 (2001) 449–467

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$N_{k+1}$ is larger than $N_k$, $k = 1, 2, \ldots, k_{max}-1$.

In some cases, $N_k \subset N_{k+1}$, $k = 1, 2, \ldots, k_{max}-1$.

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*P. Hansen, N. Mladenović / European Journal of Operational Research 130 (2001) 449–467*

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$N_1$ is used first, then $N_2$ , $N_3$ , ...

# Basic Variable Neighborhood Search (VNS) Algorithm

*P. Hansen, N. Mladenović / European Journal of Operational Research 130 (2001) 449–467*

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Local search is from a randomly selected neighbor in the current $N_k$.

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Local search run. Local solution: $x$"
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the best solution so far

If the current solution is improved, $N_k$ ==> $N_1$. If not, $N_k$ ==> $N_{k+1}$ .

**Neighborhood Structures:** $N_k$, $k = 1, 2, \ldots, k_{max}$

$N_{k+1}$ is larger than $N_k$, $k = 1, 2, \ldots, k_{max}-1$.

In some cases, $N_k \subset N_{k+1}$, $k = 1, 2, \ldots, k_{max}-1$.

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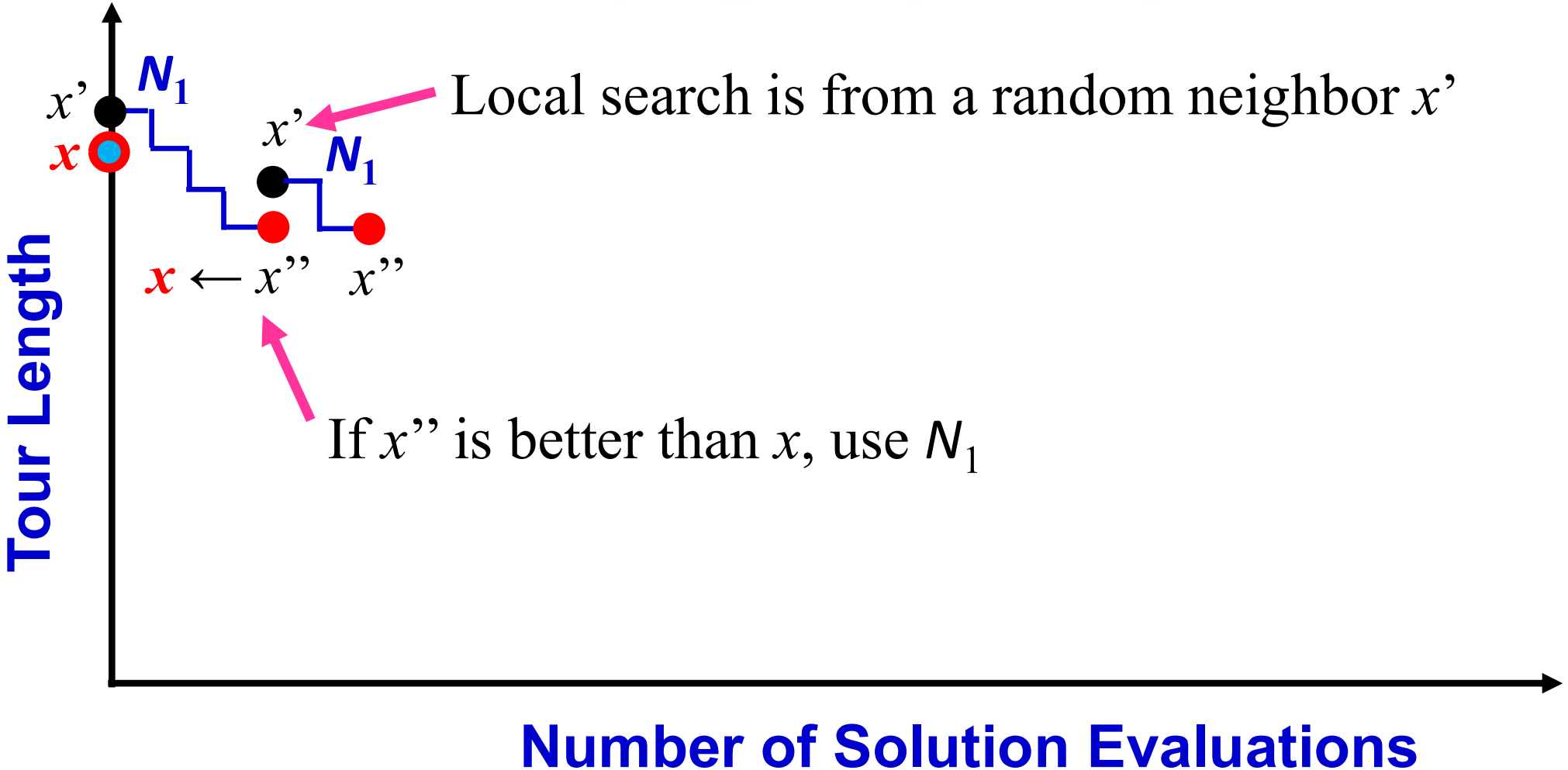

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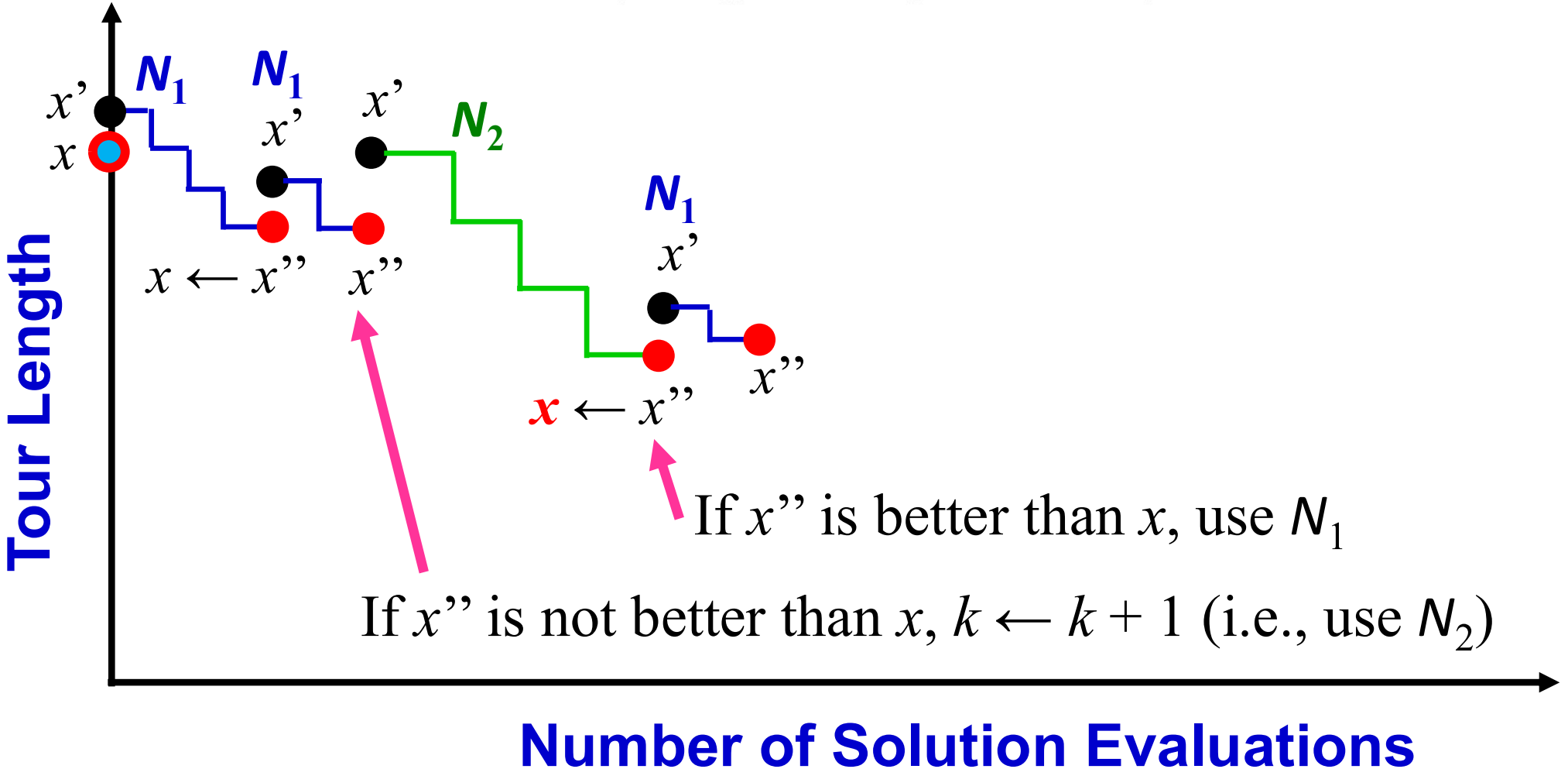

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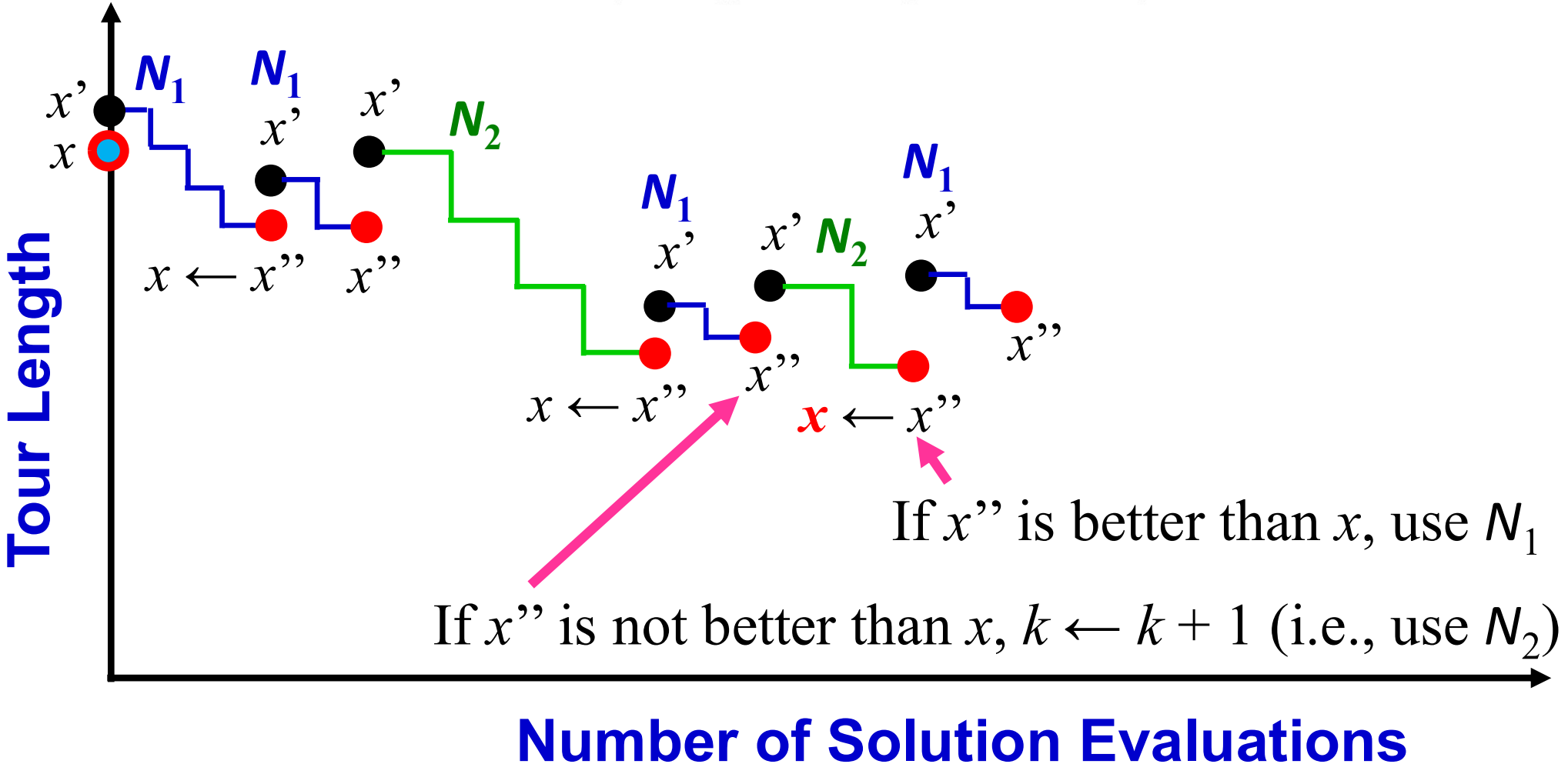

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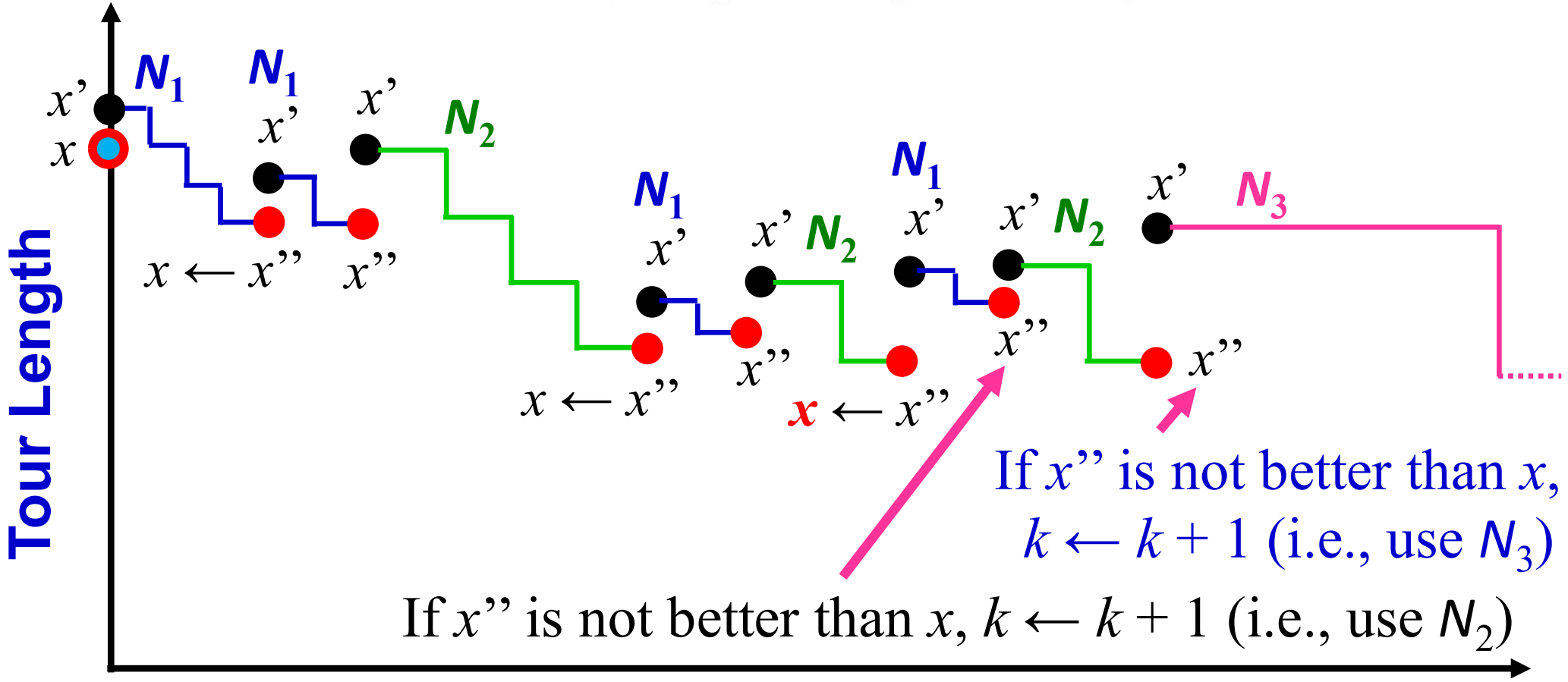

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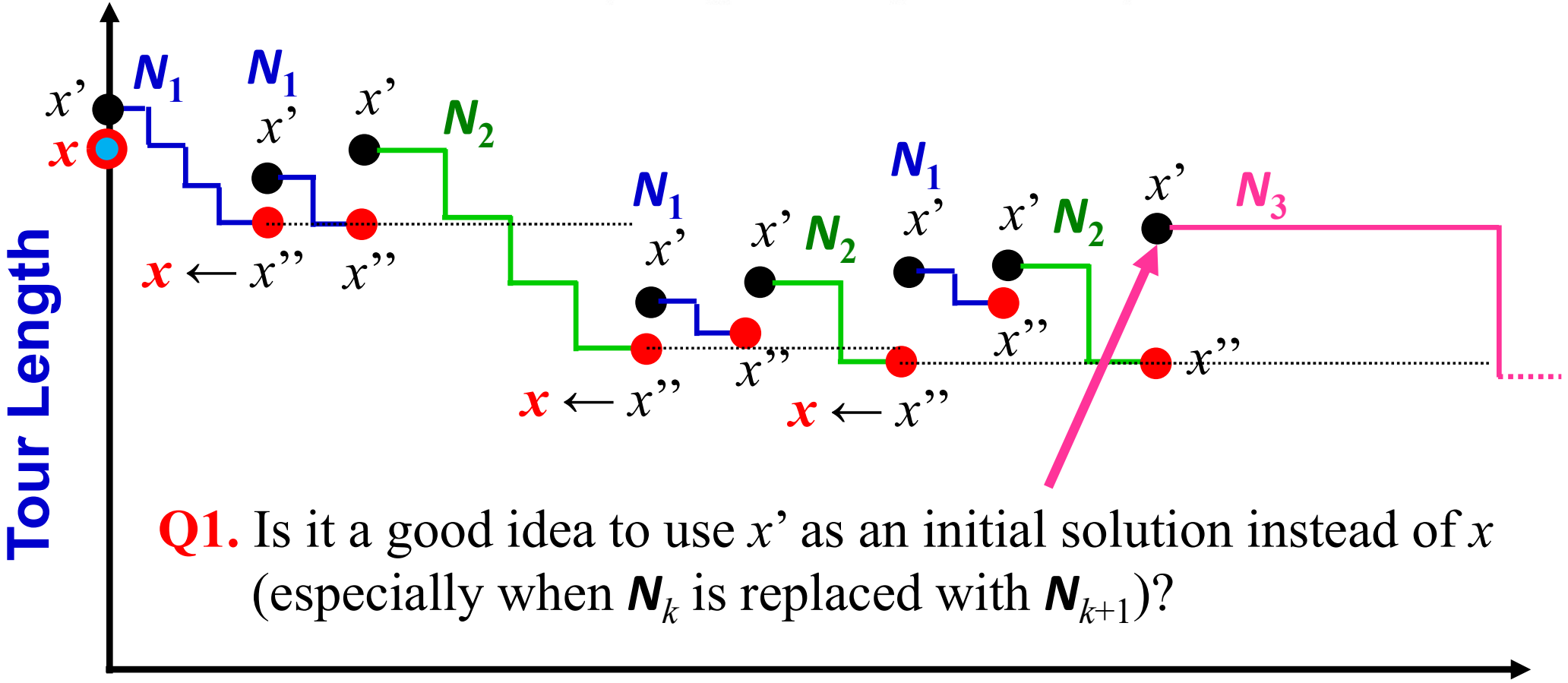


**Q1.** Is it a good idea to use $x'$ as an initial solution instead of $x$ (especially when $\boldsymbol{N}_k$ is replaced with $\boldsymbol{N}_{k+1}$)?

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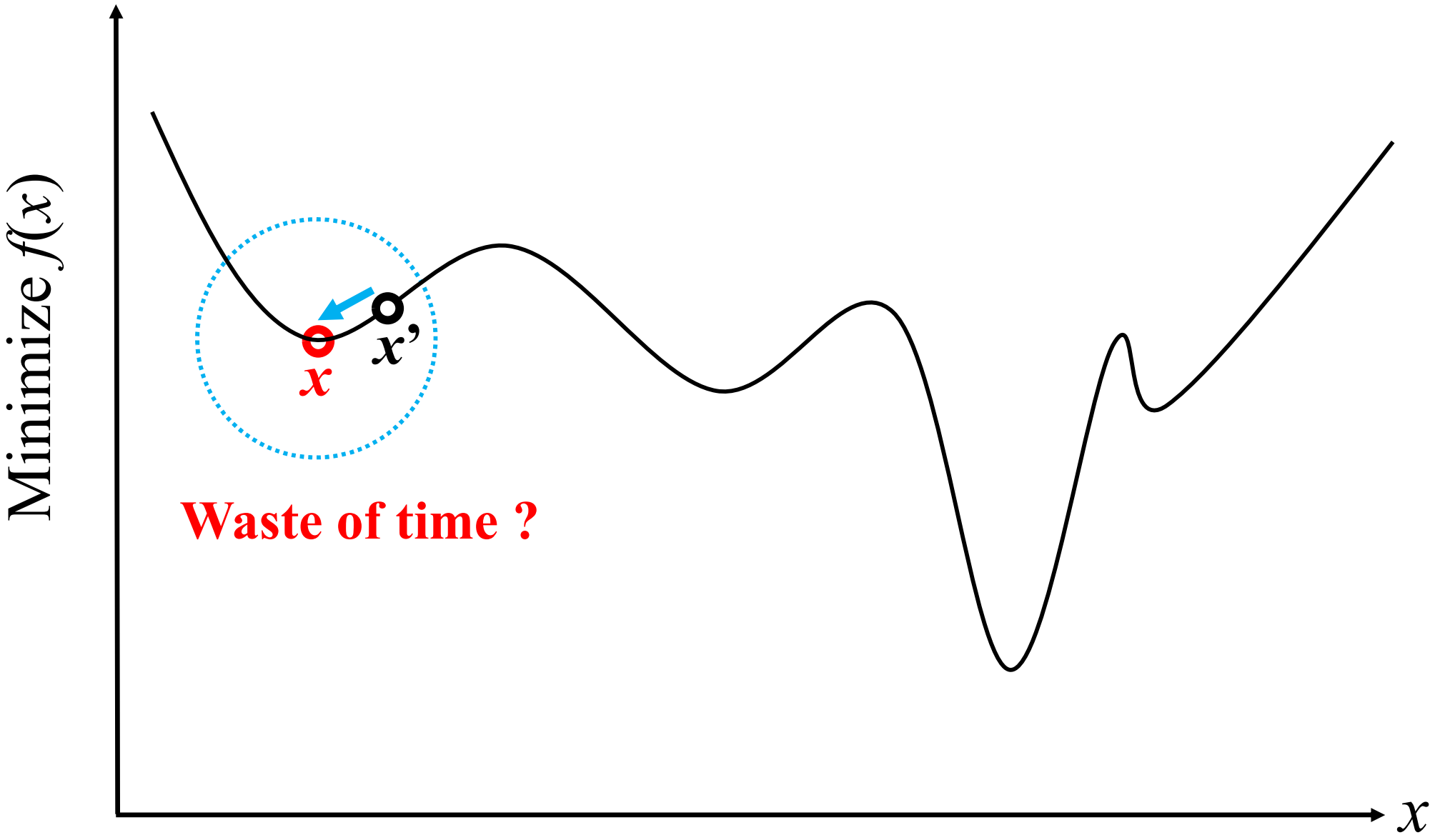

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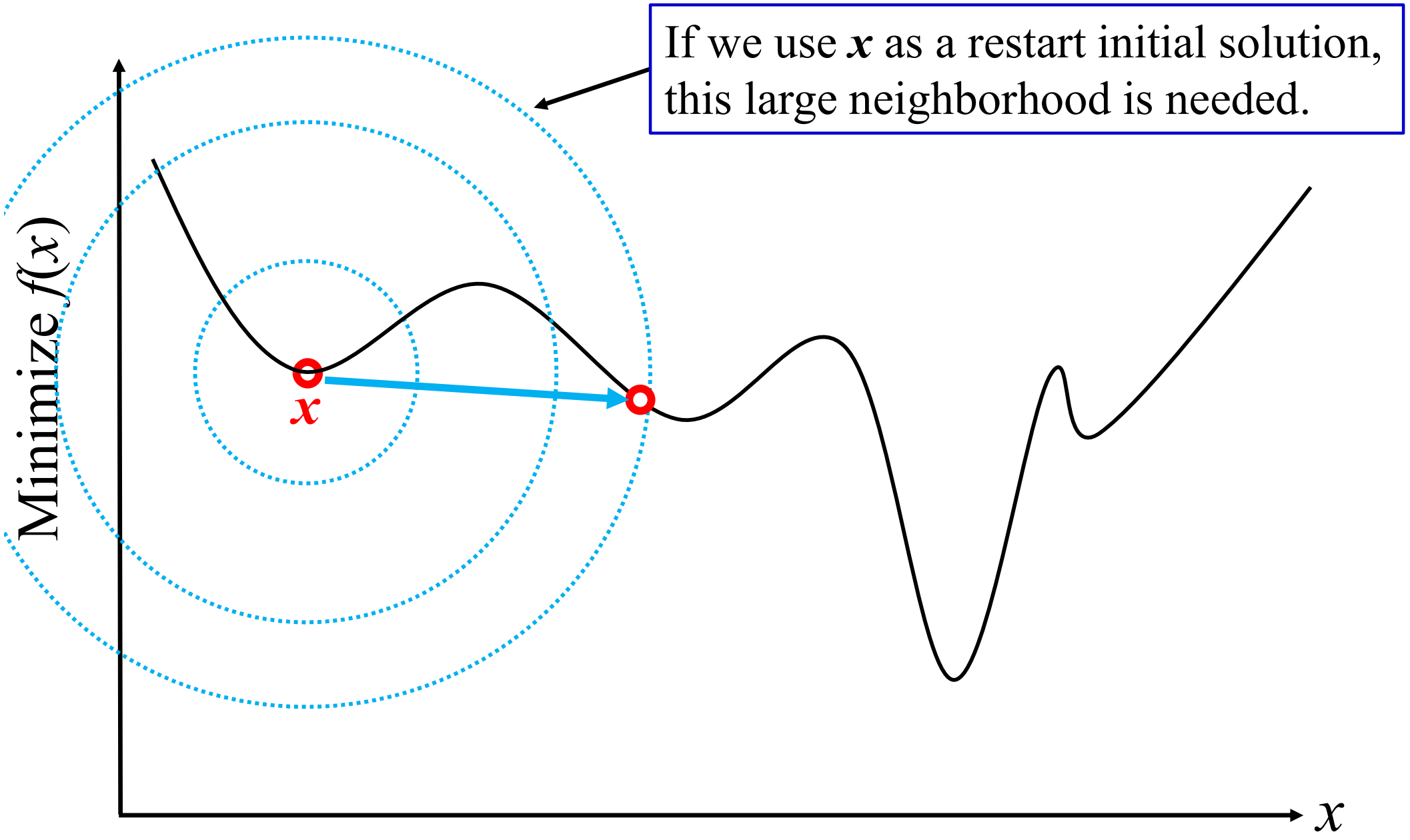

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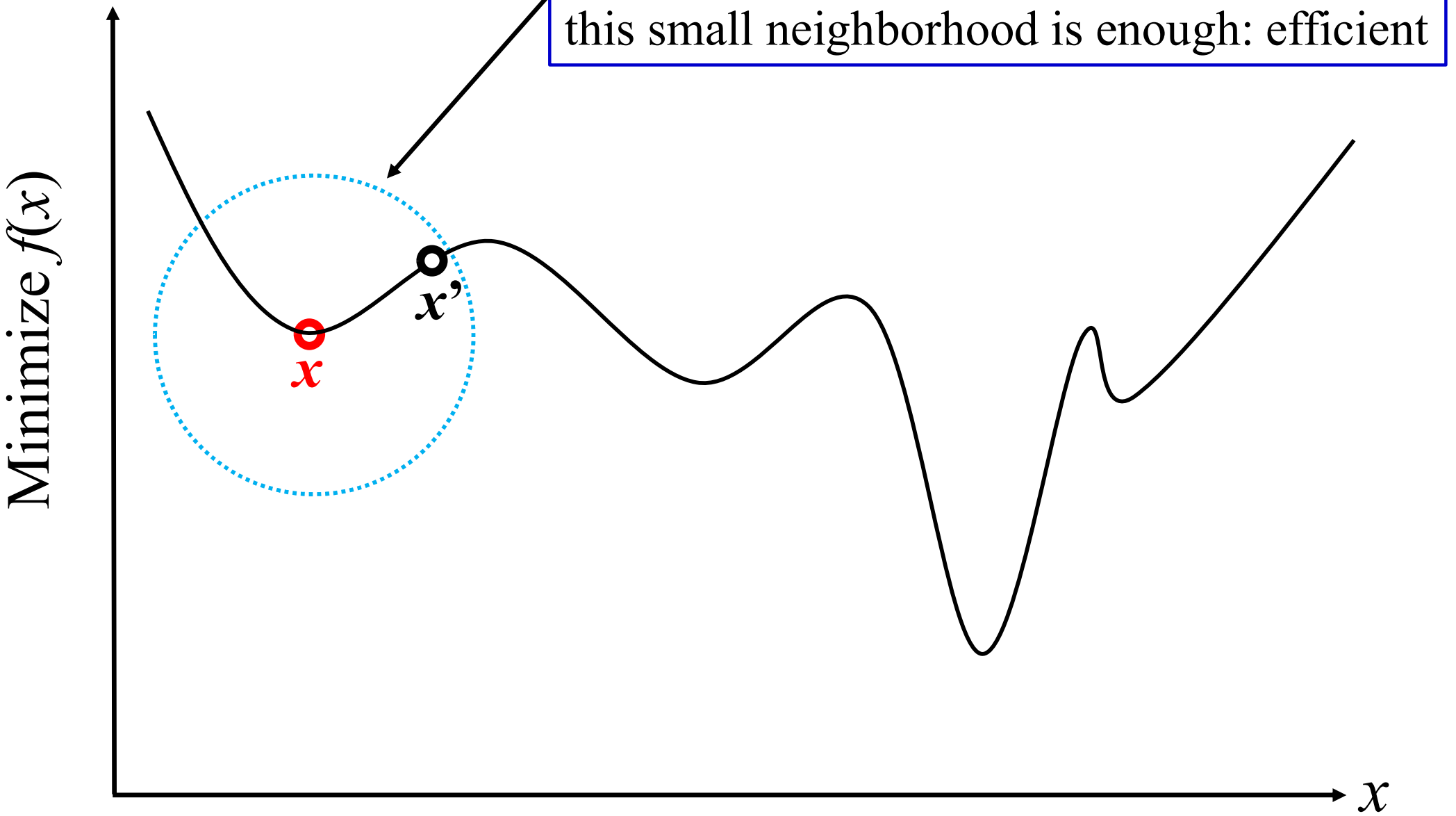
If we use ***x'*** as a restart initial solution,
this small neighborhood is enough: efficient



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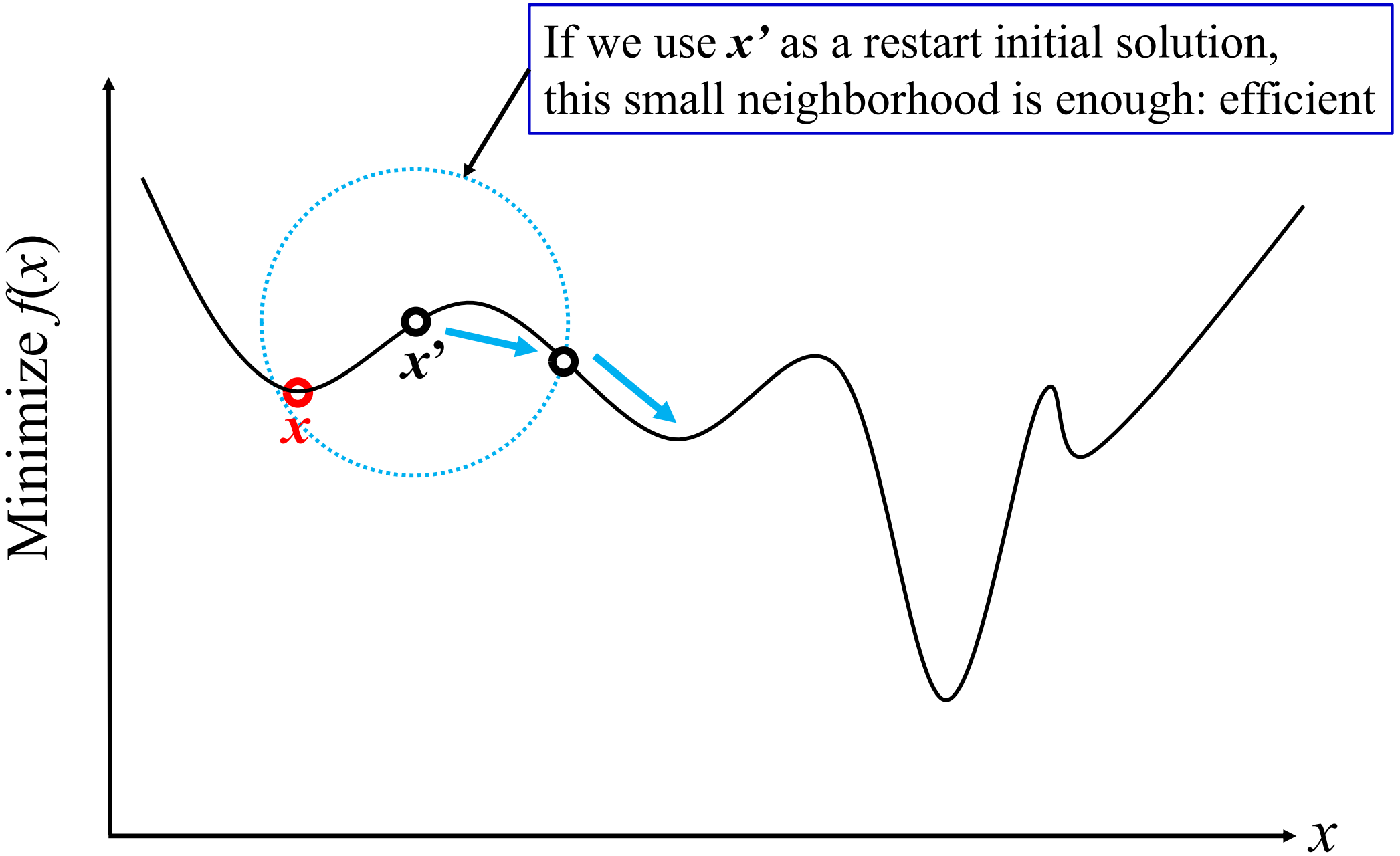

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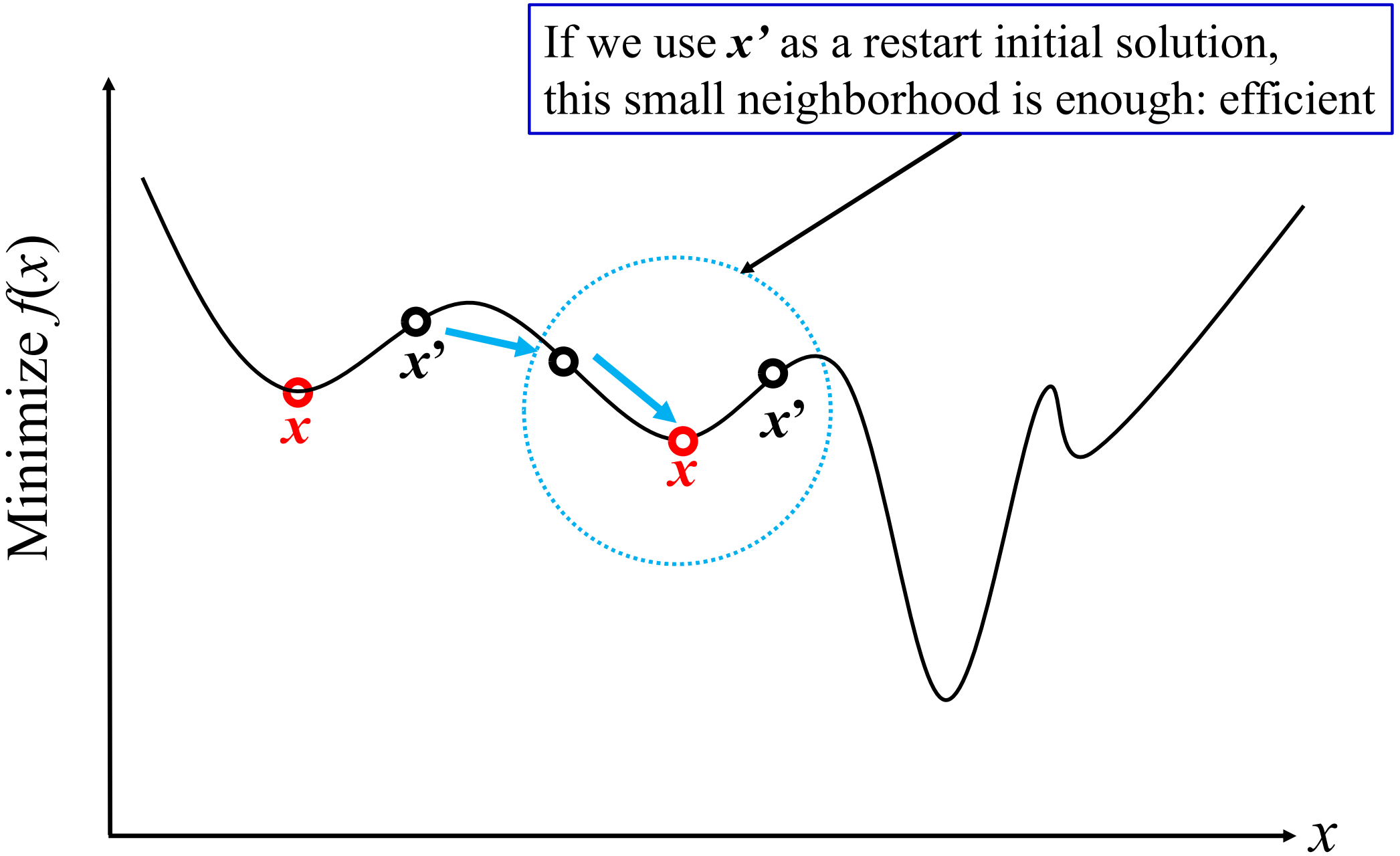

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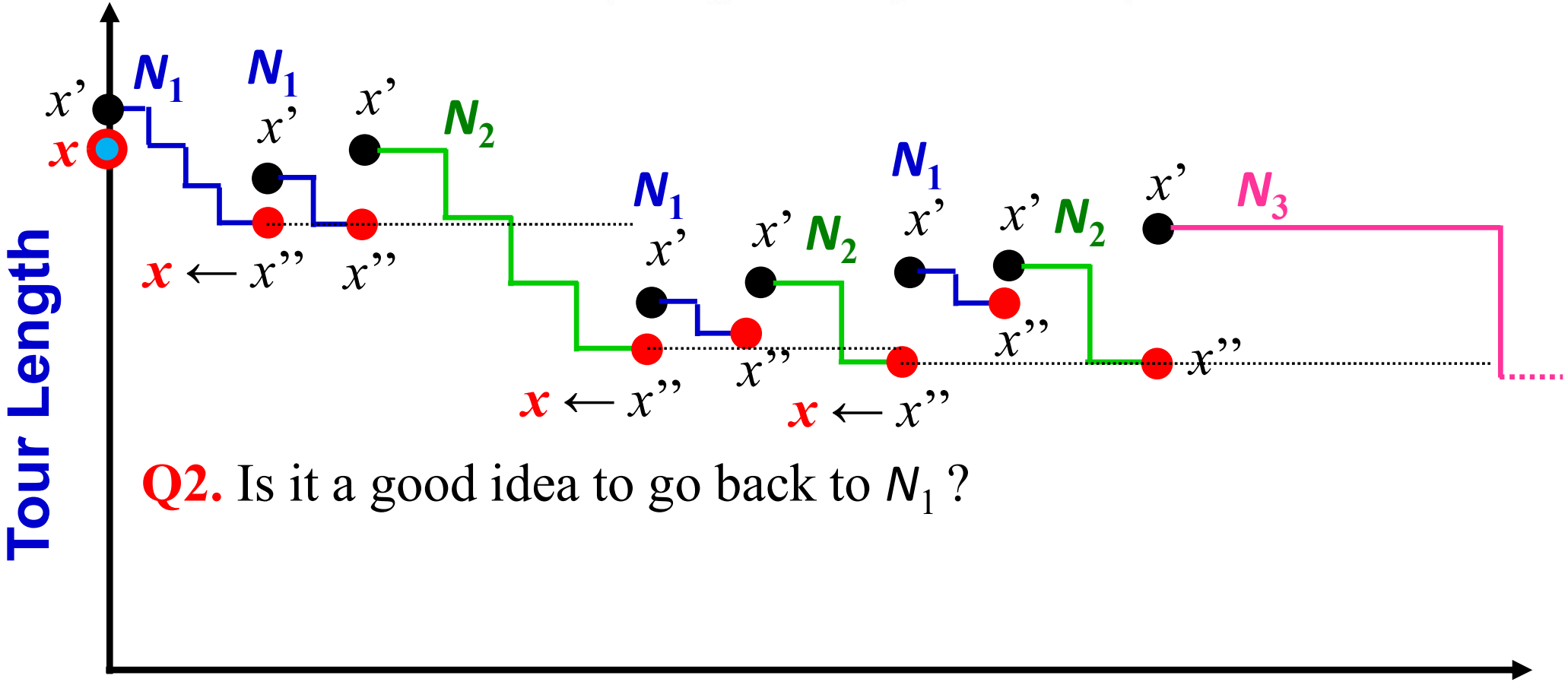


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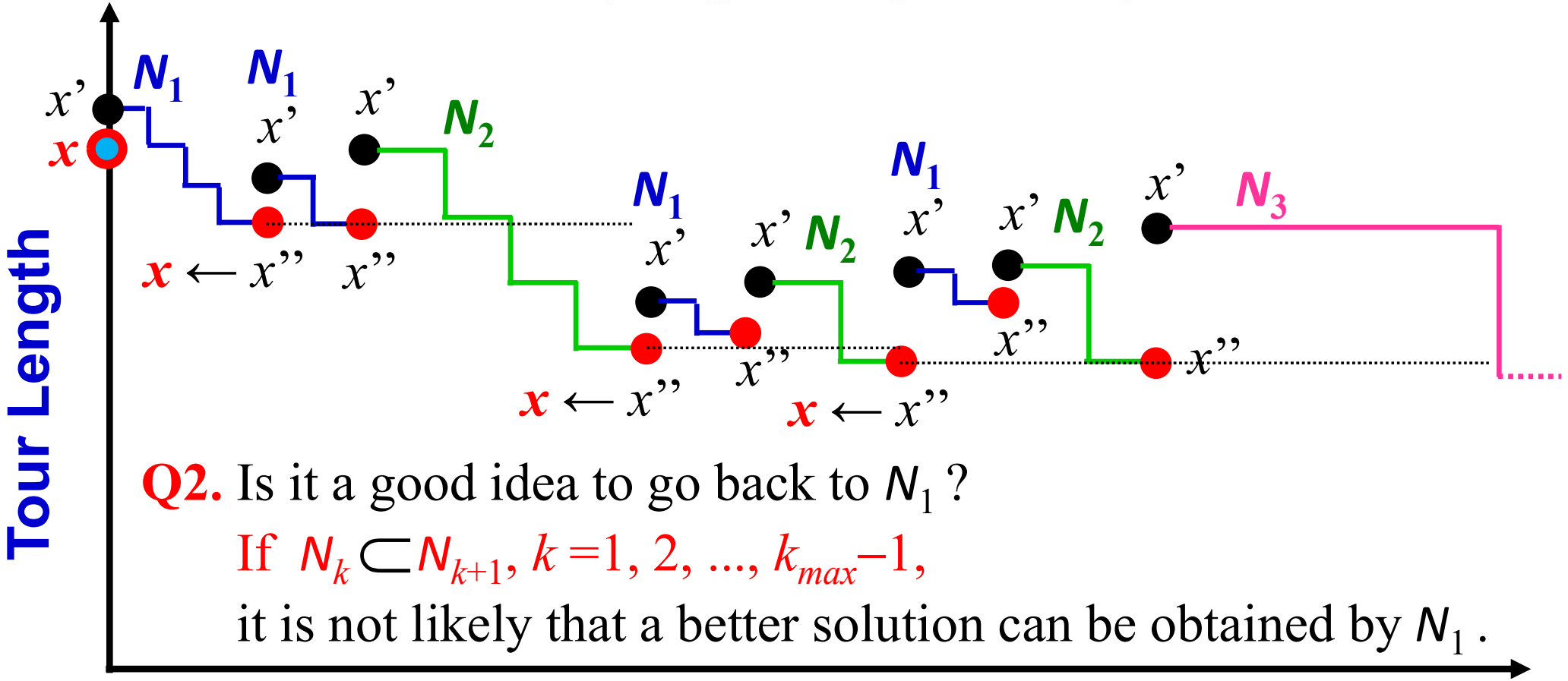


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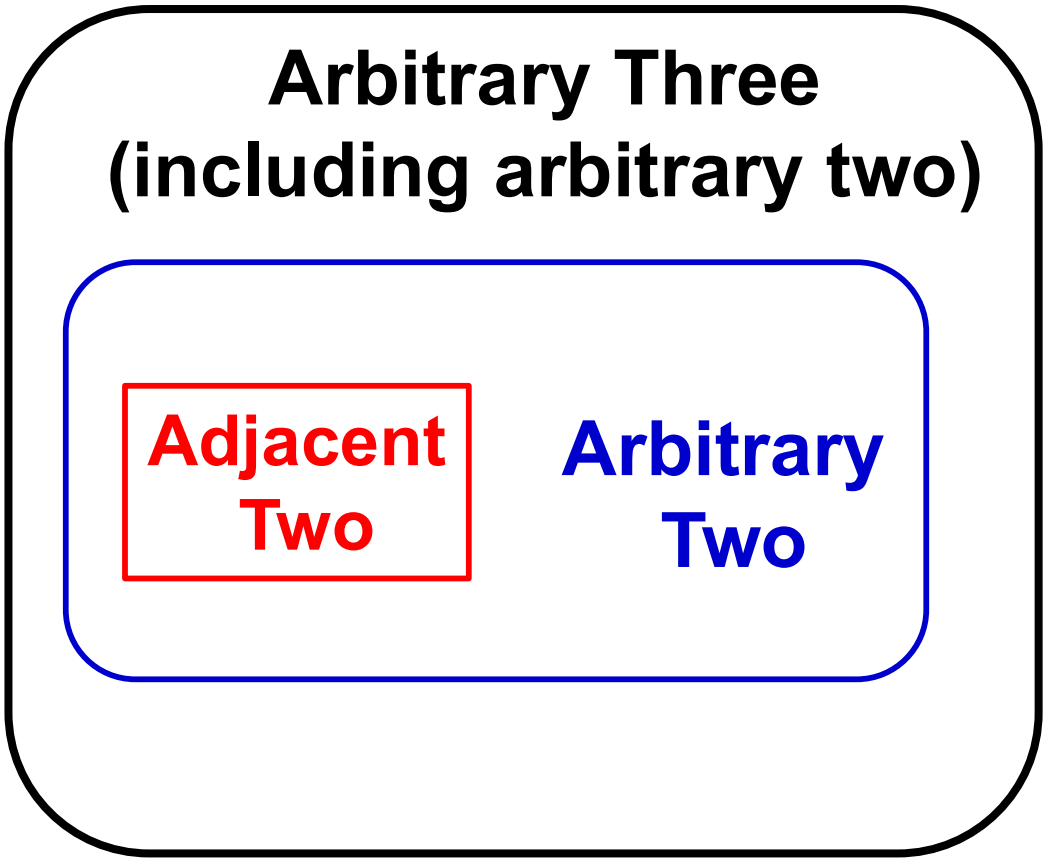


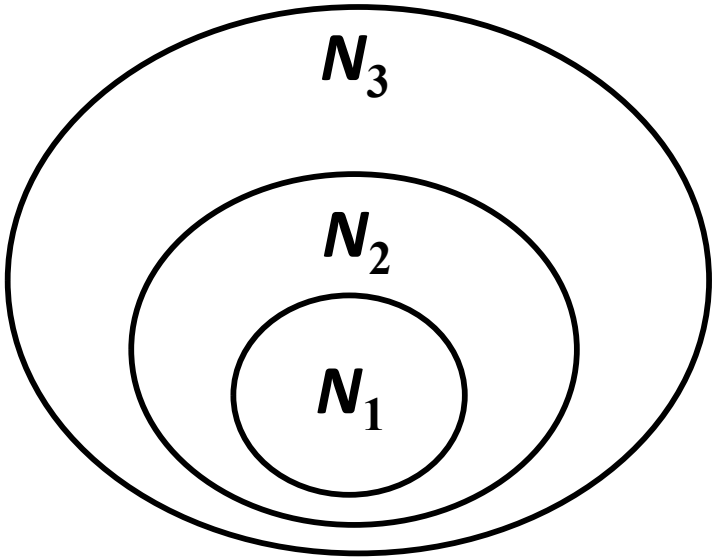

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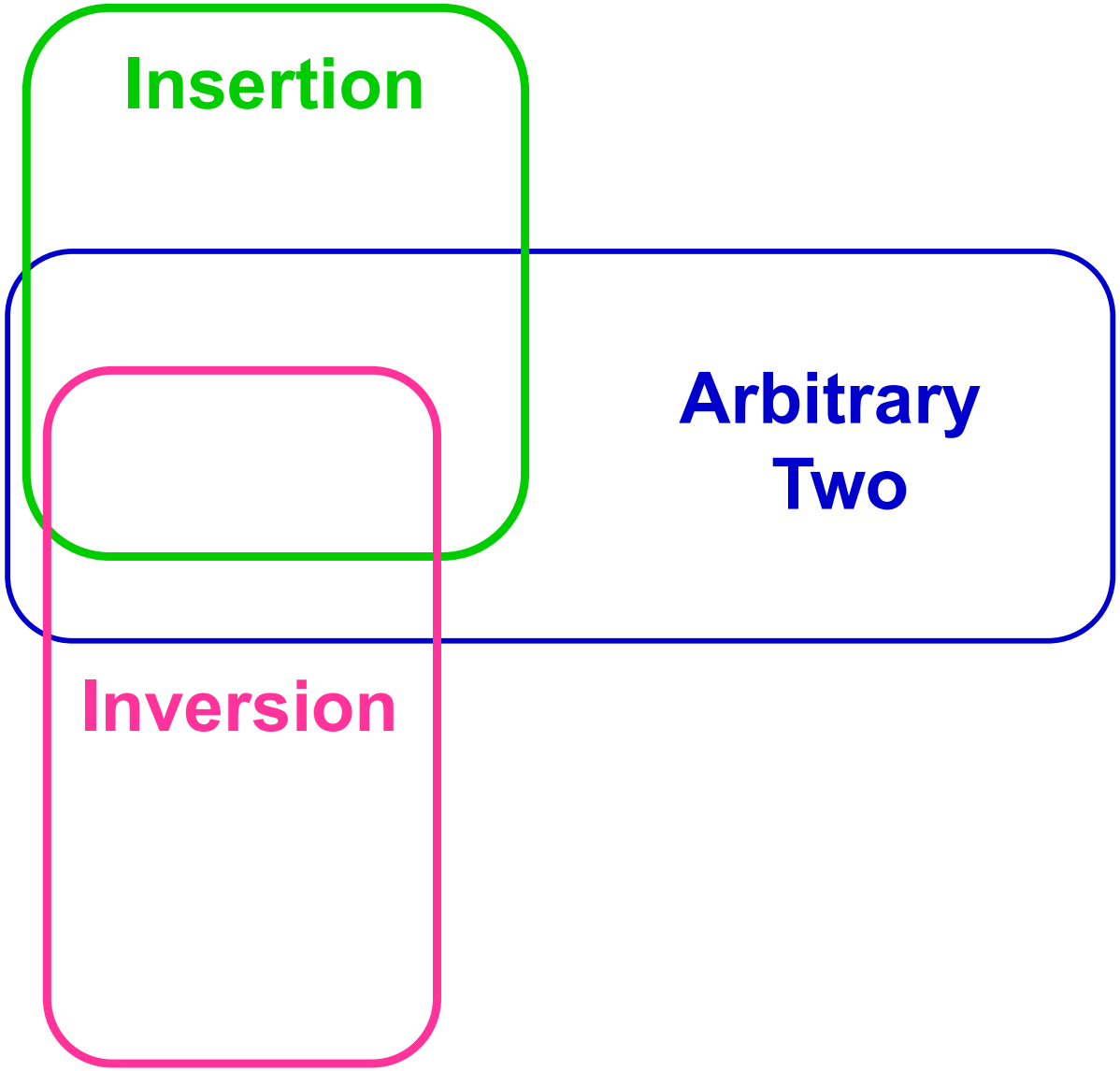
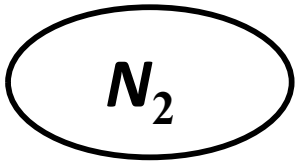
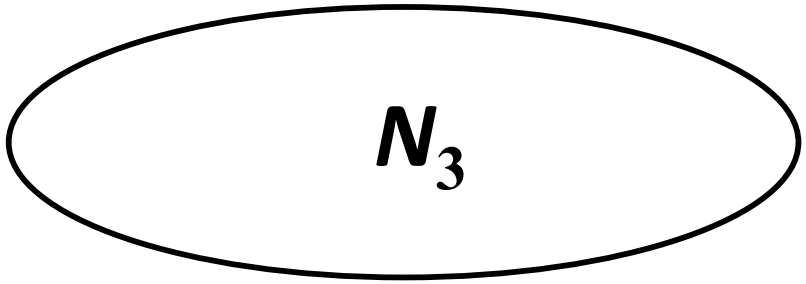


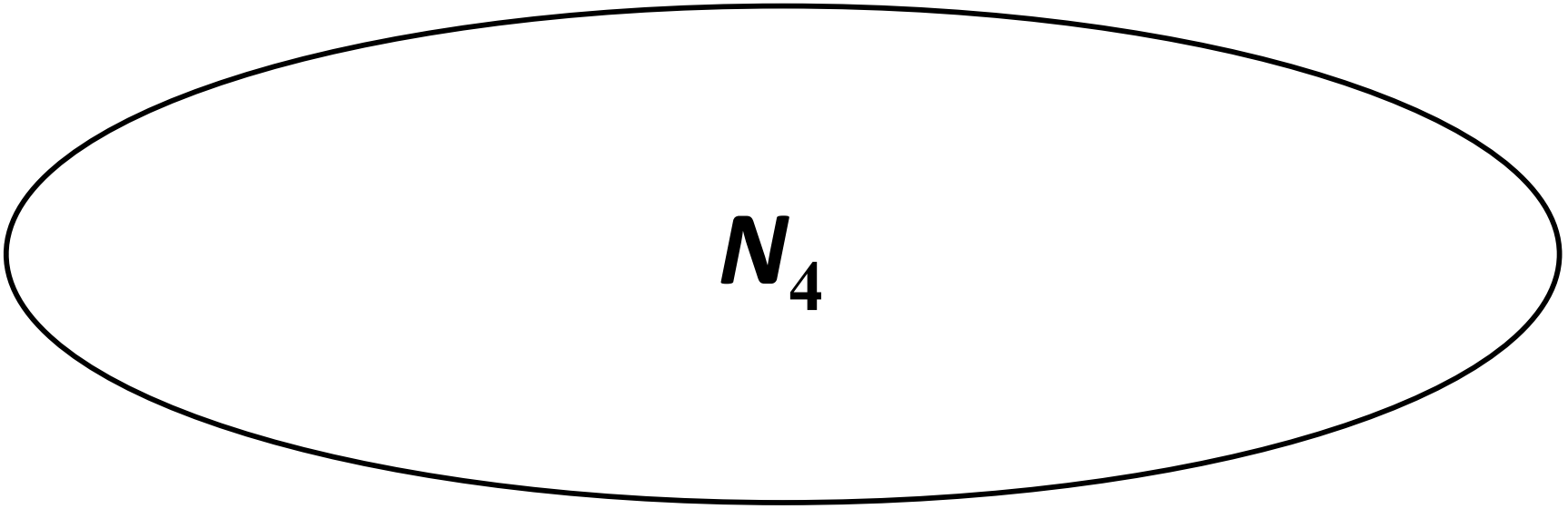
# Example of Neighborhood Structures of TSP (*n*-city problem)

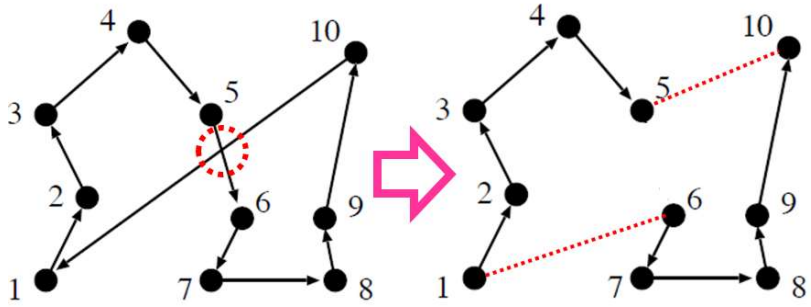
$N_k$ : *k*-edge change neighborhood ($k = 2, 3, ..., n$)

$N_1 = \emptyset$




$N_k$ : 2-edge change neighborhood

**Q2.** Is it a good idea to go back to $N_1$ ? **Yes ?** (Depends on the problem.)

$N_k$ : $k$-edge change neighborhood ($k = 2, 3, ..., n$)

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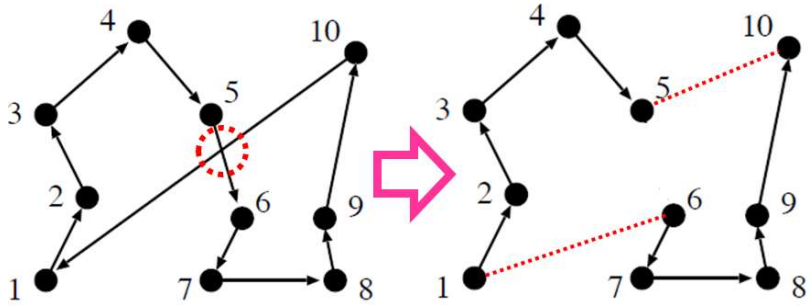
$N_3$

$N_4$

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# Variant of Variable Neighborhood Search (VNS)

## Variable Neighborhood Descent (VND)

European Journal of Operational Research 130 (2001) 449–467

Invited Review

# Variable neighborhood search: Principles and applications

Pierre Hansen *, Nenad Mladenović

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*P. Hansen, N. Mladenović / European Journal of Operational Research 130 (2001) 449–467*

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$N_1$ is used first, then $N_2$ , $N_3$ , ...

~~Local search is from a randomly selected neighbor in the current $N_k$.~~

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If no improvement, use the next neighborhood ($N_k$ ==> $N_{k+1}$ ).

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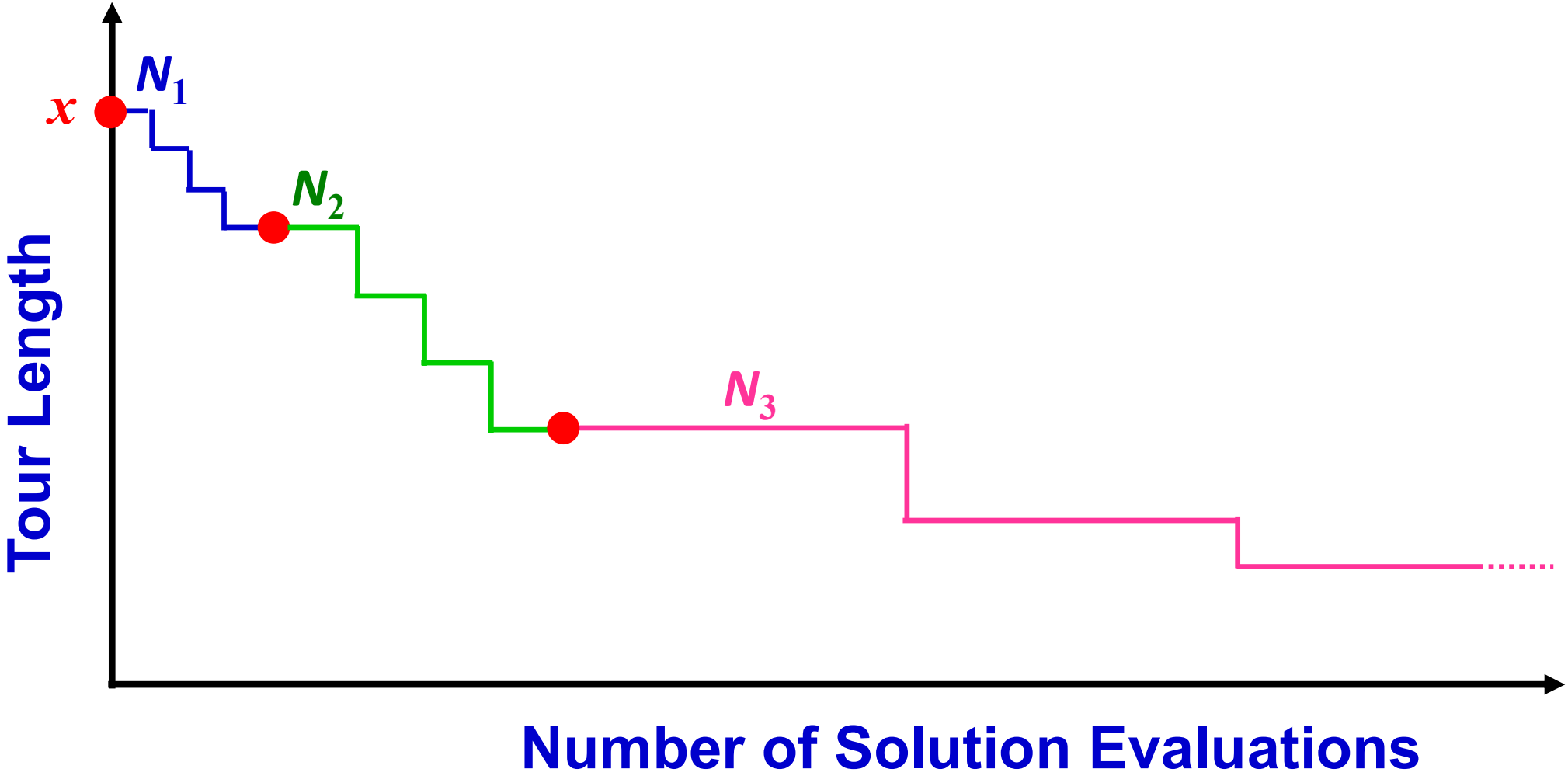

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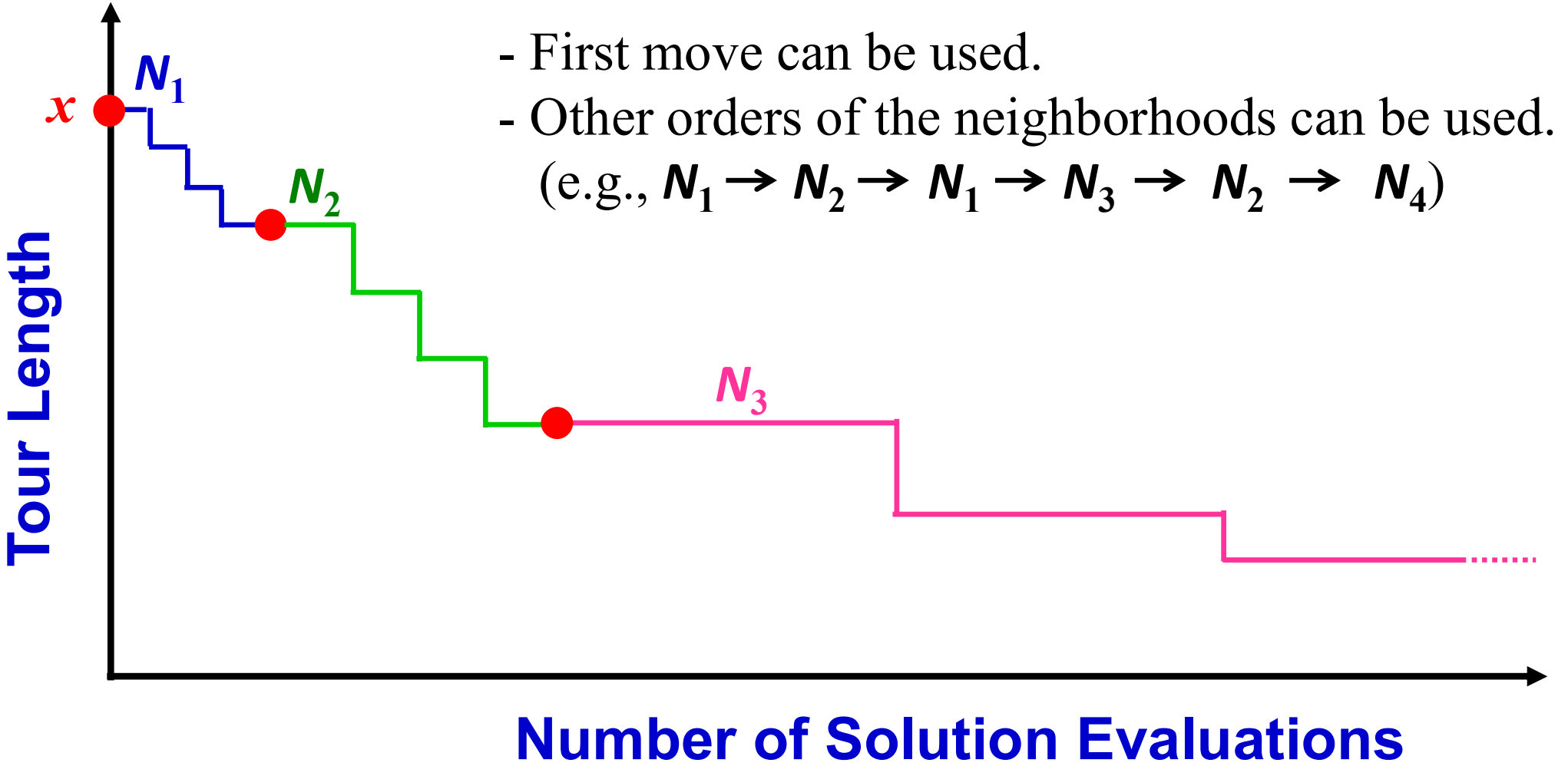


- First move can be used.
- Other orders of the neighborhoods can be used.
  (e.g., $N_1 \rightarrow N_2 \rightarrow N_1 \rightarrow N_3 \rightarrow N_2 \rightarrow N_4$)

# A Tutorial on Variable Neighborhood Search

**Pierre Hansen**
*GERAD and HEC Montreal*
**Nenad Mladenović**
*GERAD and Mathematical Institute, SANU, Belgrade*

## Basic Ideas behind Variable Neighborhood Search

**Fact 1** *A local minimum with respect to one neighborhood structure is not necessary so for another;*

**Fact 2** *A global minimum is a local minimum with respect to all possible neighborhood structures.*

**Fact 3** *For many problems local minima with respect to one or several neighborhoods are relatively close to each other.*

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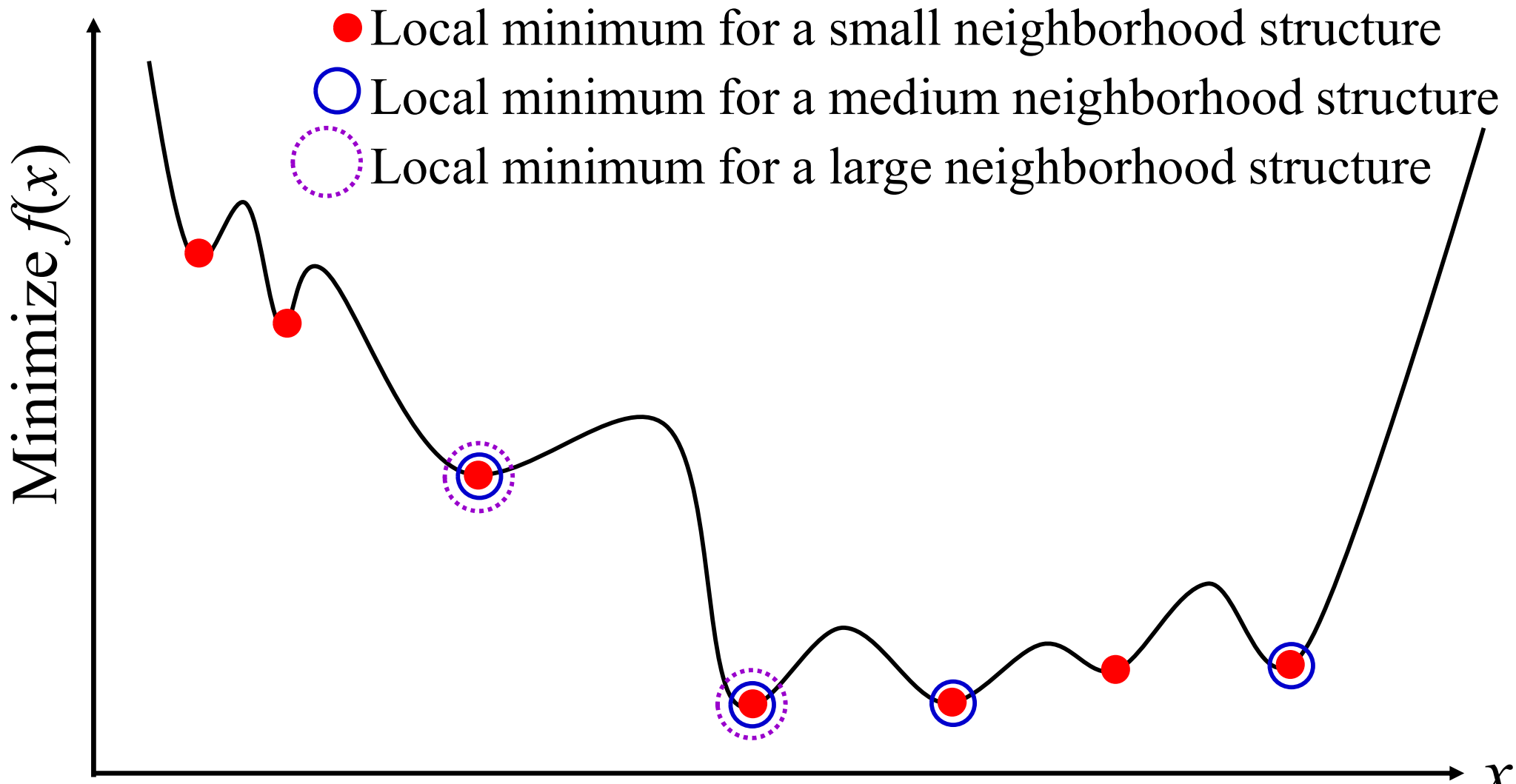

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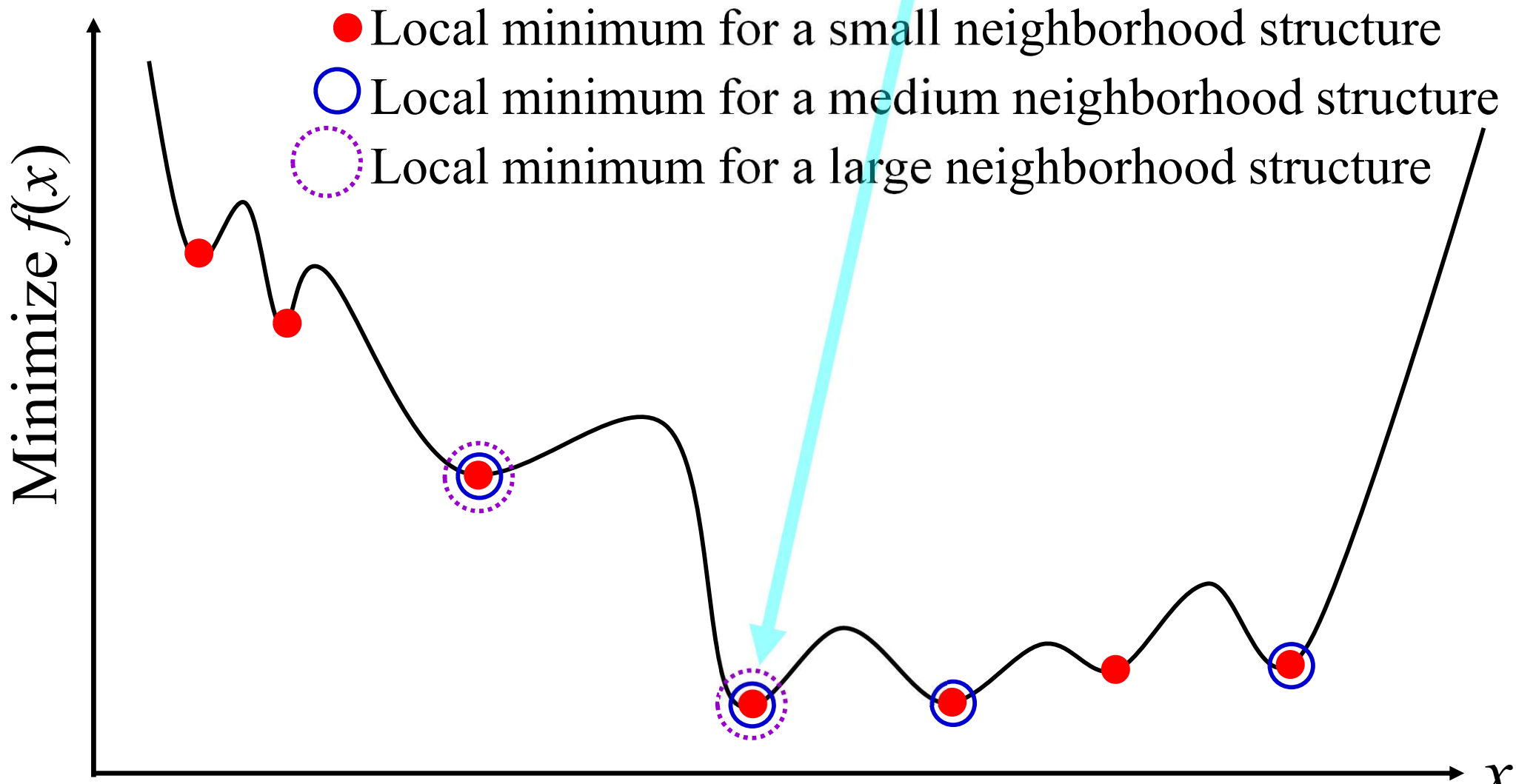

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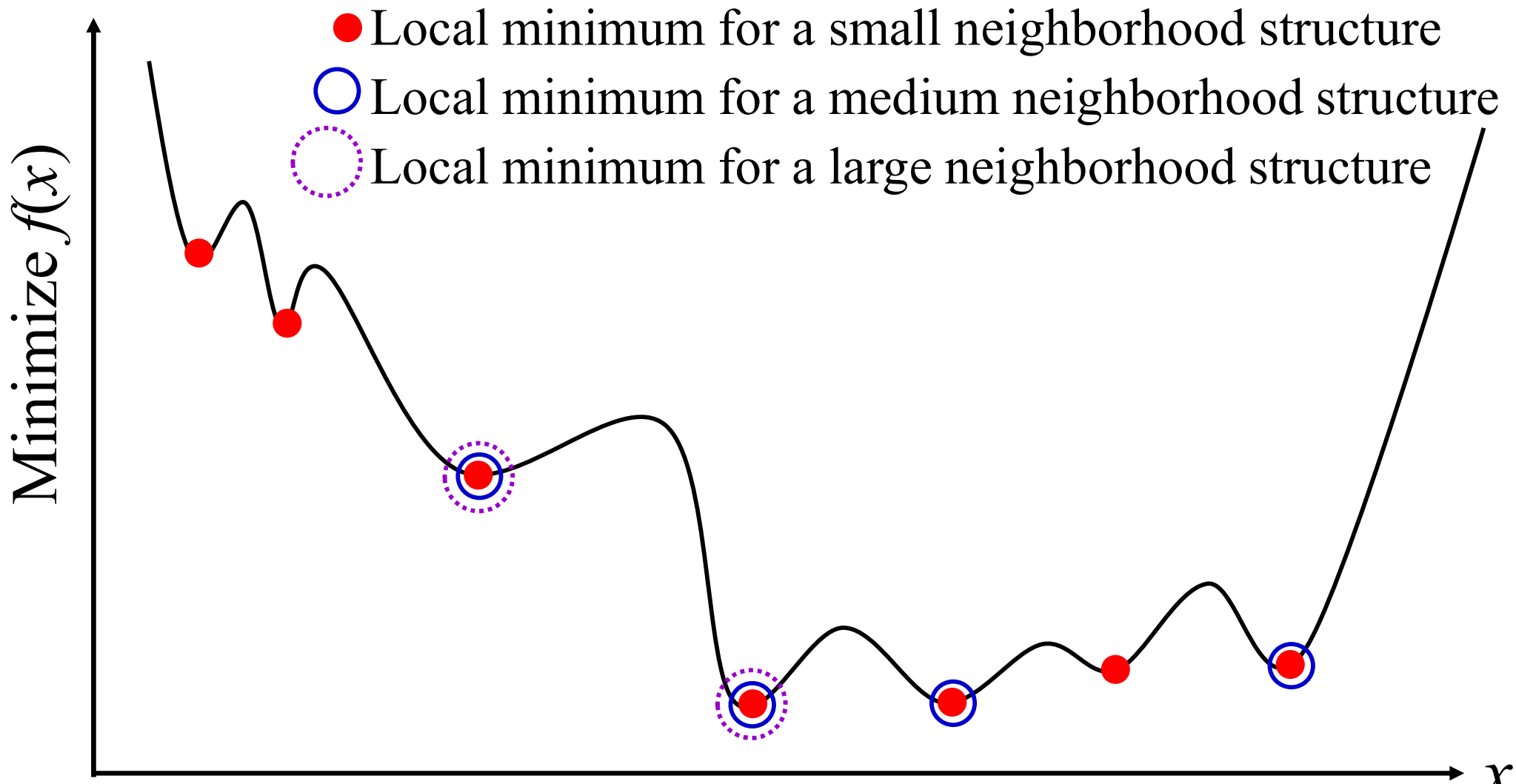

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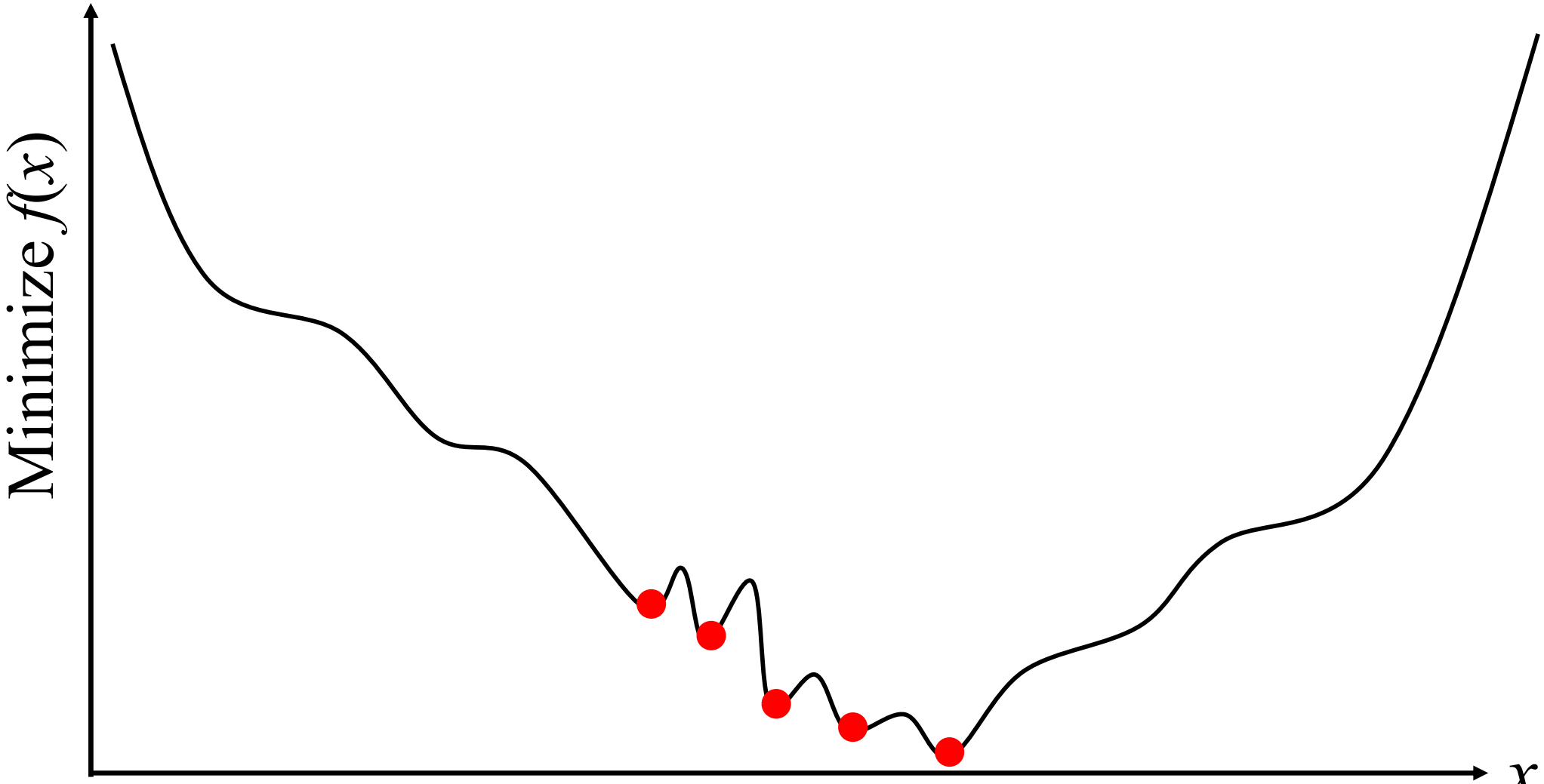

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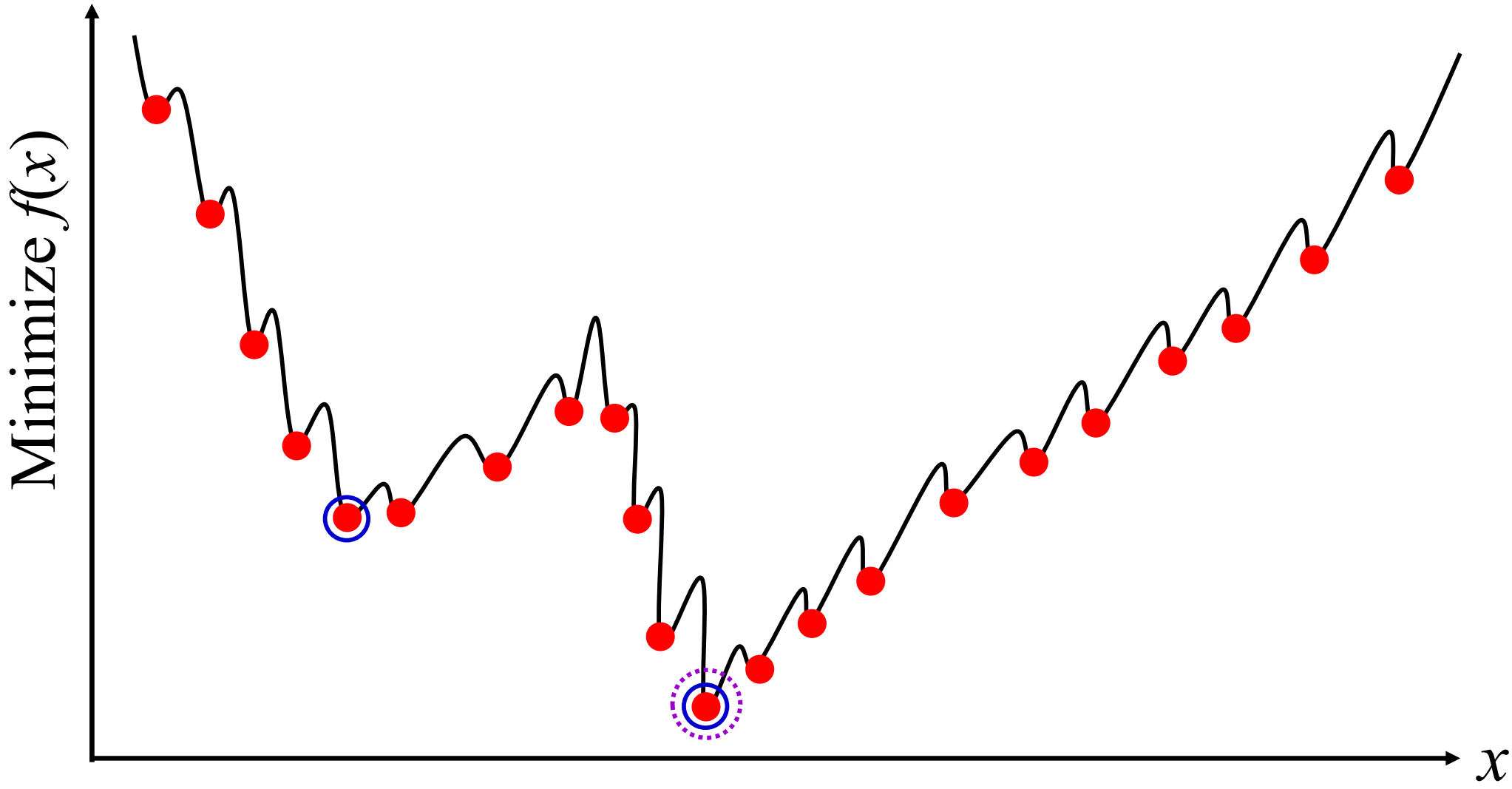

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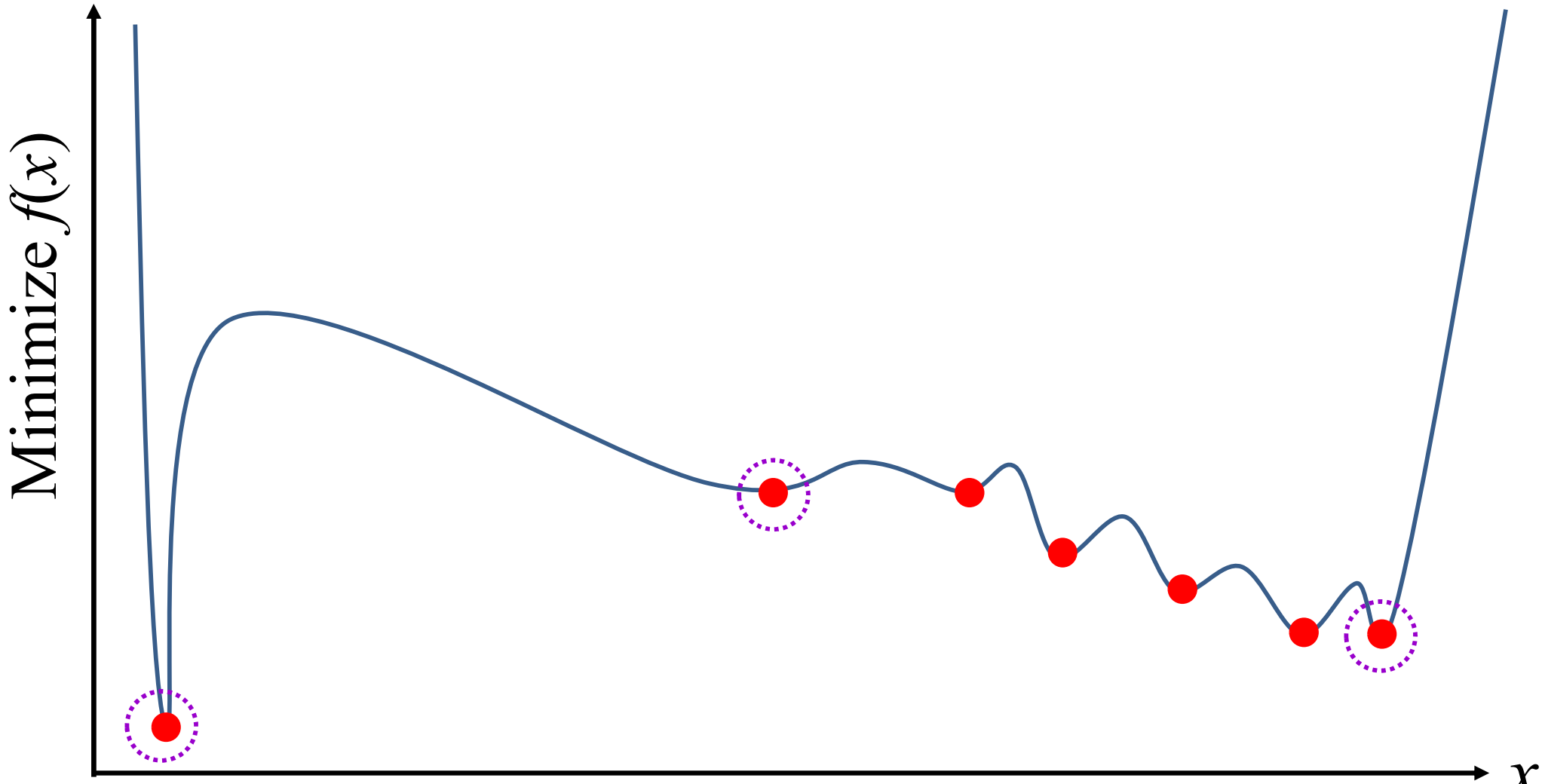
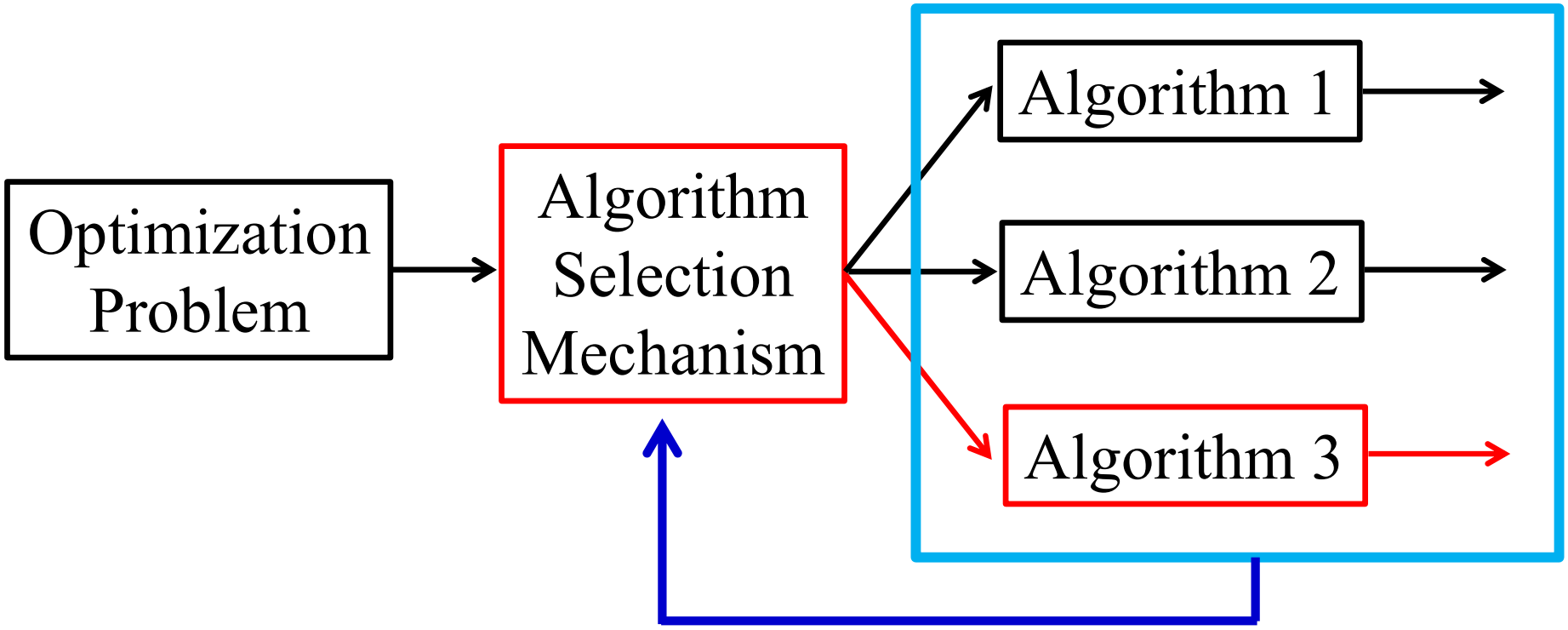


# Variable Neighborhood Search

## Algorithm Ensemble

Based on the characteristics of the problem,
an appropriate algorithm will be selected.




Depending on the progress of the search, the selection of an appropriate algorithm will be changed.

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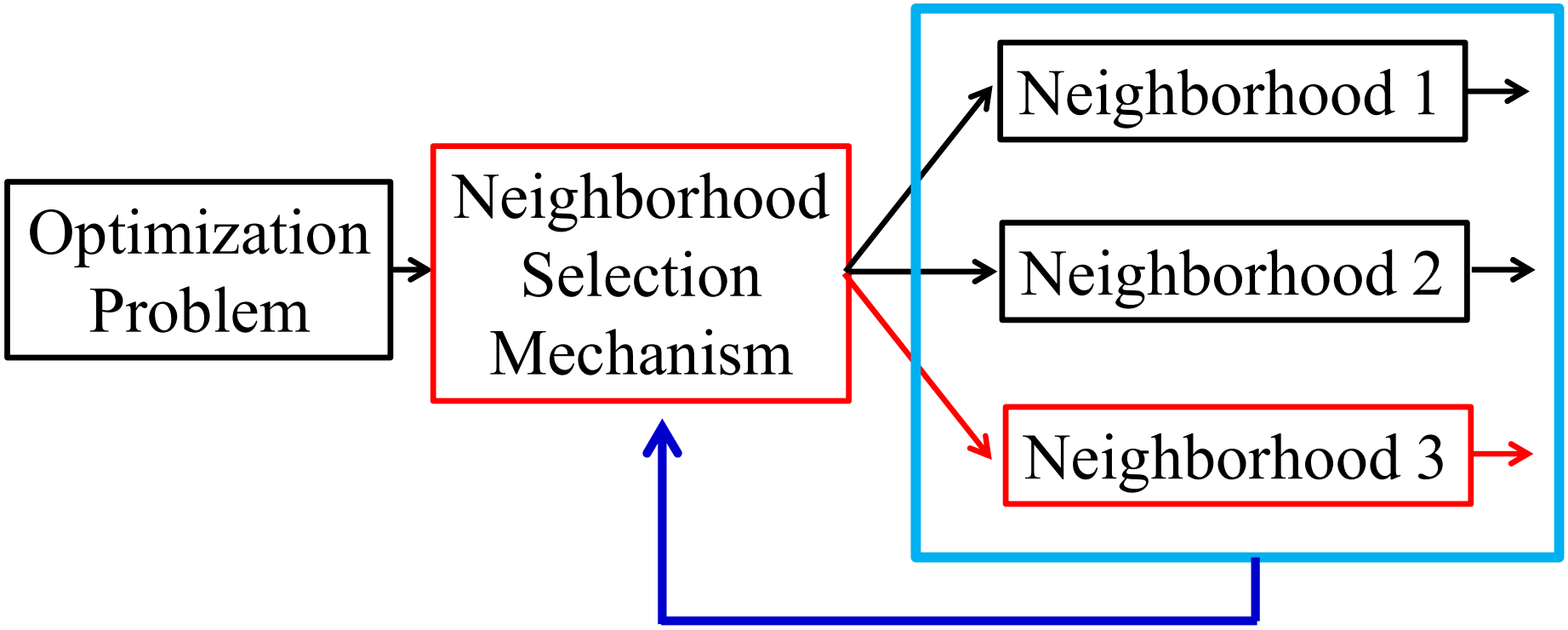


Depending on the progress of the search, the selection of an appropriate neighborhood will be changed.

# Large Neighborhood Search

**If you are interested in Large Neighborhood Search, please check the following references for detail:**

## Large Neighborhood Search

P. Shaw. Using constraint programming and local search methods to solve vehicle routing problems. In CP-98 (Fourth International Conference on Principles and Practice of Constraint Programming), volume 1520 of Lecture Notes in Computer Science, pages 417–431, 1998.

D, Pisinger, and S Ropke: Large Neighborhood Search. In Handbook of Metaheuristics, Springer (2019)

## Very Large Neighborhood Search

R. K. Ahuja, Ö. Ergun, J. B. Orlin, and A. P. Punnen. A survey of very largescale neighborhood search techniques. Discrete Applied Mathematics, 123: 75–102, 2002.

There are many other papers on “Large Neighborhood Search”.

## We may have the following situations

(i) The current solution cannot be improved by a small change.

(ii) The current solution can be efficiently improved by a large change.

**Subset selection problem**: 1,000 Items

Current solutions: 0001000100011100000010110111011 ... 0101

Hamming Distance 1: $_{1000}C_1$ = 1,000 neighbors. No improvement.

Hamming Distance 2: $_{1000}C_2$ = 499,500. No improvement.

Hamming Distance 3: $_{1000}C_3$ = 166,167,000. No improvement.

Hamming Distance 4: $_{1000}C_4$ = 41,417,124,750. **Too many.**

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It is difficult to examine the change of 4 bits by local search.

However, we can easily optimize more bits (e.g., 10 bits: 1,024 combinations)

**Current solutions:** 0001000100011100000010110111011 ... 0101

**New solutions:** 0001000001000111000010110111011 ... 0101

## We may have the following situations

(i) The current solution cannot be improved by a small change.

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**Current solutions:** 0001000100011100000101101110 11 ... 0101

**New solutions:** 0001000001000111000101101110 11 ... 0101

**Current solutions:** 0001000**0**0**1**0**00**1**1**00010110111011 ... 0101

**New solutions:** 0001000**0**0**1**0**00**1**1**00011001101011 ... 0101

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**New solutions:** 0001000010001110000101101110 11 ... 0101

**Current solutions:** 0001000010001110000101101110 11 ... 0101

**New solutions:** 0001000010001110001100110111 11 ... 0101

**Question:** Which part should be optimized ?

**Current solutions:** 000100010001110000010110111011 ... 0101

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**Current solutions:** 0001000100011100000101101111011 ... 0101

# Large Neighborhood Search

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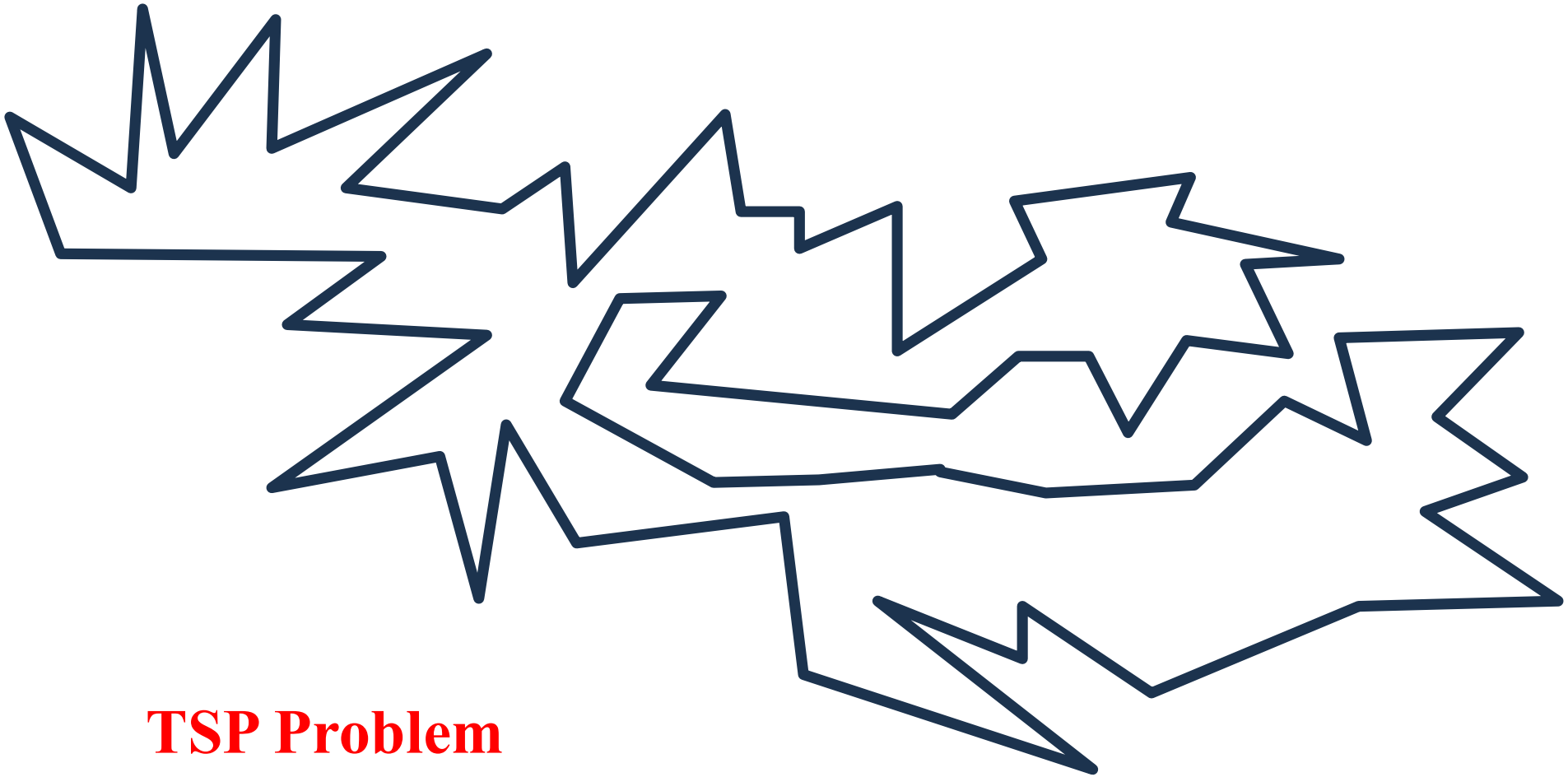

**TSP Problem**

# Large Neighborhood Search

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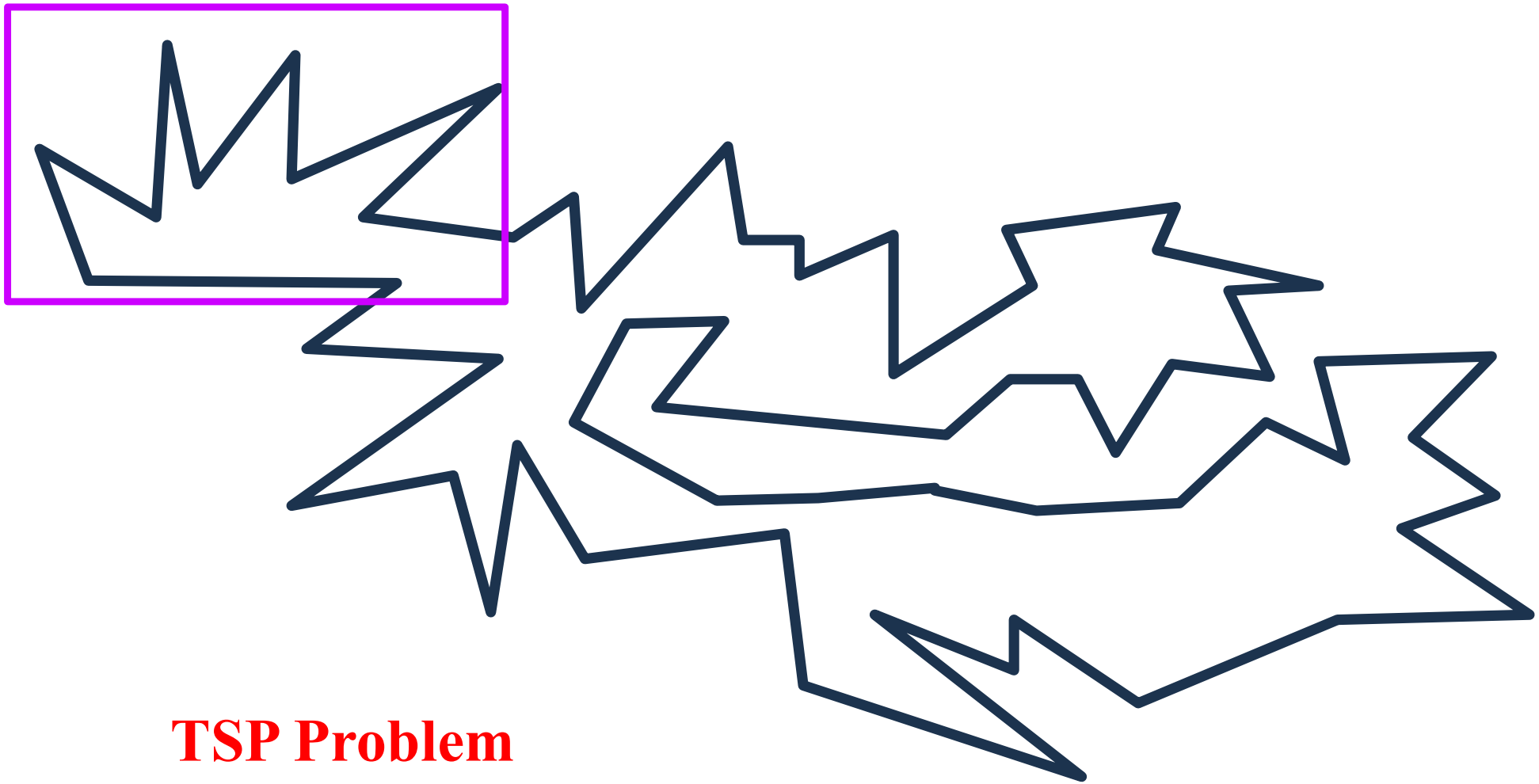

**TSP Problem**

# Large Neighborhood Search

**We may have the following situations**

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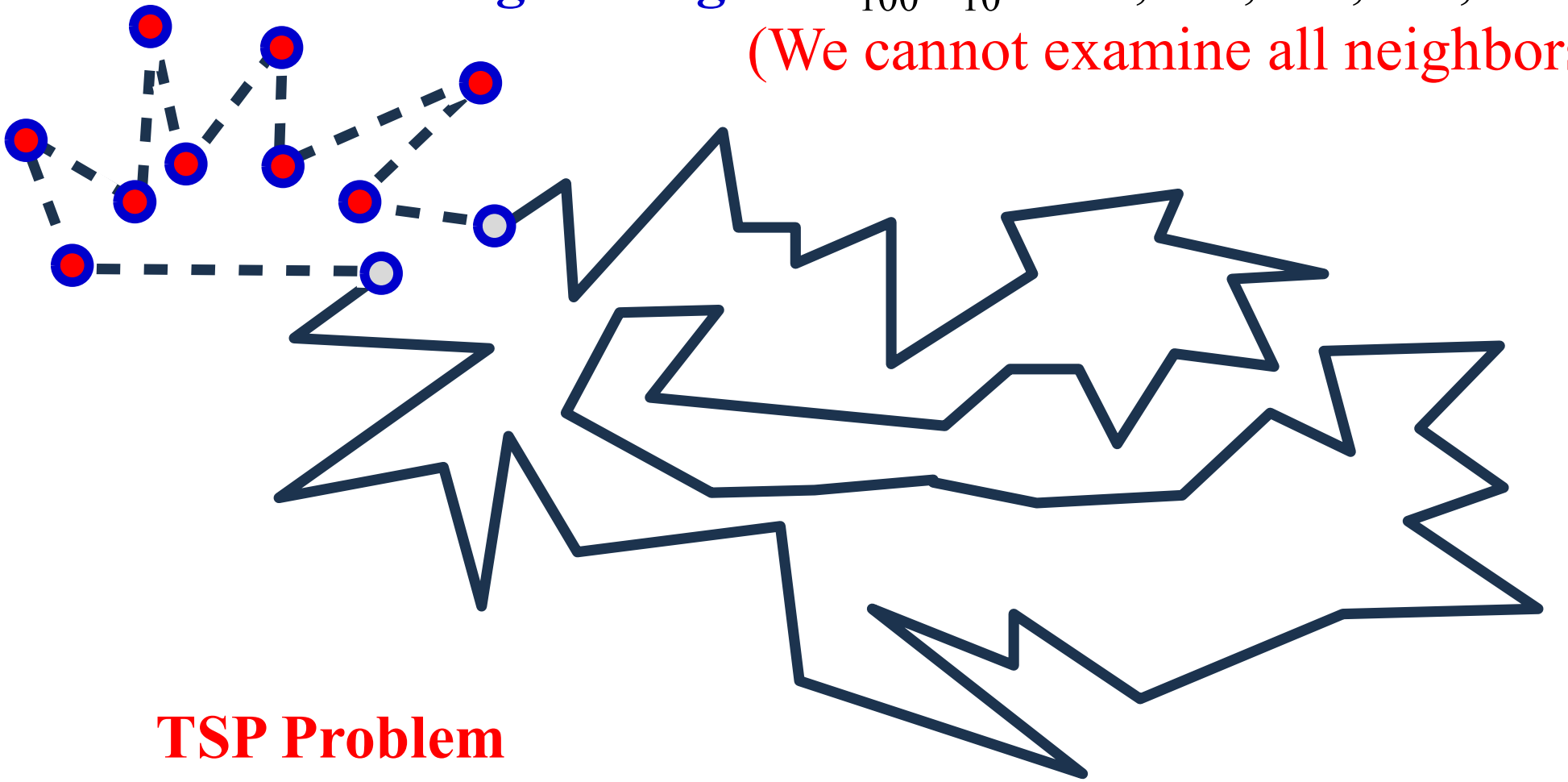
**Ten-edge changes** $\sim {}_{100}C_{10} = 17{,}120{,}086{,}275{,}600$

(We cannot examine all neighbors)



**TSP Problem**

# Large Neighborhood Search

**We may have the following situations**

(i) The current solution cannot be improved by a small change.

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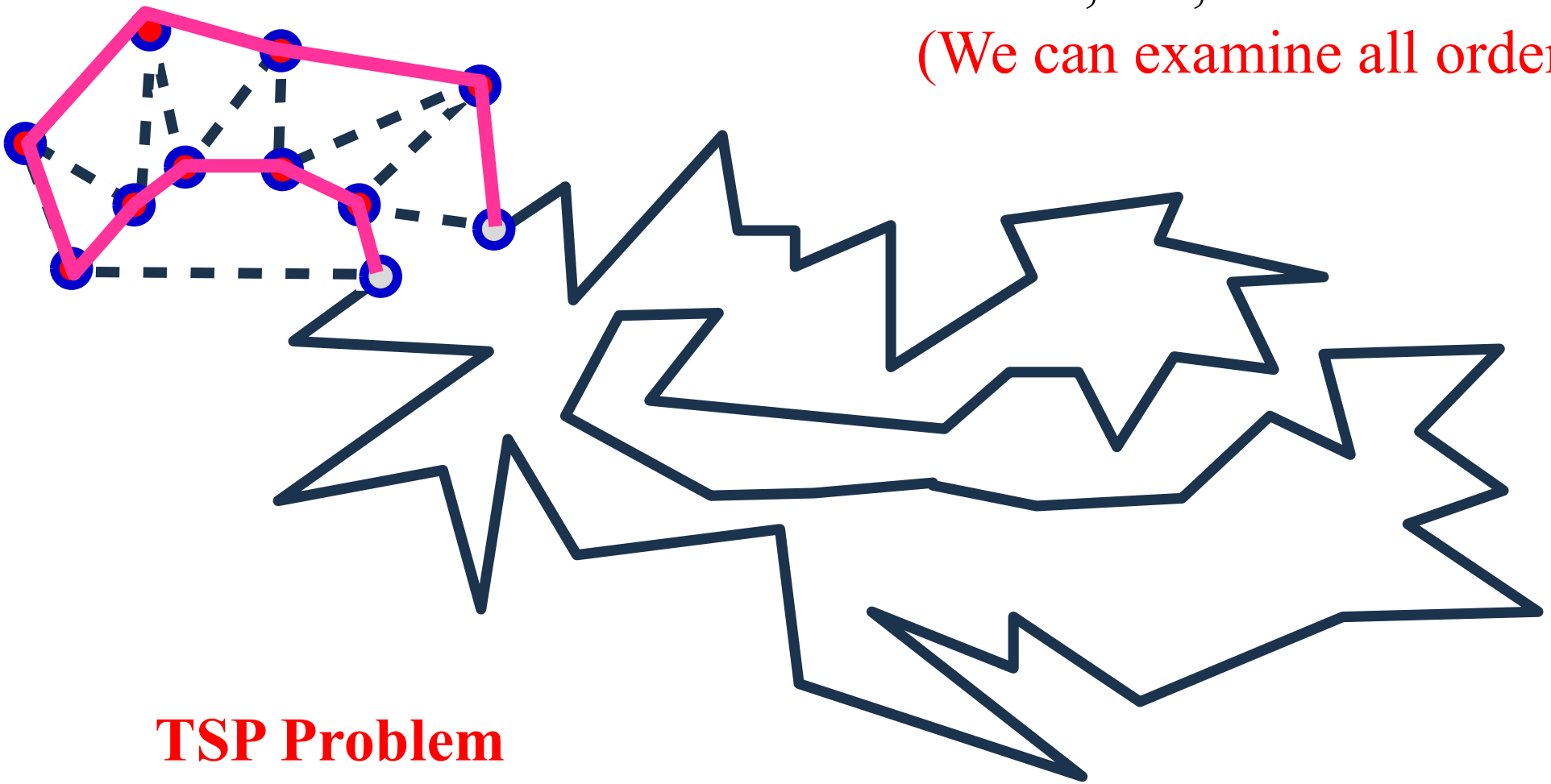
**All order of ten cities:** 10! = 3,628,800

(We can examine all orders )



**TSP Problem**

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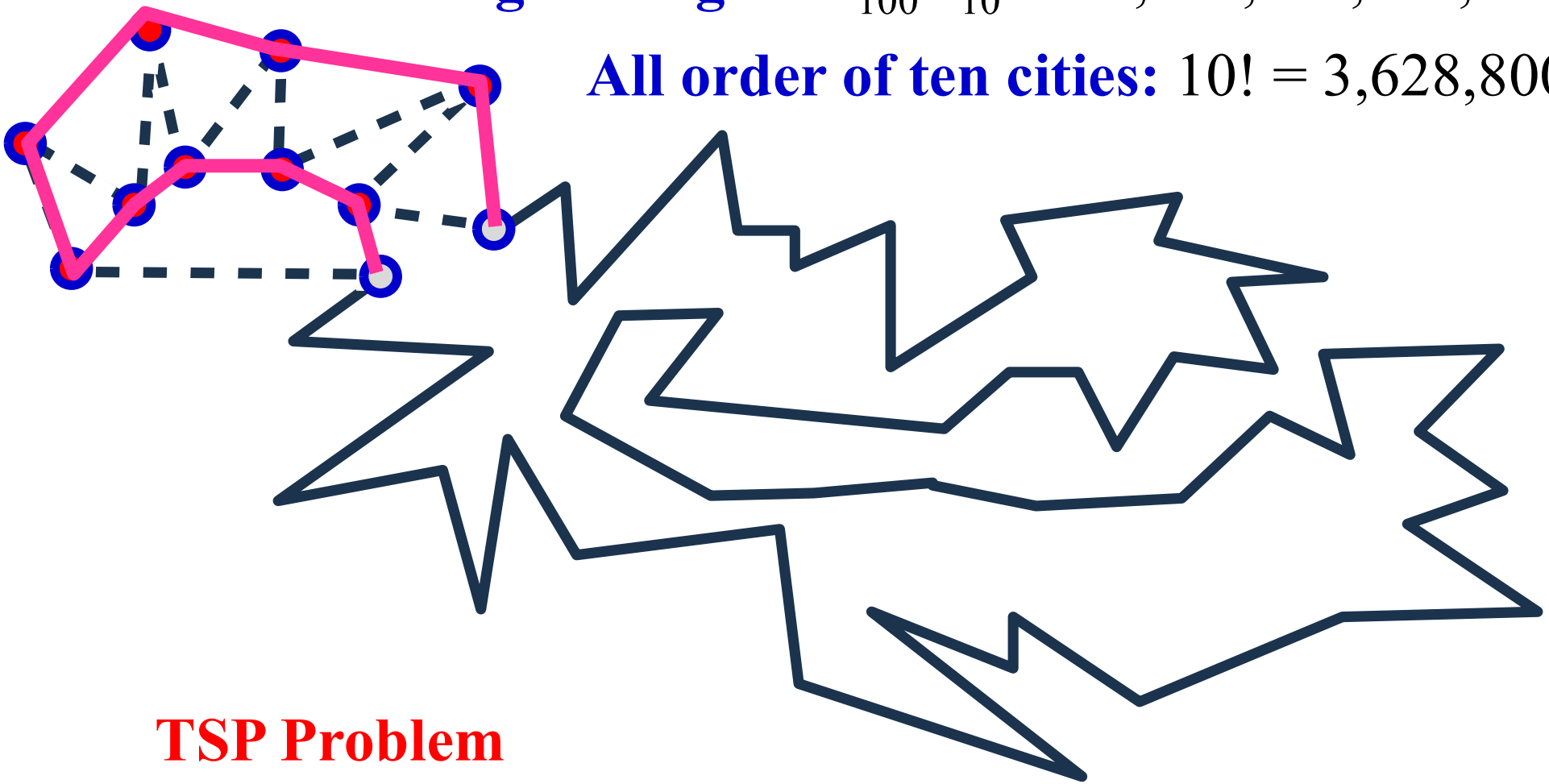
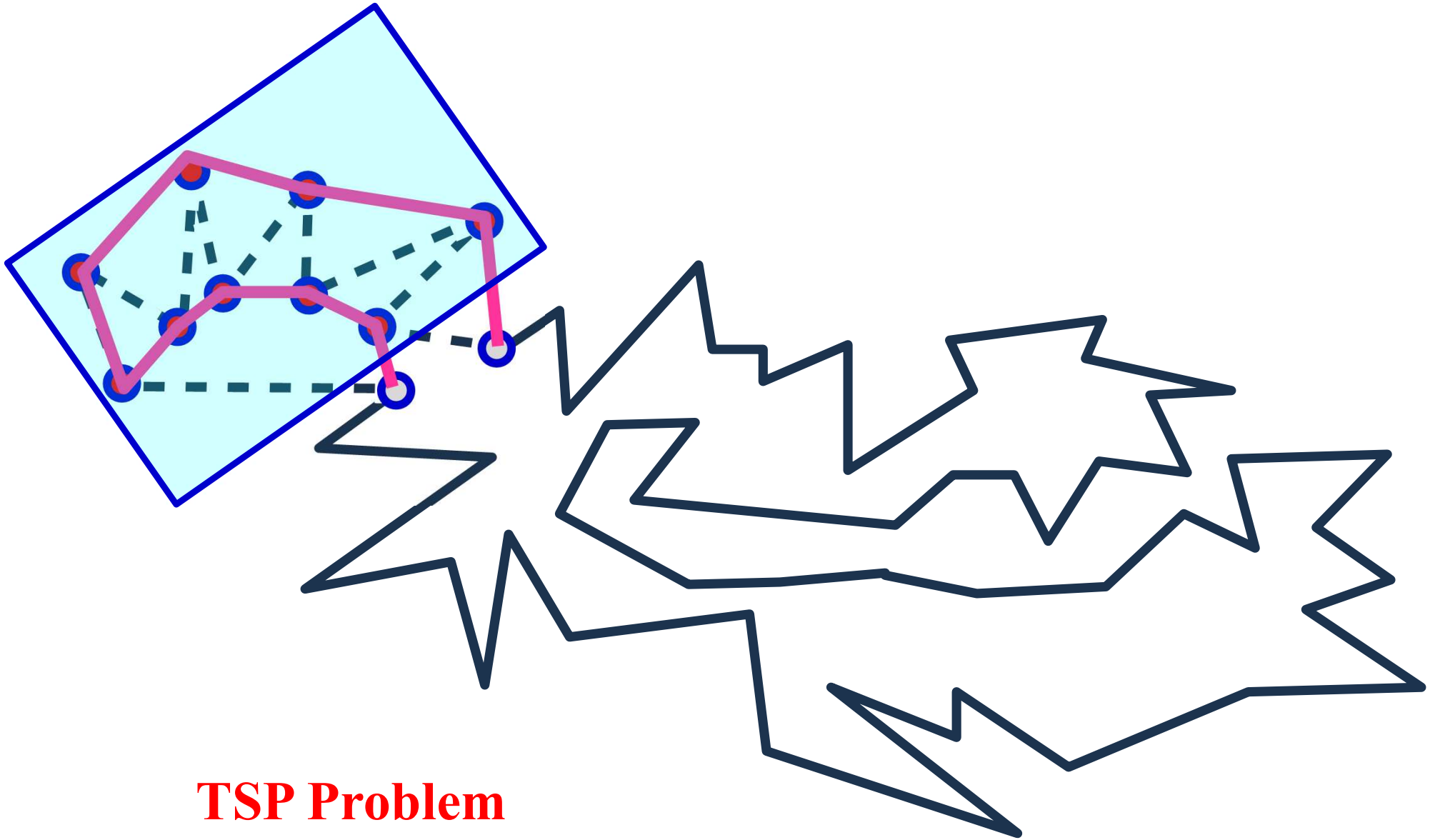


**TSP Problem**

# Large Neighborhood Search

**Step 1.** Choose a part of the current solution to be optimized.

**Step 2.** The selected part is optimized.



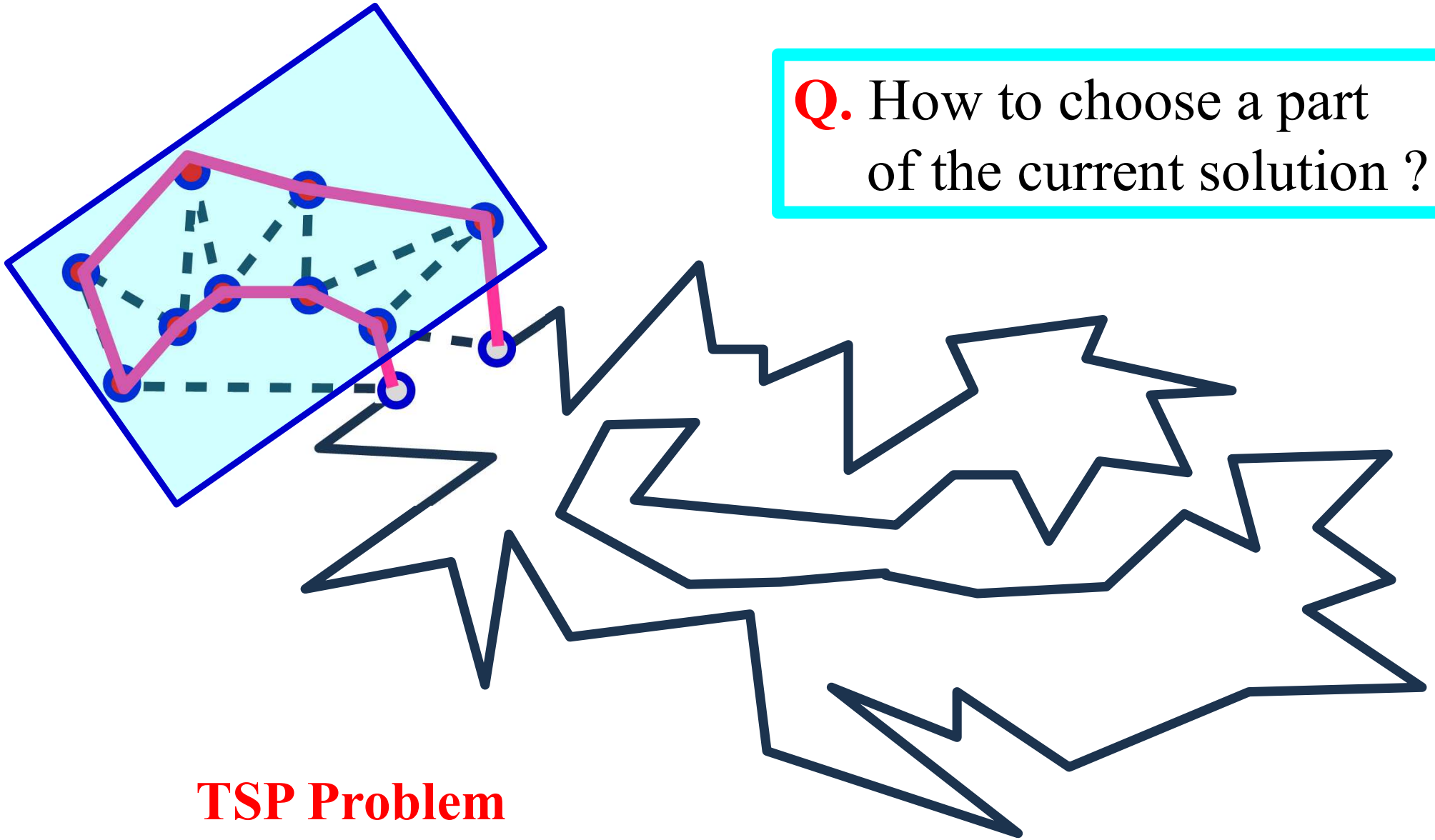
**TSP Problem**

# Large Neighborhood Search

**Step 1.** Choose a part of the current solution to be optimized.

**Step 2.** The selected part is optimized.

**Q.** How to choose a part of the current solution ?

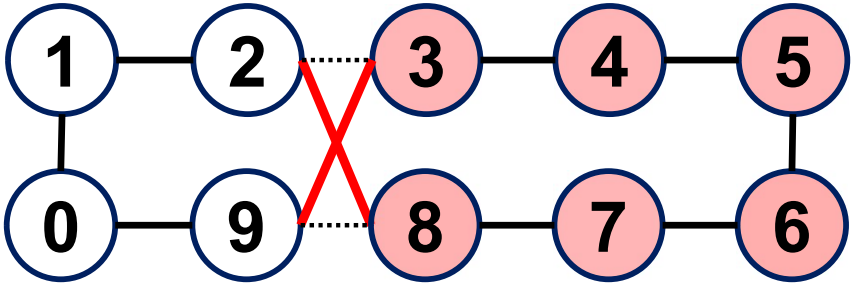


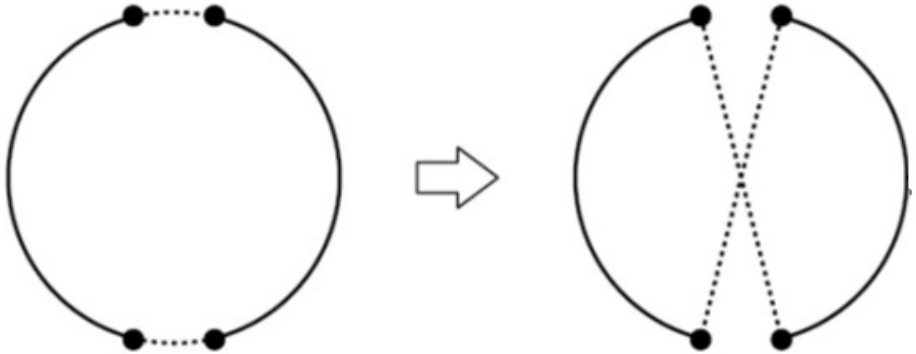
**TSP Problem**

# Lab Session Task 1

## Local search

- Neighborhood structure: Inversion (two-edge change)



## Creation of a new initial solution:

**- 1st initial solution: Greedy solution**

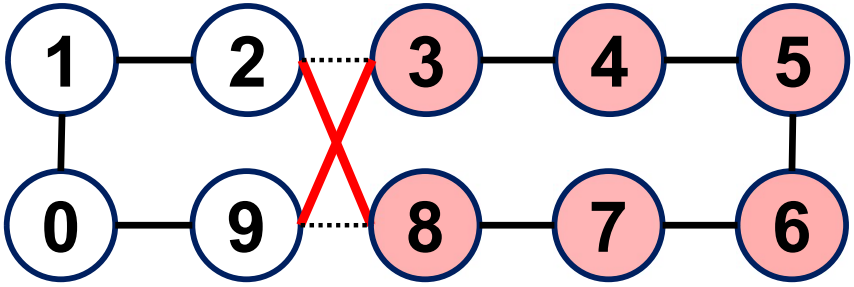
### Task 1

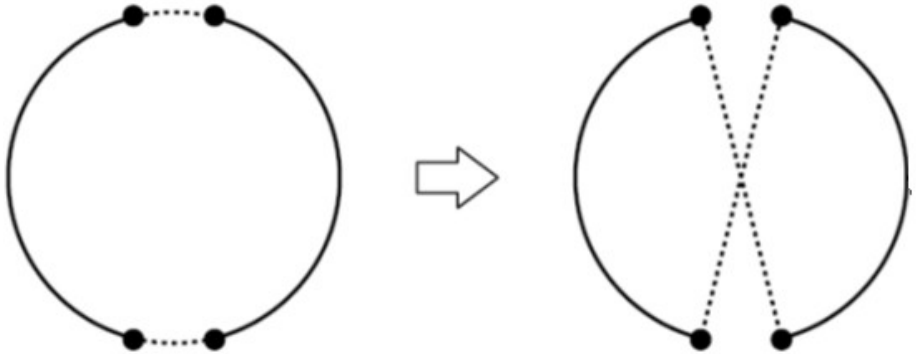
Create a TSP problem where a non-optimal greedy solution is obtained from an initial city and the obtained greedy solution cannot be improved by inversion-based local search. It is enough to show that one greedy solution cannot be improved. You do not have to show that any greedy solution from an arbitrary initial city cannot be improved.

# Lab Session Task 2

## Local search

- Neighborhood structure: Inversion (two-edge change)

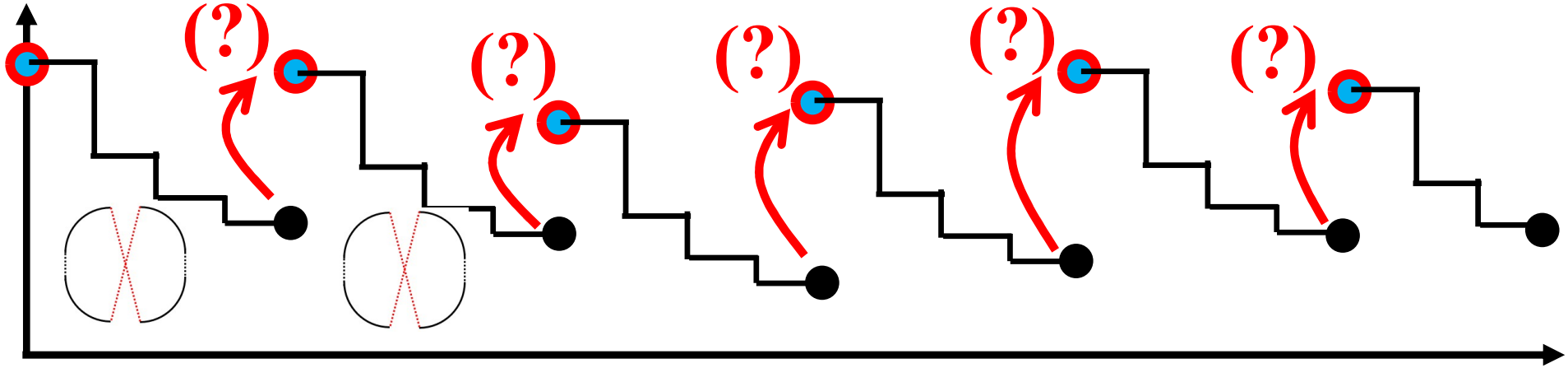


## Creation of a new initial solution:

**- 1st initial solution: Greedy solution**

**- Other initial solutions: Explain your own idea about how to create an initial solution for each local search run (Task 2).**



# Lab Session Task 3

Specify neighborhood structures $N_1$, $N_2$, $N_3$, ... for designing a Variable Neighborhood Descent (VND) algorithm for a knapsack problem.